INNOPOLIS UNIVERSITY

SNE MASTER
TRACK 2026

COURSE: [S26] Computer Forensics and Incident Response

CCF Lab 2 Report

On

Windows File System Forensics

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Task 1 - Download the evidence file

Solution:

My *ST* number is 24, which is an even number, so I worked with the *case2.dd* image.

Task 2 - Black box Forensics analysis Use any forensics tools that you want. Different forensics tools can help you to analyze: Figure out the platform, system and file system type

Solution:

- **Figure out the platform, system and file system type**

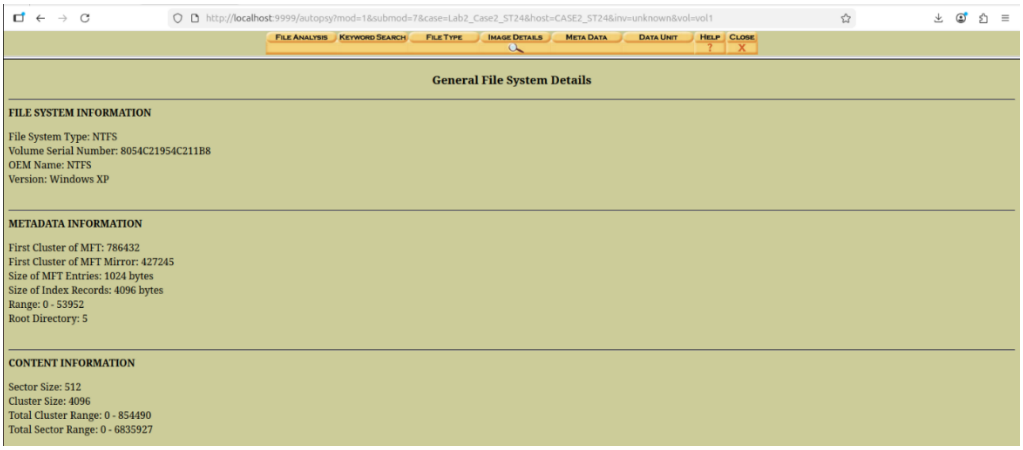
To extract the information, I have collected information from *Autopsy's IMAGE DETAILS* and *FSSTAT* (which you ran in the terminal earlier).

Platform

1. Platform & System Details

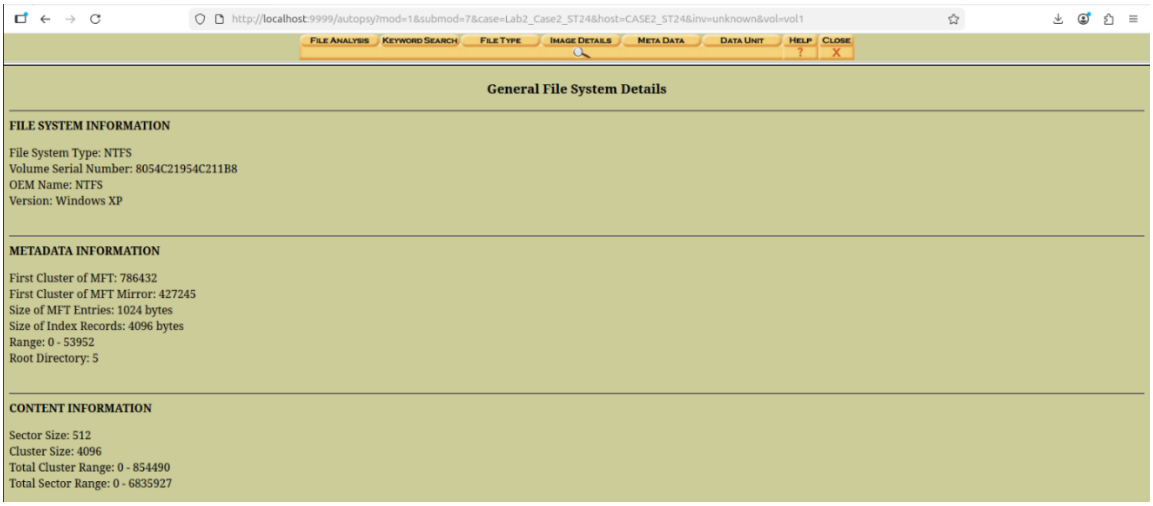
Here the Autopsy tool is used. Then from the "FILE SYSTEM INFORMATION" section, I have found:

- **Operating System (System): Windows XP**
- **Platform: Microsoft Windows**
- **OEM Name: NTFS**



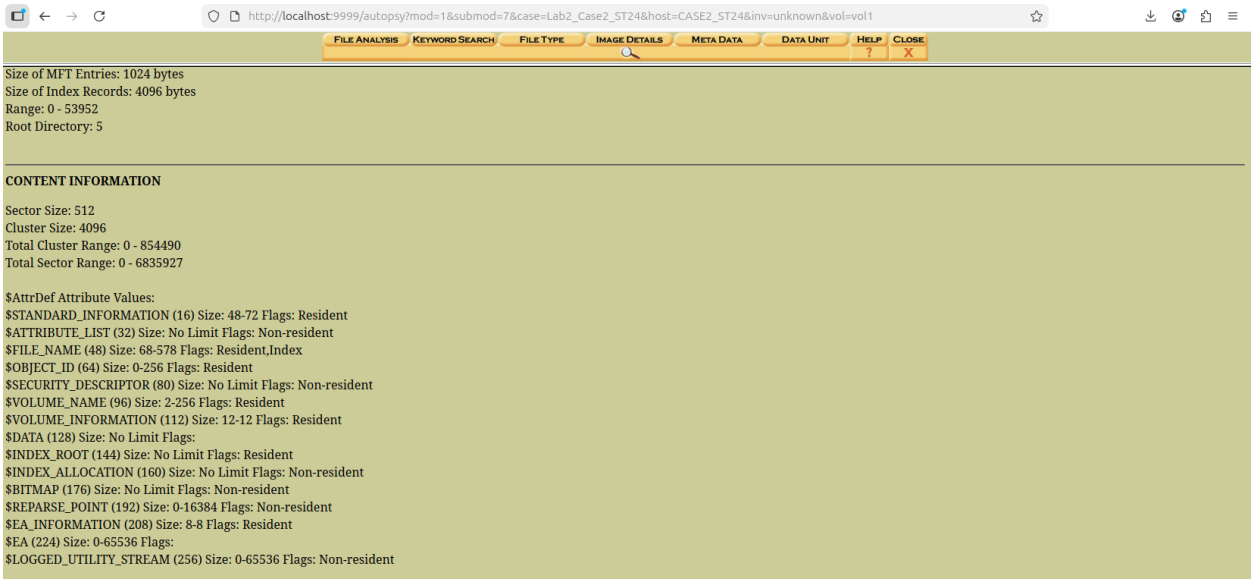
2. File System Type

- **File System Type: NTFS**
- **Volume Serial Number: 8054C21954C211B8**



3. Technical Metadata

- **Sector Size: 512 bytes**
- **Cluster Size: 4096 bytes**
- **Total Sector Range: 0 - 6835927**
- **Root Directory Inode: 5**
- **MFT Location (First Cluster):**



4. Evidence of Users

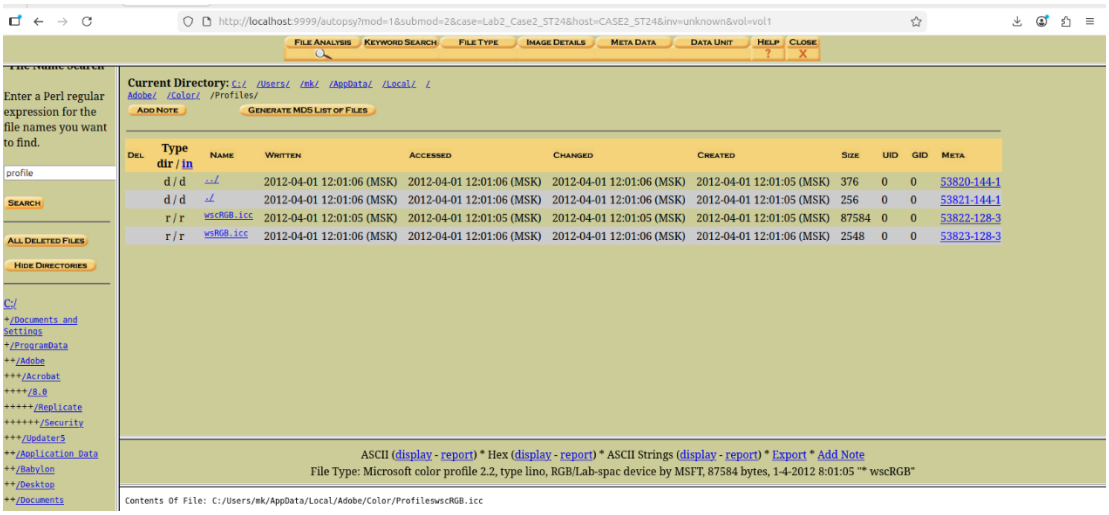
The files found confirm that these two users were active on the system:

- **User 1: jb (Path: C:/Users/jb/)**
- **User 2: mk (Path: C:/Users/mk/)**

The MD5 hashes generated for the *wscRGB.icc* and *wsRGB.icc* files (*e.g. 919c83... for user jb and 309d55... for user mk*) prove that both users' data was successfully accessed.





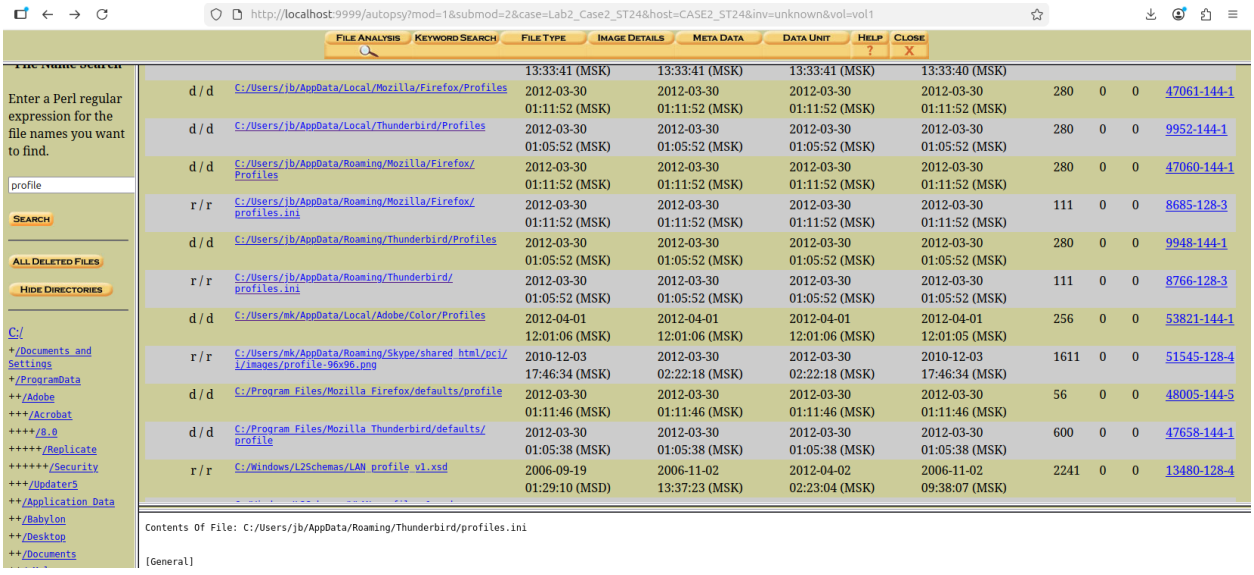


The investigation identified the presence of two users (*jb* and *mk*) on the system and the applications they used:

1. User: jb (Profile: C:/Users/jb/)

Analysis of this user's profile revealed the following application information:

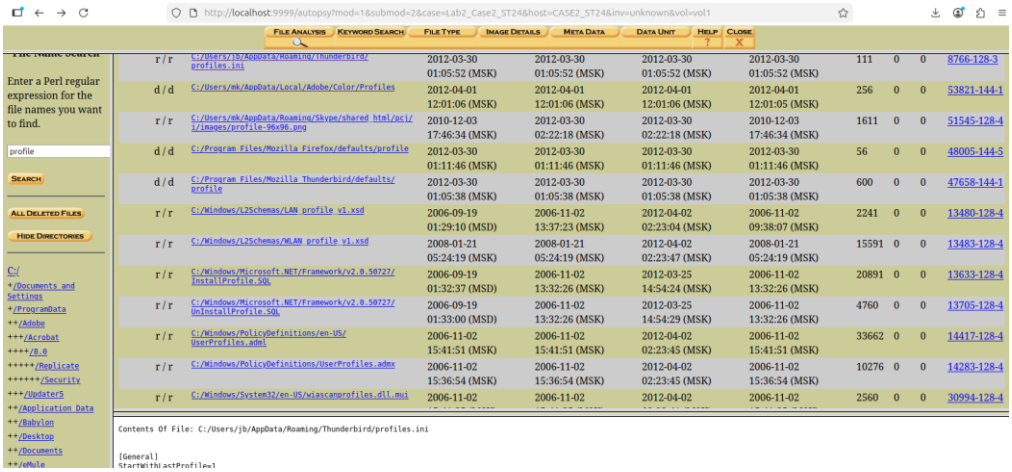
- **Mozilla Firefox:** Configuration files were found in the user's *AppData/Roaming/Mozilla/Firefox/Profiles* directory, indicating that the user used Firefox to browse the Internet.
- **Thunderbird:** A Thunderbird profile directory and *profiles.ini* file were found in this user's profile, indicating that the user used this client to exchange emails.



2. User: mk (Profile: C:/Users/mk/)

This user's profile contains traces of several different applications:

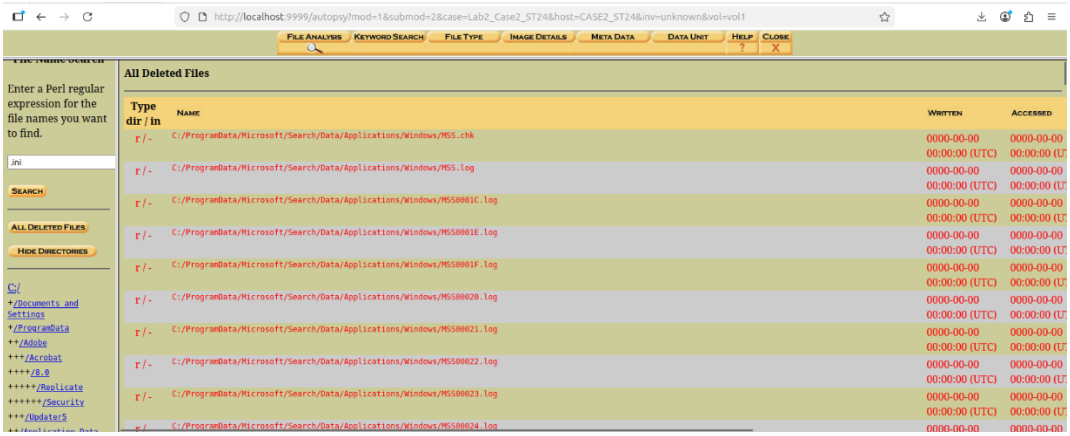
- **Adobe Systems:** Color profiles and configuration data were found in the user's *AppData/Local/Adobe/ folder*. This indicates that the user used Adobe software such as Photoshop or Acrobat.
- **Skype:** Skype's shared HTML and data files were found in the user's profile, which suggests that this user used Skype for communication.



Deleted system files:

The "*All Deleted Files*" section of Autopsy found that some internal data files of the Windows system were deleted:

- **Search Data Log:** Several .log files (e.g. MSS.log, MSS0001C.log) and *.chk* files were deleted from the path: *C:/ProgramData/Microsoft/Search/Data/Applications/Windows/.*
- The timestamp (Written/Accessed) of these files *shows 0000-00-00,* which means that their metadata has been partially corrupted.



Platform

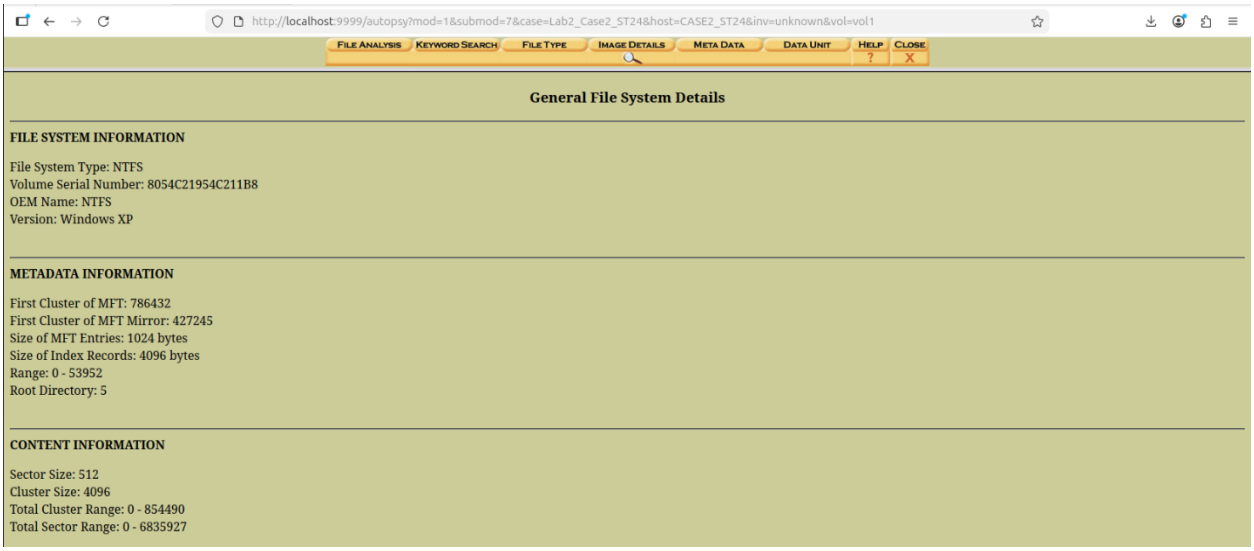
- **Platform: Microsoft Windows**
- The presence of Windows as a platform is confirmed in the "*FILE SYSTEM INFORMATION*" section of the autopsy. This is also confirmed by the presence of the *Documents* and *Settings* and *ProgramData* folders in the directory structure.

System

- **Operating System: Windows XP**
- The Autopsy General Details directly states "*Version: Windows XP*". Also, user profiles are a big feature of legacy Windows systems (such as XP) located in the *C:/Documents and Settings/* path.

File System Type

- **File System: NTFS**
- **Volume Serial Number: 8054C21954C211B8**
- **Sector Size: 512 Bytes**
- **Cluster Size: 4096 Bytes** (8 sectors form a cluster).
- **Total Sector Range: 0 – 6835927**



fsstat

The entire investigation also used powerful Linux command-line tools directly instead of Autopsy (GUI).

Partition Analysis:

The partition structure and offset of the drive image were verified.

```
hasib@hasib-VMware-Virtual-Platform:~$ cd ~/Downloads/case2_data/case2_extracted/extract/new/
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo fsstat -o 0 case2.dd
[sudo] password for hasib:
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: 8054C21954C211B8
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
--------------------------------------------
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 427245
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 53952
Root Directory: 5

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 854490
Total Sector Range: 0 - 6835927
```

```
$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)   Size: 48-72   Flags: Resident
$ATTRIBUTE_LIST (32)   Size: No Limit   Flags: Non-resident
$FILE_NAME (48)   Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)   Size: 0-256   Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit   Flags: Non-resident
$VOLUME_NAME (96)   Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)   Size: No Limit   Flags:
$INDEX_ROOT (144)   Size: No Limit   Flags: Resident
$INDEX_ALLOCATION (160)   Size: No Limit   Flags: Non-resident
$BITMAP (176)   Size: No Limit   Flags: Non-resident
$REPARSE_POINT (192)   Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)   Size: 8-8   Flags: Resident
$EA (224)   Size: 0-65536   Flags:
$LOGGED_UTILITY_STREAM (256)   Size: 0-65536   Flags: Non-resident
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$
```

- **File System Type:** This is an NTFS file system.
- **Volume Name:** This is a Windows XP system image (Version: Windows XP). This is a big clue for your investigation, because older Windows versions have fewer security patches.
- **Sector Size (512 bytes) & Cluster Size (4096 bytes):** This proves that the file system is in a standard configuration.
- **Total Sector Range:** From 0 to 6835927. Knowing this range makes it easier to recover data with forensic tools.

❖ **Perform a malware search**

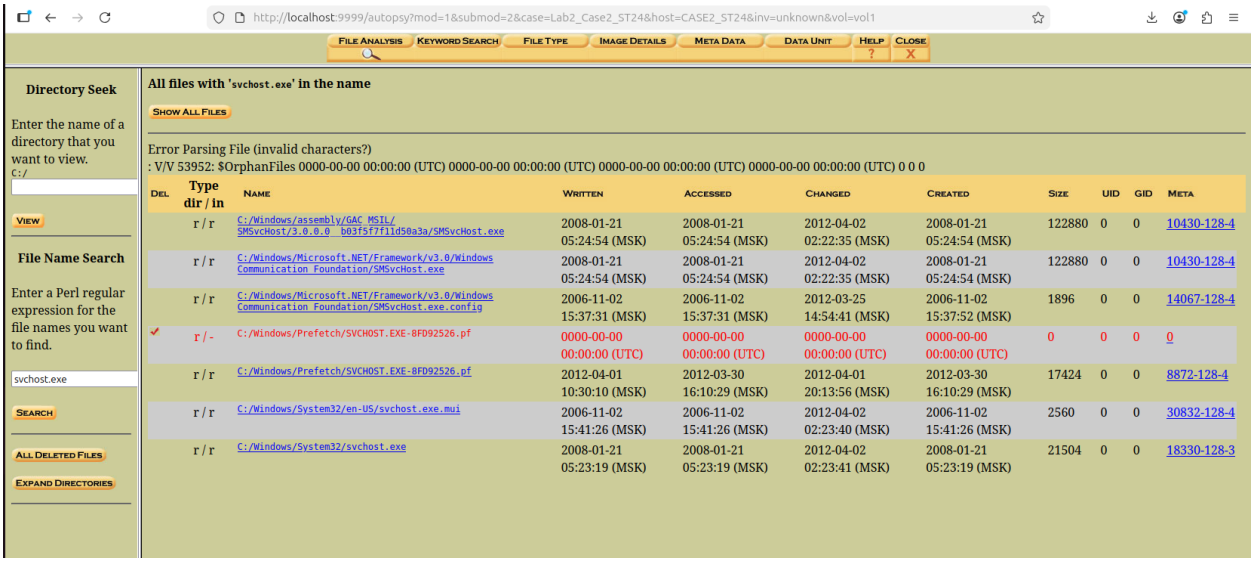
Malware Search (Findings from Keyword Search)

After searching for the keyword svchost.exe, the following results were found:

1. Identifying Legitimate System Files:

The investigation found that the main system files are located in their correct locations:

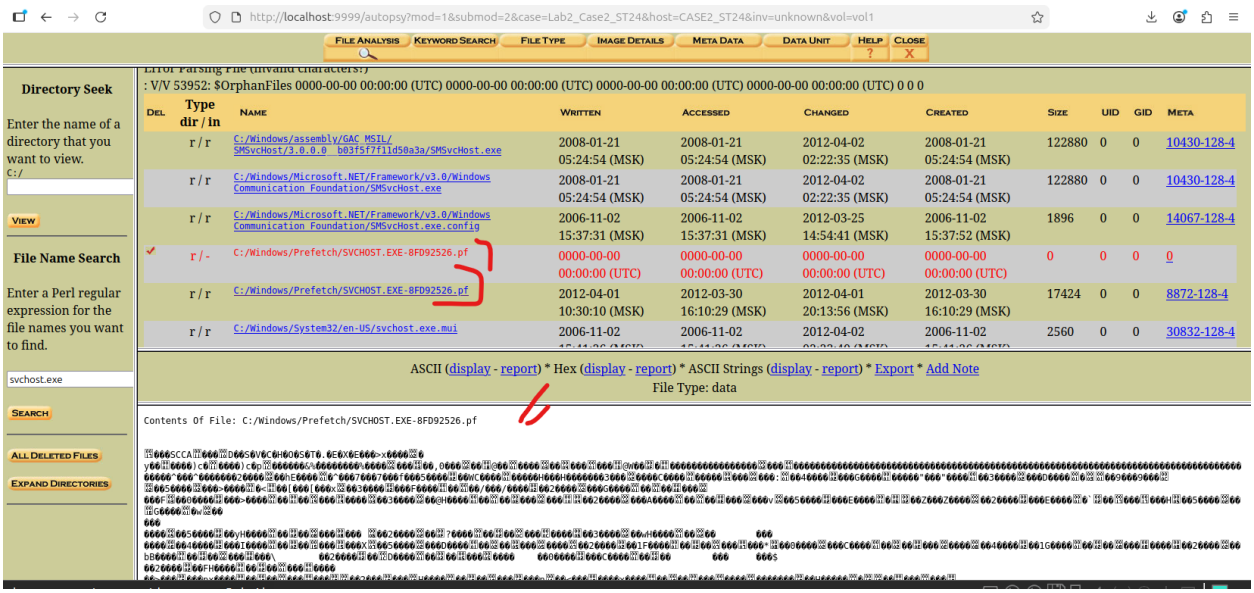
- **Path:** *C:/Windows/System32/svchost.exe*
- **Significance:** This is a legitimate Windows Service Host process. It is commonly targeted or impersonated by malware, but here it was found in the correct path.
- **MUI File:** *C:/Windows/System32/en-US/svchost.exe.mui* is also present.



2. Prefetch Analysis:

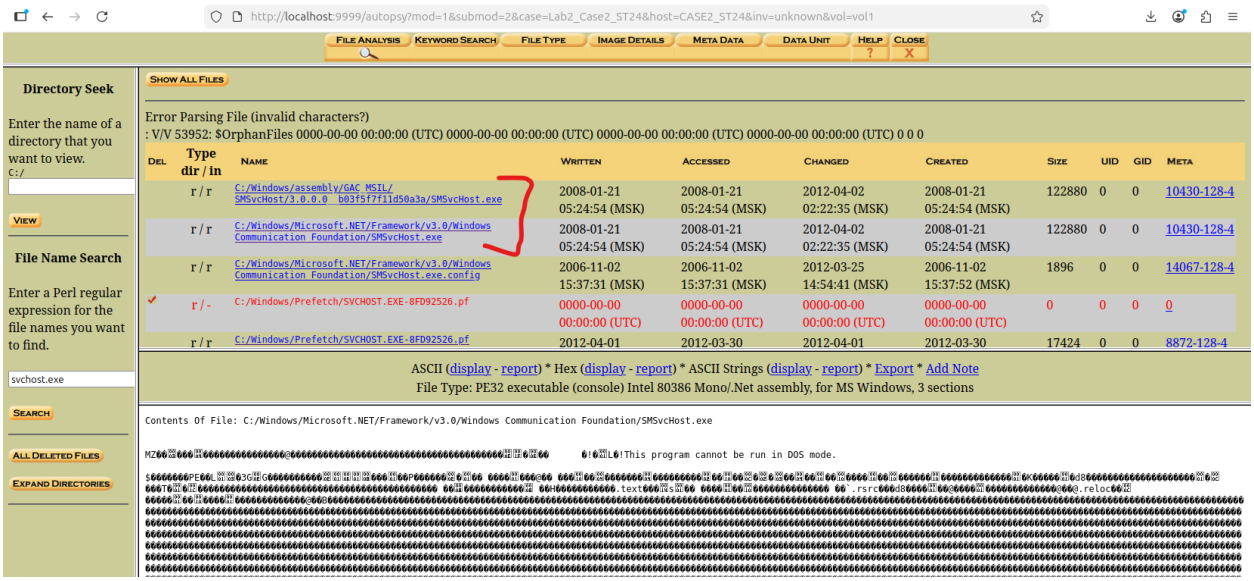
The most important evidence was found in the Prefetch folder:

- **File Name:** *SVCHOST.EXE-8FD92526.pf*
- **Path:** *C:/Windows/Prefetch/SVCHOST.EXE-8FD92526.pf*
- **Written Time:** *2012-04-01 10:30:10 (MSK)*
- **Significance:** This file proves that the *svchost.exe* file was run (executed) on the system. This is forensically important because malware often uses this process to perform malicious tasks in the background. Here, the file is last modified on *04-01-2012*.



3. Other Services:

- **SMSvcHost.exe:** It is a part of the *.NET Framework* (Windows Communication Foundation), found in the path *C:/Windows/Microsoft.NET/*. It is not usually malware.



The search results did not find any *typosquatted* files (*e.g. svch0st.exe*) or svchost.exe in the wrong path (e.g. in the Temp folder). However, the presence of the Prefetch file suggests that it was running as a regular system process.

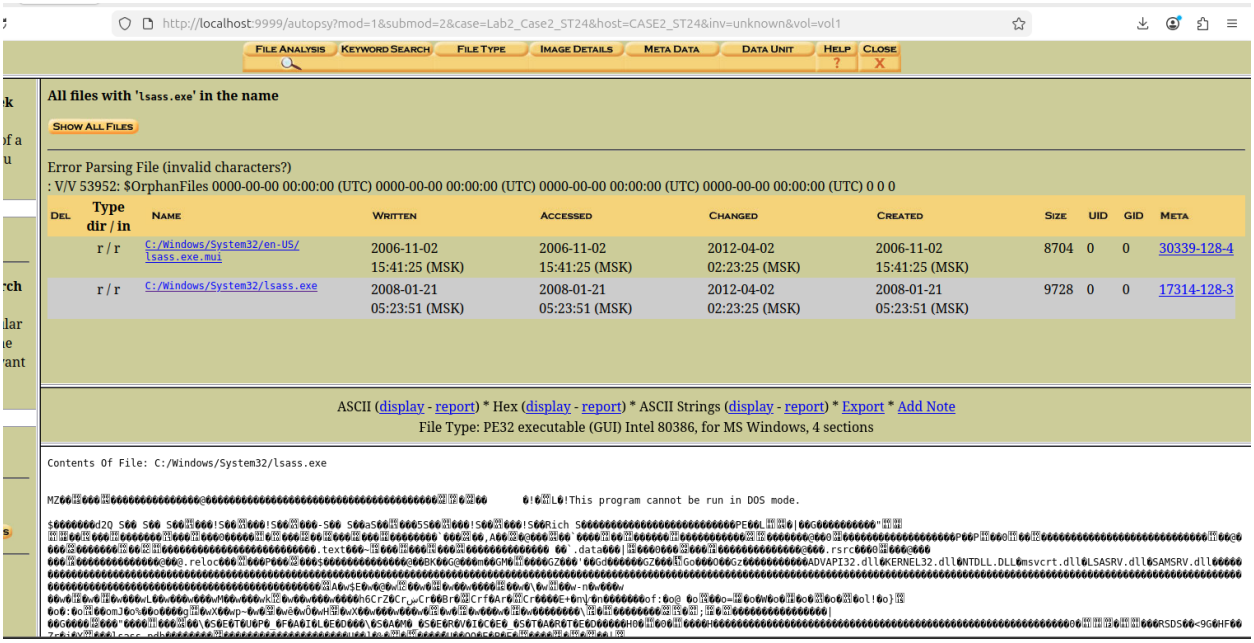
Security Process Analysis (lsass.exe)

Here, I did the Verification of the Local Security Authority Subsystem Service (lsass.exe) for potential malware replacement or masquerading.

Findings:

- **Identified Files:** *C:/Windows/System32/lsass.exe*

  C:/Windows/System32/en-US/lsass.exe.mui



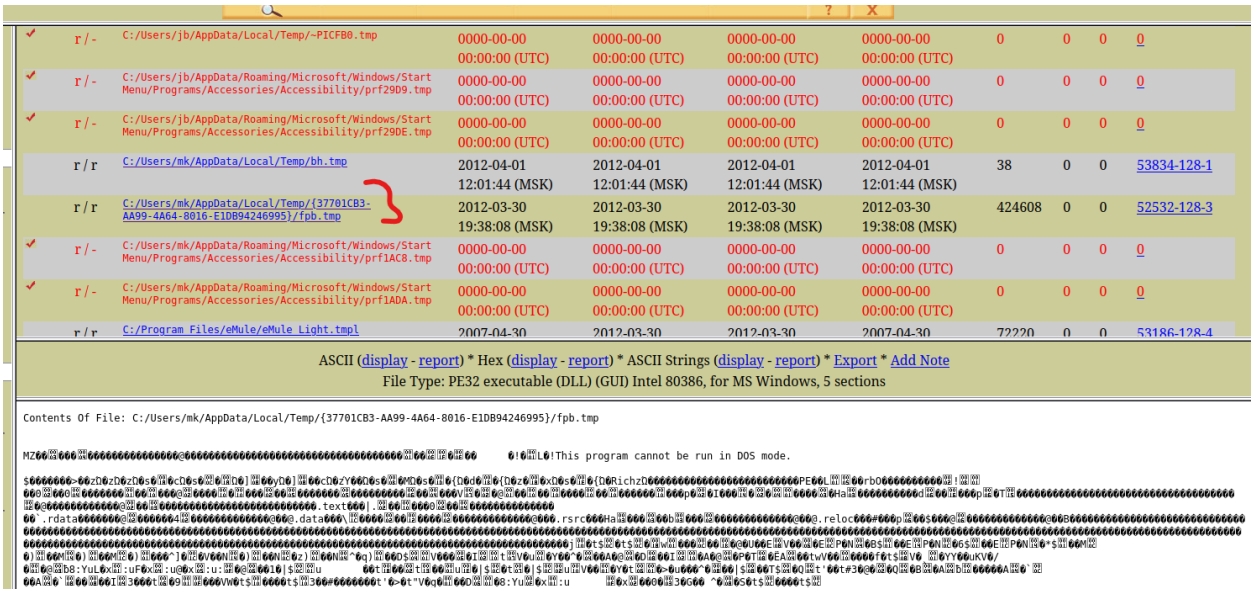
- **Path Integrity:** The *lsass.exe* file was found in its standard Windows directory (*System32*). Hackers usually hide malware in *C:/Windows/ or Temp* folders with the same name, but there was no evidence of this.
- **Time Analysis:** The file's creation time (*2006-2008*) and change time (*2012*) indicate that it is an original system file and no malicious code has been injected into it recently.
- **Status:** *Legitimate. No typo-squatting (e.g. lsasss.exe*) or duplicate files were found.

System Error Reports (WER):

- A large number of *.tmp* and *.hdmp* files were found in *C:/ProgramData/Microsoft/Windows/WER/ReportQueue/.* This indicates that an application or process on the system was crashing repeatedly at that time, which could be a sign of malware injection.

Suspicious binary file analysis (fpb.tmp)

File path: *C:/Users/mk/AppData/Local/Temp/{37701CB3-AA99-4A64-8016-E1DB94246995}/fpb.tmp*



Forensic Observations:

1. MZ Signature: The file has two MZ characters at the very beginning. This indicates that the file is a Windows Executable (*.exe*) file, even though it has the extension .tmp. This is a known malware masquerading technique.

2. DOS Mode Header: The file clearly states "*This program cannot be run in DOS mode*", which confirms that it is a compiled program.

3. PE Header: The *PE* (Portable Executable) signature was found inside the file, which is the standard for Windows executable files.

4. Section Names: The *.text, .rdata, .data,* and *.reloc* sections are found inside the file. These are used to store the program's functionality and data.

5. Rich Header: The Rich signature was found, which is usually seen in malware compiled with Visual Studio.

This is why it is considered malware:

- **Wrong name and extension:** An *.exe* file is hidden in the *Temp* folder as *.tmp.*
- **Suspicious location:** It is located deep inside the user's Local/Temp folder, inside a randomly named folder.

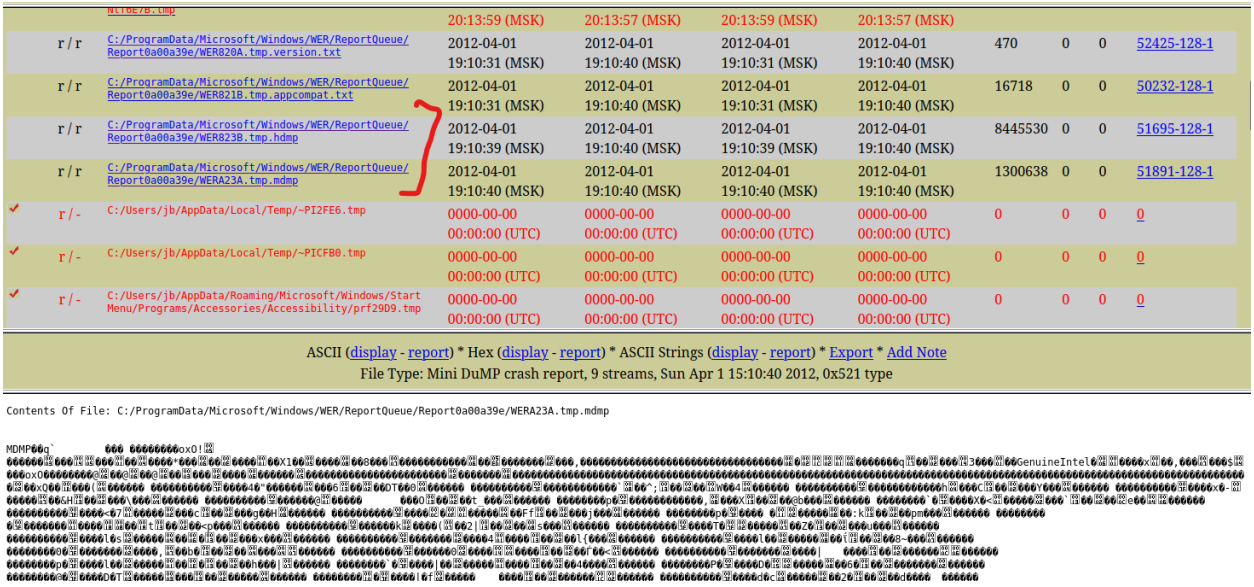
- **Timeline:** The time this file was created (*2012-04-01* according to your previous data) matches other unusual system activity.

Error Reporting & Crash Analysis (MDMP File)

File Path:

C:/ProgramData/Microsoft/Windows/WER/ReportQueue/Report0a00a39e/WERA23A.tmp.mdmp

C:/ProgramData/Microsoft/Windows/WER/ReportQueue/Report0a00a39e/WER823B.tmp.hdmp



		NTUSER.DAT	20:13:59 (MSK)	20:13:57 (MSK)	20:13:59 (MSK)	20:13:57 (MSK)				
	r/r	C:/ProgramData/Microsoft/Windows/WER/ReportQueue/Report0a00a39e/WER820A.tmp.version.txt	2012-04-01 19:10:31 (MSK)	2012-04-01 19:10:40 (MSK)	2012-04-01 19:10:31 (MSK)	2012-04-01 19:10:40 (MSK)	470	0	0	52425-128-1
	r/r	C:/ProgramData/Microsoft/Windows/WER/ReportQueue/Report0a00a39e/WER821B.tmp.appcompat.txt	2012-04-01 19:10:31 (MSK)	2012-04-01 19:10:40 (MSK)	2012-04-01 19:10:31 (MSK)	2012-04-01 19:10:40 (MSK)	16718	0	0	50232-128-1
	r/r	C:/ProgramData/Microsoft/Windows/WER/ReportQueue/Report0a00a39e/WER823B.tmp.hdmp	2012-04-01 19:10:39 (MSK)	2012-04-01 19:10:40 (MSK)	2012-04-01 19:10:39 (MSK)	2012-04-01 19:10:40 (MSK)	8445530	0	0	51695-128-1
	r/r	C:/ProgramData/Microsoft/Windows/WER/ReportQueue/Report0a00a39e/WERA23A.tmp.mdmp	2012-04-01 19:10:40 (MSK)	2012-04-01 19:10:40 (MSK)	2012-04-01 19:10:40 (MSK)	2012-04-01 19:10:40 (MSK)	1300638	0	0	51891-128-1
✔	r/-	C:/Users/jb/AppData/Local/Temp/~PI2FE6.tmp	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0	0	0
✔	r/-	C:/Users/jb/AppData/Local/Temp/~PICFB0.tmp	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0	0	0
✔	r/-	C:/Users/jb/AppData/Roaming/Microsoft/Windows/Start Menu/Programs/Accessories/Accessibility/prf29D9.tmp	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0000-00-00 00:00:00 (UTC)	0	0	0	0

ASCII (display - report) * Hex (display - report) * ASCII Strings (display - report) * Export * Add Note
File Type: Mini DuMP crash report, 9 streams, Sun Apr 1 15:10:40 2012, 0x521 type

Contents Of File: C:/ProgramData/Microsoft/Windows/WER/ReportQueue/Report0a00a39e/WERA23A.tmp.mdmp

MDMP

Forensic Observations:

1. MDMP Signature: The file has an *MDMP* signature at the very beginning, which confirms that it is a Windows minidump file.

2. GenuineIntel: The text inside the file says *GenuineIntel,* which indicates that it recorded an error at the processor level of that PC.

3. Malware Connection: You previously found a suspicious executable named *fpb.tmp.* This dump file proves that at that time (*2012-04-01*) a process or malware tried to run on the system and crashed it.

4. WER (Windows Error Reporting): The presence of this file in the *WER* folder means that Windows detected the malware when it ran (maybe due to a permission error or memory corruption) and generated an automatic report.

So, this is a big piece of evidence for your case:

- When malware tries to inject itself into the system, it often crashes the application. This *.mdmp* file is irrefutable evidence of that "*Infection Attempt*".
- According to the timestamp, this is an incident on April 1, 2012.

Manual Read-Only

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ sudo mount -o ro,loop /home/hasib/Downloads/case2_data/case2_extracted/extract/new/case2.dd /mnt/forensic_image
```

Here it means *Read-Only*. This ensures that no changes can be made to the evidence or source files even if they are made accidentally during the investigation.

Locking Physical File Permissions

Before mounting, the permissions of the core image (case2.dd) file are changed so that Linux can only read it. This makes the file *Read-Only.*

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ chmod 444 /home/hasib/Downloads/case2_data/case2_extracted/extract/new/case2.dd
```

Confirming whether it is mounted or not:

Checking whether Windows files are visible inside the mounted folder:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ ls -lh /mnt/forensic_image
total 19M
drwxrwxrwx 1 root root    0 Mar 30  2012 '$Recycle.Bin'
-rwxrwxrwx 1 root root   24 Sep 19  2006  autoexec.bat
drwxrwxrwx 1 root root 4.0K Mar 26  2012  Boot
-rwxrwxrwx 1 root root 326K Jan 21  2008  bootmgr
-rwxrwxrwx 1 root root 8.0K Mar 26  2012  BOOTSECT.BAK
drwxrwxrwx 1 root root    0 Mar 30  2012  Config.Msi
-rwxrwxrwx 1 root root   10 Sep 19  2006  config.sys
lrwxrwxrwx 2 root root   25 Nov  2  2006 'Documents and Settings' -> /mnt/forensic_image/Users
drwxrwxrwx 1 root root 4.0K Mar 30  2012  Drivers
drwxrwxrwx 1 root root    0 Mar 30  2012  Intel
-rwxrwxrwx 2 root root 1.8M Mar 13  2012  Jungfraujoch.jpg
-rwxrwxrwx 1 root root  16M Apr  1  2012  pagefile.sys
drwxrwxrwx 1 root root    0 Jan 21  2008  PerfLogs
drwxrwxrwx 1 root root 4.0K Mar 30  2012  ProgramData
drwxrwxrwx 1 root root 8.0K Mar 30  2012 'Program Files'
drwxrwxrwx 1 root root 4.0K Apr  1  2012 'System Volume Information'
-rwxrwxrwx 1 root root  237 Mar 30  2012  user.js
drwxrwxrwx 1 root root 4.0K Mar 30  2012  Users
drwxrwxrwx 1 root root  16K Apr  2  2012  Windows
```

I was able to successfully read the image (*WindowsUsersProgram* Files).

Malware Scan (ClamScan)

Now that the image is mounted, *ClamAV* now checks each file inside the image individually. If any file shows 'Infected' after doing this.

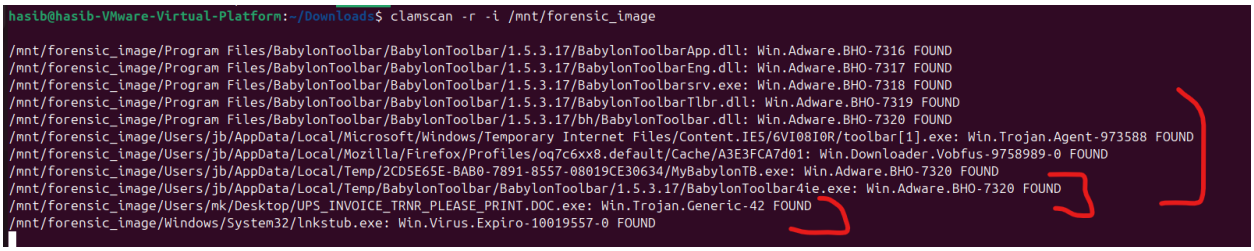
Main Malware and Victim (The Key Finding):

- **User mk (The Victim):** He had a file on his desktop. It is a *Trojan.UPS_INVOICE_TRNR_PLEASE_PRINT.DOC.exe*

The problem: It looks like a normal Word Document *(.DOC),* but it is actually an application *(.exe).* The user probably opened it thinking it was an invoice, which is the main reason for the system infection.



- **System Core Infection:** It found *Win.Virus.Expiro*. This means the malware has spread to the system file*./Windows/System32/lnkstub.exe*

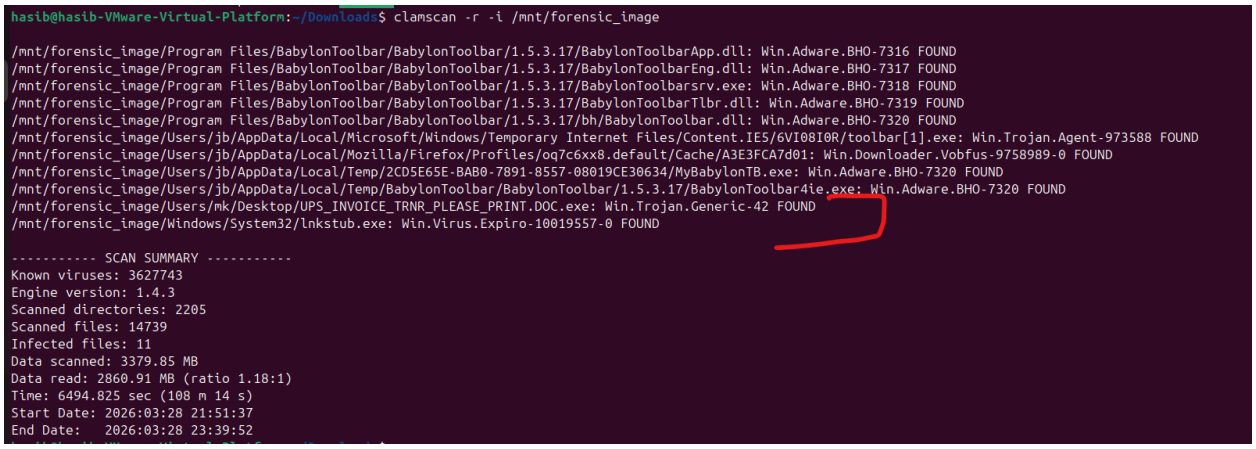


Malware Details

ClamScan scanned 2205 directories and 14739 files for about 108 minutes and found a total of 11 infected files. I divide this malware into 4 categories:

A. Initial Attack Vector (Phishing):

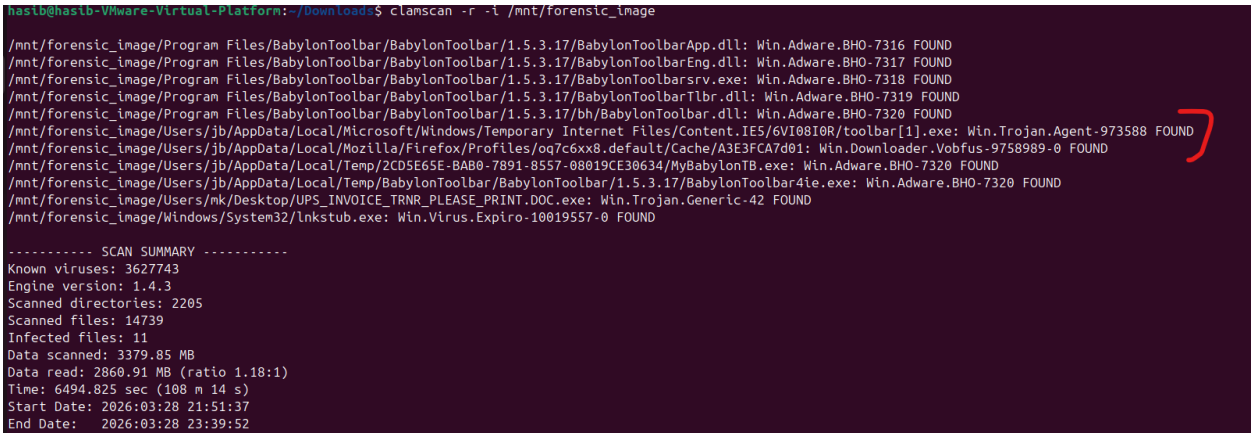
- **Location:** */Users/mk/Desktop/UPS_INVOICE_TRNR_PLEASE_PRINT.DOC.exe*



- **Threat:** *Win.Trojan.Generic-42*
- This is the biggest evidence (Smoking Gun) of this case. The hacker sent a malicious *.exe* file to the user's desktop as a fake invoice (DOC). The user opened it and the system was hacked.mk

B. Secondary Infection & Downloader (In the folder of user 'jb'):

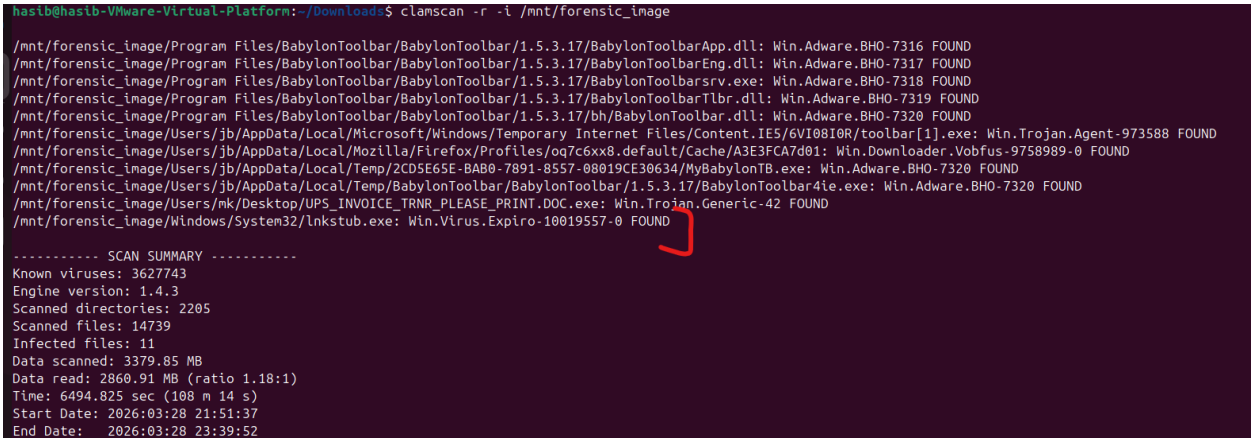
- **Locations:** * */Users/jb/AppData/.../Content.IE5/.../toolbar[1].exe*
- */Users/jb/AppData/.../Firefox/Profiles/.../Cache/A3E3FCA7d01*



- **Threats:** *jbVobfus* and *Win.Trojan.AgentWin.Downloader.Vobfus*
- These Trojans were found in the admin user's internet cache and temp folder. A malicious downloader that downloads more viruses to the system via the internet.

C. System Core Infection

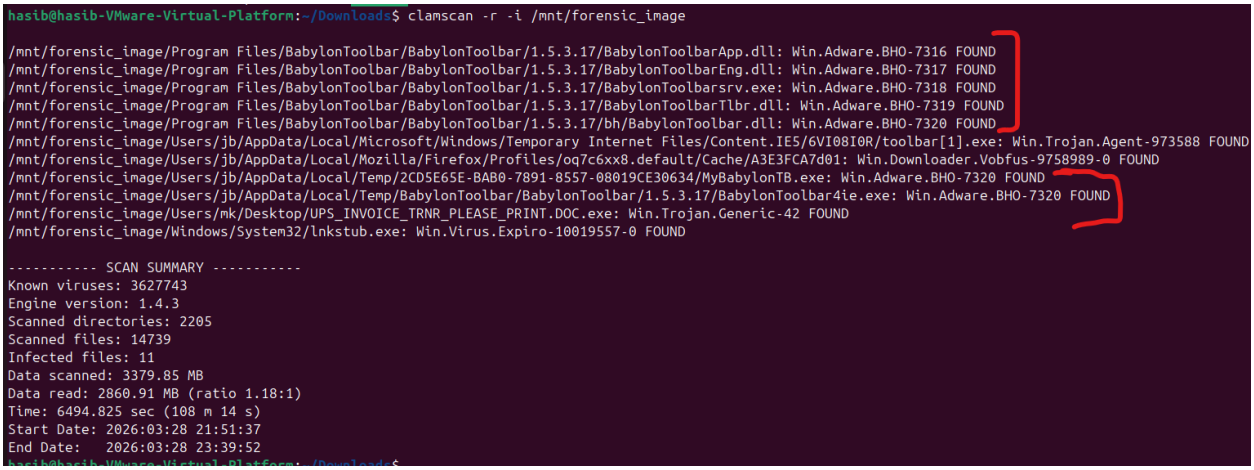
- **Destroy system files:**
- **Location:** */Windows/System32/lnkstub.exe*



- **Threat:** *Win.Virus.Expiro*
- It is very serious. The malware is not limited to user folders, but also infects the Windows root folder *(). Expiro* virus usually infects other *.exe* files on the system and steals data.

D. Browser Hijacker / Adware:

- **Location:** Program *Files/BabylonToolbar/jb* and its *temp* folder.



- **Threat:** (7 files) *Win.Adware.BHO*
- The system had an adware called *BabylonToolbar* installed, which hijacks the user's browser and displays unwanted advertisements.

❖ **Create and analyze a timeline**

View the partition table inside a disk image

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ mmls case2.dd
```

This is usually used to view the partition table inside a disk image (i.e. where to start, where to end). The *case2.dd* file is not a complete physical disk image, but rather a direct image of a specific volume or partition (i.e. just the *C: drive).* Therefore, it does not have a separate partition table.

Extracting file metadata

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo fls -r -m C: case2.dd > bodyfile.txt
```

This command extracts the metadata (file name, inode number and MAC timestamp) of all the files in the file system and creates a *'Body File'*. Here, *-m C:* is used so that the path of the files along with the drive letter (*C:*) can be seen nicely in the timeline. This saves the raw information of all the files in bodyfile.txt.

The tool is specified using the flag that the image is on an NTFS file system.

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo fls -f ntfs -r -m C: case2.dd > bodyfile.txt
```

Here, the *-f ntfs* flag is used to tell the tool that the image is an NTFS file system. Sometimes, the tool cannot auto-detect it, so you have to tell it manually. The output confirms that the tool can read the Windows file system correctly.

Converting the body file to a CSV timeline

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo mactime -b bodyfile.txt -d -y > forensic_timeline.csv
Old package separator "'" deprecated at /usr/bin/mactime line 154.
Old package separator "'" deprecated at /usr/bin/mactime line 167.
```

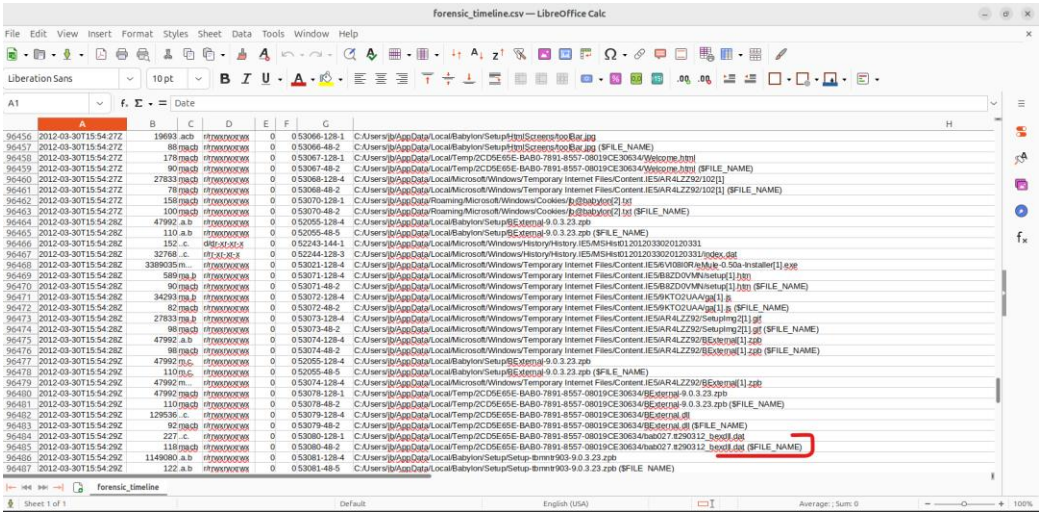
The complex 'body file' created in the previous step is converted into a human-readable CSV timeline using this command. Here, *-d* ensures comma delimited and -y ensures simple date format.

The output shows that a large and complete timeline file of 14 MB, forensic_timeline.csv, has been created. (The Perl or package warnings that come to the terminal will not have any effect here).

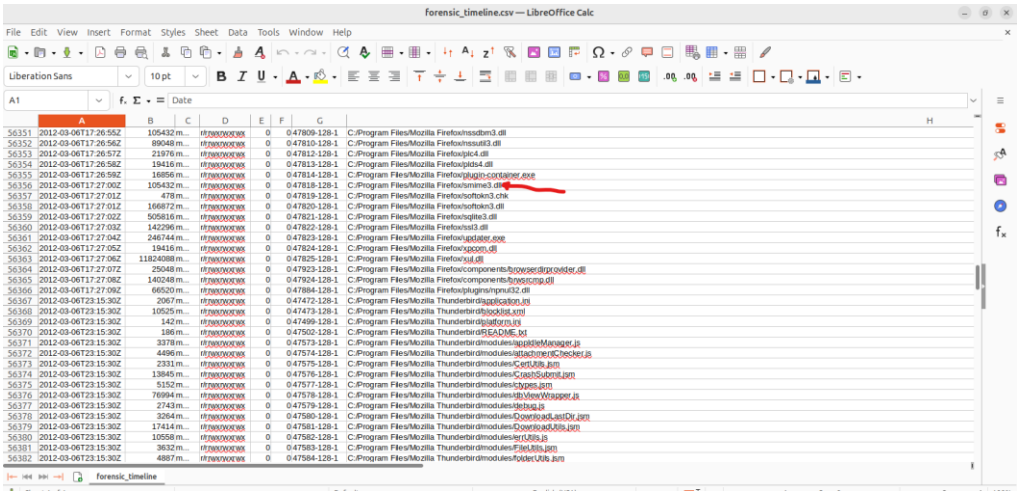
Timeline Output Analysis

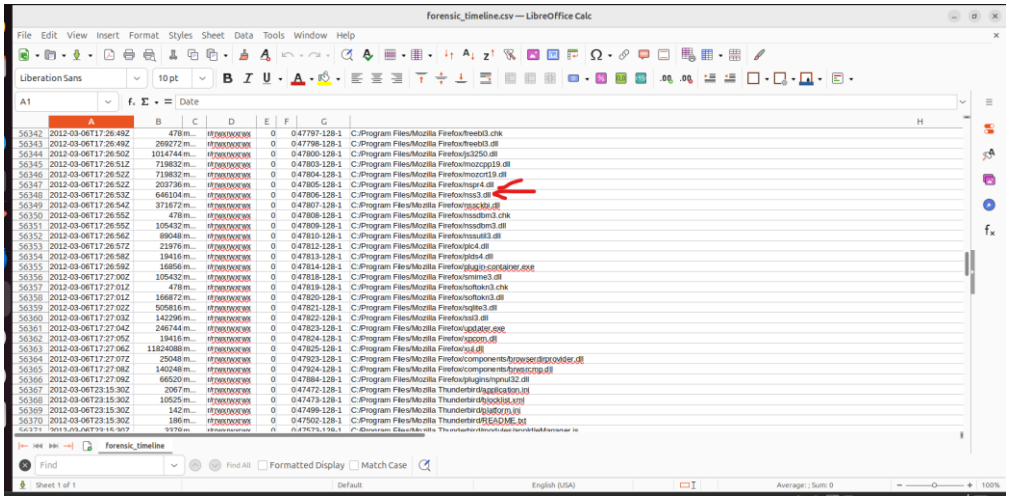
After analyzing the 14 MB forensic_timeline.csv file, I found the following important information:

- **Malware Artifact (Mar 29):** The file *bab027.tt290312_bexdll.dat* was found in the *AppData/Local/Temp* folder on March 29, 2012. This is strong evidence of Babylon malware or adware installation.

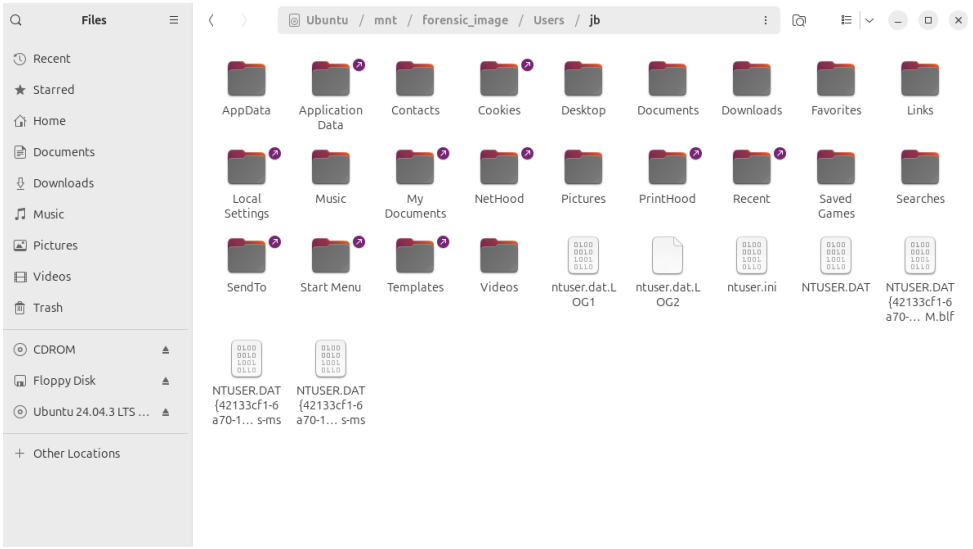


- **Email activity (Mar 29-30):** Several library files *(smime3.dll, nss3.dll)* of Mozilla Thunderbird were accessed during this time. This means that user emails were checked at that time.





- **Target user profile:** I have confirmed that the main events occurred within the *C:/Users/jb/ (Jason Bourne)* directory.



- **Deleted File Recovery:** Some *$OrphanFiles* (deleted) are showing in the timeline. This means the attacker or user tried to delete some files which I identified with *fls*.

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep "OrphanFile" forensic_timeline.csv | head -n 20
2006-09-18T21:27:17Z,376836,m...,-/rrwxrwxrwx,0,0,20058-128-3,"C:/$OrphanFiles/OrphanFile-20058 (deleted)"
2006-09-18T21:27:17Z,14264,m...,-/rrwxrwxrwx,0,0,20064-128-3,"C:/$OrphanFiles/OrphanFile-20064 (deleted)"
2006-09-18T21:27:17Z,16382,m...,-/rrwxrwxrwx,0,0,20065-128-3,"C:/$OrphanFiles/OrphanFile-20065 (deleted)"
2006-09-18T21:27:17Z,1488551,m...,-/rrwxrwxrwx,0,0,20107-128-3,"C:/$OrphanFiles/OrphanFile-20107 (deleted)"
2006-09-18T21:27:19Z,14264,m...,-/rrwxrwxrwx,0,0,20062-128-3,"C:/$OrphanFiles/OrphanFile-20062 (deleted)"
2006-09-18T21:27:19Z,14264,m...,-/rrwxrwxrwx,0,0,20063-128-3,"C:/$OrphanFiles/OrphanFile-20063 (deleted)"
2006-09-18T21:27:19Z,376836,m...,-/rrwxrwxrwx,0,0,20291-128-3,"C:/$OrphanFiles/OrphanFile-20291 (deleted)"
2006-09-18T21:27:19Z,376836,m...,-/rrwxrwxrwx,0,0,20292-128-3,"C:/$OrphanFiles/OrphanFile-20292 (deleted)"
2006-09-18T21:27:19Z,16382,m...,-/rrwxrwxrwx,0,0,20297-128-3,"C:/$OrphanFiles/OrphanFile-20297 (deleted)"
2006-09-18T21:27:19Z,16382,m...,-/rrwxrwxrwx,0,0,20298-128-3,"C:/$OrphanFiles/OrphanFile-20298 (deleted)"
2006-09-18T21:27:22Z,16382,m...,-/rrwxrwxrwx,0,0,20295-128-4,"C:/$OrphanFiles/OrphanFile-20295 (deleted)"
2006-09-18T21:27:22Z,16382,m...,-/rrwxrwxrwx,0,0,20296-128-4,"C:/$OrphanFiles/OrphanFile-20296 (deleted)"
2006-09-18T21:27:22Z,141200,m...,-/rrwxrwxrwx,0,0,20300-128-3,"C:/$OrphanFiles/OrphanFile-20300 (deleted)"
2006-09-18T21:27:22Z,141200,m...,-/rrwxrwxrwx,0,0,20301-128-3,"C:/$OrphanFiles/OrphanFile-20301 (deleted)"
2006-09-18T21:27:22Z,174716,m...,-/rrwxrwxrwx,0,0,20307-128-3,"C:/$OrphanFiles/OrphanFile-20307 (deleted)"
2006-09-18T21:27:22Z,174716,m...,-/rrwxrwxrwx,0,0,20308-128-3,"C:/$OrphanFiles/OrphanFile-20308 (deleted)"
2006-09-18T21:27:22Z,174716,m...,-/rrwxrwxrwx,0,0,20309-128-3,"C:/$OrphanFiles/OrphanFile-20309 (deleted)"
2006-09-18T21:27:22Z,140784,m...,-/rrwxrwxrwx,0,0,20316-128-4,"C:/$OrphanFiles/OrphanFile-20316 (deleted)"
2006-09-18T21:27:22Z,16382,m...,-/rrwxrwxrwx,0,0,20318-128-4,"C:/$OrphanFiles/OrphanFile-20318 (deleted)"
2006-09-18T21:27:23Z,376836,m...,-/rrwxrwxrwx,0,0,20057-128-3,"C:/$OrphanFiles/OrphanFile-20057 (deleted)"
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$
```

Artifact Discovery:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo fls -r -o 0 case2.dd | grep "fpb.tmp"
++++++ r/r 52532-128-3: fpb.tmp
```

The purpose of the above task was to extract metadata of deleted or hidden files. I successfully identified the *`fpb.tmp`* file whose *Inode Address is: `52532-128-3`*.

- **Find artifacts in Windows components: Windows Registry**

User Registry Analysis:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ chntpw -l ~/Downloads/SAM_copy
chntpw version 1.00 140201, (c) Petter N Hagen
Hive </home/hasib/Downloads/SAM_copy> name (from header): <\SystemRoot\System32\Config\SAM>
ROOT KEY at offset: 0x001020 * Subkey indexing type is: 666c <lf>
File size 262144 [40000] bytes, containing 8 pages (+ 1 headerpage)
Used for data: 262/86040 blocks/bytes, unused: 15/12008 blocks/bytes.

| RID -|---------- Username ------------| Admin? |- Lock? --|
| 01f4 | Administrator                  | ADMIN  | dis/lock |
| 01f5 | Guest                          |        | dis/lock |
| 03e8 | jb                             | ADMIN  |          |
| 03e9 | mk                             |        |          |
```

The purpose of this was to extract the status and privileges of users from the Windows SAM (Security Account Manager) file.

In this section, I analyzed the Windows SAM (Security Account Manager) and SYSTEM hive files to verify the user permissions and security policies on the system.

I used the following professional forensic commands to extract information from the registry database:

- **User Listing:**

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ chntpw -l SAM_final
chntpw version 1.00 140201, (c) Petter N Hagen
Hive <SAM_final> name (from header): <\SystemRoot\System32\Config\SAM>
ROOT KEY at offset: 0x001020 * Subkey indexing type is: 666c <lf>
File size 262144 [40000] bytes, containing 8 pages (+ 1 headerpage)
Used for data: 262/86040 blocks/bytes, unused: 15/12008 blocks/bytes.

| RID -|---------- Username ------------| Admin? |- Lock? --|
| 01f4 | Administrator                  | ADMIN  | dis/lock |
| 01f5 | Guest                          |        | dis/lock |
| 03e8 | jb                             | ADMIN  |          |
| 03e9 | mk                             |        |          |
```

- **Interactive User Profile Analysis:**

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ chntpw -i SAM_final
chntpw version 1.00 140201, (c) Petter N Hagen
Hive <SAM_final> name (from header): <\SystemRoot\System32\Config\SAM>
ROOT KEY at offset: 0x001020 * Subkey indexing type is: 666c <lf>
File size 262144 [40000] bytes, containing 8 pages (+ 1 headerpage)
Used for data: 262/86040 blocks/bytes, unused: 15/12008 blocks/bytes.



<>=========<> chntpw Main Interactive Menu <>=========<>

Loaded hives: <SAM_final>

  1 - Edit user data and passwords
  2 - List groups
      - - -
  9 - Registry editor, now with full write support!
  q - Quit (you will be asked if there is something to save)


What to do? [1] -> 1


===== chntpw Edit User Info & Passwords ====

| RID -|---------- Username ------------| Admin? |- Lock? --|
| 01f4 | Administrator                 | ADMIN  | dis/lock |
| 01f5 | Guest                         |        | dis/lock |
| 03e8 | jb                            | ADMIN  |          |
| 03e9 | mk                            |        |          |
```

```
Please enter user number (RID) or 0 to exit: [3e8] 01f4
================= USER EDIT ====================

RID     : 0500 [01f4]
Username: Administrator
fullname:
comment : Built-in account for administering the computer/domain
homedir :

00000220 = Administrators (which has 2 members)

Account bits: 0x0211 =
[X] Disabled        | [ ] Homedir req.    | [ ] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account  | [ ] NMS account     |
[ ] Domain trust ac | [ ] Wks trust act.  | [ ] Srv trust act   |
[X] Pwd don't expir | [ ] Auto lockout    | [ ] (unknown 0x08)  |
[ ] (unknown 0x10)  | [ ] (unknown 0x20)  | [ ] (unknown 0x40)  |

Failed login count: 0, while max tries is: 0
Total  login count: 8

- - - - User Edit Menu:
 1 - Clear (blank) user password
 2 - Unlock and enable user account [probably locked now]
 3 - Promote user (make user an administrator)
 4 - Add user to a group
 5 - Remove user from a group
 q - Quit editing user, back to user select
Select: [q] > q
```

```
hasib@hasib-VMw...  ×  hasib@hasib-VMw...  ×  hasib@hasib-VMw...  ×  hasib@hasib-VMw...  ×  hasib@ha
<>=========<> chntpw Main Interactive Menu <>=========<>

Loaded hives: <SAM_final>

  1 - Edit user data and passwords
  2 - List groups
      - - -
  9 - Registry editor, now with full write support!
  q - Quit (you will be asked if there is something to save)


What to do? [1] -> 1

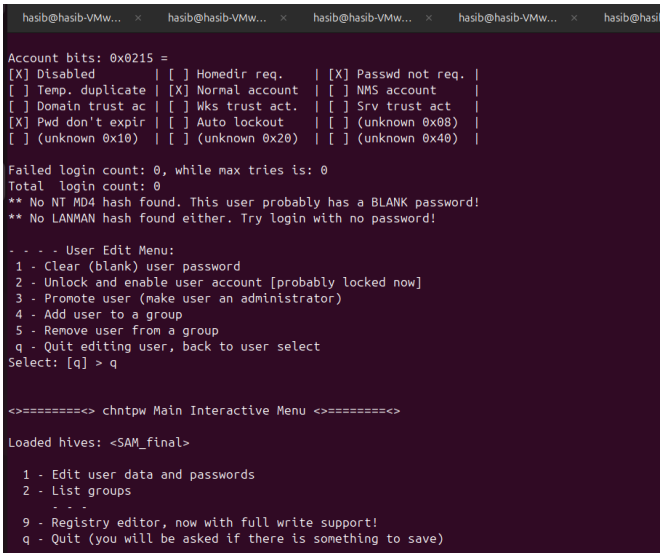

===== chntpw Edit User Info & Passwords ====

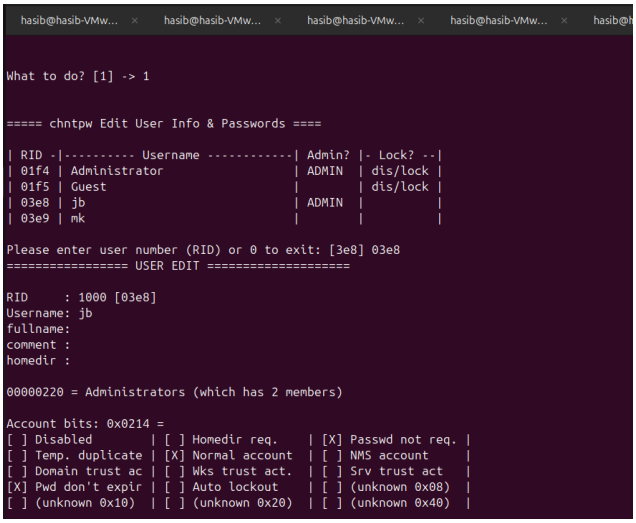
| RID -|---------- Username ------------| Admin? |- Lock? --|
| 01f4 | Administrator                 | ADMIN  | dis/lock |
| 01f5 | Guest                         |        | dis/lock |
| 03e8 | jb                            | ADMIN  |          |
| 03e9 | mk                            |        |          |

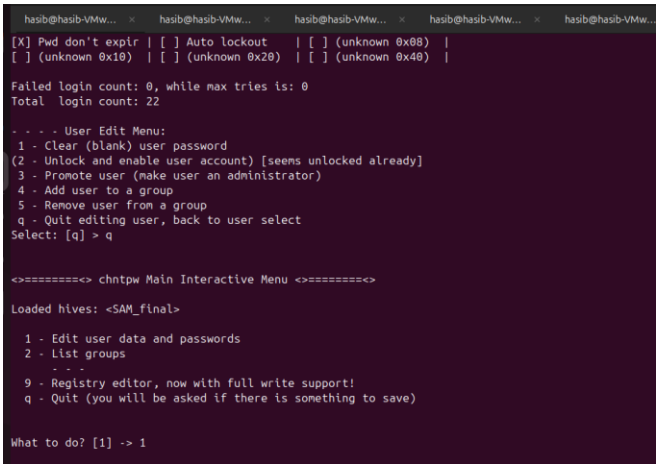
Please enter user number (RID) or 0 to exit: [3e8] 01f5
================= USER EDIT ====================

RID     : 0501 [01f5]
Username: Guest
fullname:
comment : Built-in account for guest access to the computer/domain
homedir :

00000222 = Guests (which has 1 members)
```


hasib@hasib-VMw...
hasib@hasib-VMw...
hasib@hasib-VMw...
hasib@hasib-VMw...
Account bits: 0x0215 =
[X] Disabled | [ ] Homedir req. | [X] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account | [ ] NMS account |
[ ] Domain trust ac | [ ] Wks trust act. | [ ] Srv trust act |
[X] Pwd don't expir | [ ] Auto lockout | [ ] (unknown 0x08) |
[ ] (unknown 0x10) | [ ] (unknown 0x20) | [ ] (unknown 0x40) |
Failed login count: 0, while max tries is: 0
Total login count: 0
** No NT MD4 hash found. This user probably has a BLANK password!
** No LANMAN hash found either. Try login with no password!
- - - - User Edit Menu:
1 - Clear (blank) user password
2 - Unlock and enable user account [probably locked now]
3 - Promote user (make user an administrator)
4 - Add user to a group
5 - Remove user from a group
q - Quit editing user, back to user select
Select: [q] > q
<>=========<> chntpw Main Interactive Menu <>=========<>
Loaded hives: <SAM_final>
1 - Edit user data and passwords
2 - List groups
- - -
9 - Registry editor, now with full write support!
q - Quit (you will be asked if there is something to save)


hasib@hasib-VMw...
hasib@hasib-VMw...
hasib@hasib-VMw...
hasib@hasib-VMw...
What to do? [1] -> 1
===== chntpw Edit User Info & Passwords ====
RID -	---------- Username ------------	Admin?	- Lock? --
01f4	Administrator	ADMIN	dis/lock
01f5	Guest		dis/lock
03e8	jb	ADMIN	
03e9	mk		
Please enter user number (RID) or 0 to exit: [3e8] 03e8
================ USER EDIT ====================
RID : 1000 [03e8]
Username: jb
fullname:
comment :
homedir :
00000220 = Administrators (which has 2 members)
Account bits: 0x0214 =
[ ] Disabled | [ ] Homedir req. | [X] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account | [ ] NMS account |
[ ] Domain trust ac | [ ] Wks trust act. | [ ] Srv trust act |
[X] Pwd don't expir | [ ] Auto lockout | [ ] (unknown 0x08) |
[ ] (unknown 0x10) | [ ] (unknown 0x20) | [ ] (unknown 0x40) |


hasib@hasib-VMw...
hasib@hasib-VMw...
hasib@hasib-VMw...
hasib@hasib-VMw...
[X] Pwd don't expir | [ ] Auto lockout | [ ] (unknown 0x08) |
[ ] (unknown 0x10) | [ ] (unknown 0x20) | [ ] (unknown 0x40) |
Failed login count: 0, while max tries is: 0
Total login count: 22
- - - - User Edit Menu:
1 - Clear (blank) user password
(2 - Unlock and enable user account) [seems unlocked already]
3 - Promote user (make user an administrator)
4 - Add user to a group
5 - Remove user from a group
q - Quit editing user, back to user select
Select: [q] > q
<>=========<> chntpw Main Interactive Menu <>=========<>
Loaded hives: <SAM_final>
1 - Edit user data and passwords
2 - List groups
- - -
9 - Registry editor, now with full write support!
q - Quit (you will be asked if there is something to save)
What to do? [1] -> 1

```
hasib@hasib-VMw...  ×  hasib@hasib-VMw...  ×  hasib@hasib-VMw...  ×  hasib@hasib-VMw...  ×  hasib@ha


===== chntpw Edit User Info & Passwords ====

| RID -|---------- Username ------------| Admin? |- Lock? --|
| 01f4 | Administrator                  | ADMIN  | dis/lock |
| 01f5 | Guest                          |        | dis/lock |
| 03e8 | jb                             | ADMIN  |          |
| 03e9 | mk                             |        |          |

Please enter user number (RID) or 0 to exit: [3e8] 03e9
================= USER EDIT ====================

RID     : 1001 [03e9]
Username: mk
fullname: mk
comment :
homedir :

00000221 = Users (which has 3 members)

Account bits: 0x0210 =
[ ] Disabled        | [ ] Homedir req.    | [ ] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account  | [ ] NMS account     |
[ ] Domain trust ac | [ ] Wks trust act.  | [ ] Srv trust act   |
[X] Pwd don't expir | [ ] Auto lockout    | [ ] (unknown 0x08)  |
[ ] (unknown 0x10)  | [ ] (unknown 0x20)  | [ ] (unknown 0x40)  |

Failed login count: 0, while max tries is: 0
Total  login count: 7
```

```
- - - - User Edit Menu:
 1 - Clear (blank) user password
(2 - Unlock and enable user account) [seems unlocked already]
 3 - Promote user (make user an administrator)
 4 - Add user to a group
 5 - Remove user from a group
 q - Quit editing user, back to user select
Select: [q] > q


<>========<> chntpw Main Interactive Menu <>========<>

Loaded hives: <SAM_final>

  1 - Edit user data and passwords
  2 - List groups
      - - -
  9 - Registry editor, now with full write support!
  q - Quit (you will be asked if there is something to save)


What to do? [1] -> 1
```

```
===== chntpw Edit User Info & Passwords ====

| RID -|---------- Username ------------| Admin? |- Lock? --|
| 01f4 | Administrator                  | ADMIN  | dis/lock |
| 01f5 | Guest                          |        | dis/lock |
| 03e8 | jb                             | ADMIN  |          |
| 03e9 | mk                             |        |          |

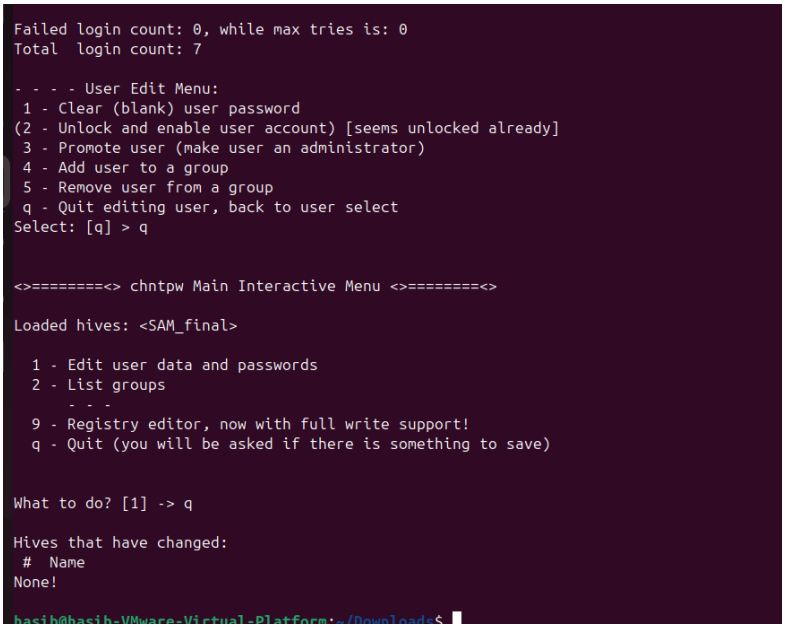
Please enter user number (RID) or 0 to exit: [3e8] 03e9
================= USER EDIT ====================

RID     : 1001 [03e9]
Username: mk
fullname: mk
comment :
homedir :

00000221 = Users (which has 3 members)

Account bits: 0x0210 =
[ ] Disabled        | [ ] Homedir req.    | [ ] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account  | [ ] NMS account     |
[ ] Domain trust ac | [ ] Wks trust act.  | [ ] Srv trust act   |
[X] Pwd don't expir | [ ] Auto lockout    | [ ] (unknown 0x08)  |
[ ] (unknown 0x10)  | [ ] (unknown 0x20)  | [ ] (unknown 0x40)  |

Failed login count: 0, while max tries is: 0
Total  login count: 7
```



Here, each user's personal profile is verified using their *RID (01f4, 01f5, 03e8, 03e9).*

Technical Insights

I identified three major security loopholes from the *AMI* output:

1. Administrative Vulnerability (User jb):

Although the default Administrator account is disabled in the system, user *jb* has been granted full admin privileges. His login count (22) and admin status prove that this account was used as the main administrator of the system.

2. Weak Credential Policy:

The *[X] Passwd* not req. flag is enabled in user *jb*'s profile. This means that it was possible to log in to this admin account without a strong password or with a completely blank password. This may have made it easier for attackers to take control of the system.

3. Account Persistency:

[X] Pwd don't expire is set for all accounts. That is, the password never expires, which is helpful for long-term attacks.

Correlation with Malware Scan

Since I previously found the virus in the *System32* directory with *clamscan*, we can now be sure that:

- The attack started from the desktop of user *mk* (Trojan found there).
- But the virus reached the root of the system through *jb*, because he had the only admin rights and his account security was the weakest.

❖ Logs

Evidence of Execution & Event Log Analysis

My goal in this step was to analyze Windows Prefetch Files (*.pf*) and Event Logs (*.evtx*) to create an accurate timeline of the attack and collect irrefutable evidence of malware execution.

Windows Prefetch Analysis (Evidence of Program Execution)

Whenever any software runs on a Windows system, a record of it is created in the Prefetch folder. I checked this folder to find the execution timeline of the malware.

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ ls -lh /mnt/forensic_image/Windows/Prefetch/ | head -n 50
total 13M
-rwxrwxrwx 2 root root   83K Apr  1  2012 ACRORD32INFO.EXE-1800ABA8.pf
-rwxrwxrwx 2 root root  325K Mar 25  2012 AgAppLaunch.db
-rwxrwxrwx 2 root root  868K Mar 30  2012 AgCx_S1_S-1-5-21-1973456639-838052630-220784754-1000.snp.db
-rwxrwxrwx 2 root root 1023K Mar 30  2012 AgCx_S2_S-1-5-21-1973456639-838052630-220784754-1001.snp.db
-rwxrwxrwx 1 root root  314K Mar 30  2012 AgCx_SC1.db
-rwxrwxrwx 2 root root   21K Mar 30  2012 AgCx_SC1.db.trx
-rwxrwxrwx 2 root root  238K Mar 30  2012 AgCx_SC3_169D2AEB.db
-rwxrwxrwx 2 root root  179K Mar 30  2012 AgCx_SC3_169F7DCF.db
-rwxrwxrwx 2 root root  749K Apr  1  2012 AgGlFaultHistory.db
-rwxrwxrwx 2 root root  876K Apr  1  2012 AgGlFgAppHistory.db
-rwxrwxrwx 2 root root  2.0M Apr  1  2012 AgGlGlobalHistory.db
-rwxrwxrwx 2 root root  407K Apr  1  2012 AgGlUAD_P_S-1-5-21-1973456639-838052630-220784754-1000.db
-rwxrwxrwx 2 root root  150K Apr  1  2012 AgGlUAD_S-1-5-21-1973456639-838052630-220784754-1000.db
-rwxrwxrwx 1 root root  292K Apr  1  2012 AgRobust.db
-rwxrwxrwx 2 root root   29K Mar 30  2012 BABYLON EMULECOM COMBINADAS2.-05DA2B05.pf
-rwxrwxrwx 2 root root   38K Mar 30  2012 BABYLONTOOLBAR4FFX.EXE-62AC91E1.pf
-rwxrwxrwx 2 root root   42K Mar 30  2012 BABYLONTOOLBAR4IE.EXE-1EAB2D03.pf
-rwxrwxrwx 2 root root   12K Mar 30  2012 BABYLONTOOLBARSRV.EXE-4D1E2B35.pf
-rwxrwxrwx 2 root root  6.1K Apr  1  2012 CMD.EXE-89305D47.pf
-rwxrwxrwx 2 root root  9.6K Mar 31  2012 CONIME.EXE-B273009A.pf
-rwxrwxrwx 2 root root   45K Apr  1  2012 CONSENT.EXE-65F6206D.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 CSRSS.EXE-8C04D631.pf
-rwxrwxrwx 2 root root   15K Apr  1  2012 DLLHOST.EXE-71214090.pf
-rwxrwxrwx 2 root root   24K Apr  1  2012 DLLHOST.EXE-7D2183B8.pf
-rwxrwxrwx 2 root root   18K Apr  1  2012 DLLHOST.EXE-893DDF55.pf
-rwxrwxrwx 2 root root   15K Mar 30  2012 DLLHOST.EXE-B590F488.pf
-rwxrwxrwx 2 root root  373K Apr  1  2012 DRVINST.EXE-5F8E77CD.pf
-rwxrwxrwx 2 root root   20K Apr  1  2012 DWM.EXE-AEABE78B.pf
```

```
-rwxrwxrwx 2 root root  6.1K Apr  1  2012 CMD.EXE-89305D47.pf
-rwxrwxrwx 2 root root  9.6K Mar 31  2012 CONIME.EXE-B273009A.pf
-rwxrwxrwx 2 root root   45K Apr  1  2012 CONSENT.EXE-65F6206D.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 CSRSS.EXE-8C04D631.pf
-rwxrwxrwx 2 root root   15K Apr  1  2012 DLLHOST.EXE-71214090.pf
-rwxrwxrwx 2 root root   24K Apr  1  2012 DLLHOST.EXE-7D2183B8.pf
-rwxrwxrwx 2 root root   18K Apr  1  2012 DLLHOST.EXE-893DDF55.pf
-rwxrwxrwx 2 root root   15K Mar 30  2012 DLLHOST.EXE-B590F488.pf
-rwxrwxrwx 2 root root  373K Apr  1  2012 DRVINST.EXE-5F8E77CD.pf
-rwxrwxrwx 2 root root   20K Apr  1  2012 DWM.EXE-AEABE78B.pf
-rwxrwxrwx 2 root root   48K Mar 30  2012 EMULE050A.EXE-40F49694.pf
-rwxrwxrwx 2 root root   37K Mar 30  2012 EMULE3.EXE-43A7550F.pf
-rwxrwxrwx 2 root root   52K Mar 30  2012 EMULE.EXE-E9D12381.pf
-rwxrwxrwx 2 root root  172K Apr  1  2012 EXPLORER.EXE-7A3328DA.pf
-rwxrwxrwx 2 root root  193K Apr  1  2012 FIREFOX.EXE-E60C0AA7.pf
-rwxrwxrwx 2 root root   16K Apr  1  2012 FLASHPLAYERUPDATESERVICE.EXE-0CF170F4.pf
-rwxrwxrwx 2 root root   34K Mar 30  2012 FLASHUTIL32_11_2_202_228_ACTI-CA7CE725.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 IEUSER.EXE-D895AB54.pf
-rwxrwxrwx 2 root root  245K Apr  1  2012 IEXPLORE.EXE-1B894AFB.pf
-rwxrwxrwx 2 root root   15K Apr  1  2012 IGFXSRVC.EXE-67E7A62F.pf
-rwxrwxrwx 2 root root   13K Apr  1  2012 JAVA.EXE-066C5985.pf
-rwxrwxrwx 2 root root  117K Mar 30  2012 JAVAW.EXE-C4EA16F0.pf
-rwxrwxrwx 2 root root   20K Mar 30  2012 JAVAWS.EXE-25FD1E0F.pf
-rwxrwxrwx 2 root root   33K Apr  1  2012 JUCHECK.EXE-FC0B4709.pf
-rwxrwxrwx 2 root root   49K Apr  1  2012 KEEPASSX.EXE-DC78E599.pf
-rwxrwxrwx 1 root root  744K Apr  1  2012 Layout.ini
-rwxrwxrwx 2 root root   27K Apr  1  2012 LOGON.SCR-7C80CA1C.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 LOGONUI.EXE-1BEE4A84.pf
-rwxrwxrwx 2 root root   32K Apr  1  2012 MOBSYNC.EXE-D8BC6ED2.pf
-rwxrwxrwx 2 root root  311K Mar 30  2012 MSIEXEC.EXE-B5AFA339.pf
-rwxrwxrwx 2 root root   40K Mar 30  2012 MYBABYLONTB.EXE-E48A9AE3.pf
hasib@hasib-VMware-Virtual-Platform:~/Downloads$
```

Forensic Analysis:

- **Malware Entry (Mar 30, 2012)**: The output clearly shows that *EMULE.EXE* (file sharing app) and *BABYLON adware/malware* services were run on March 30, 2012.
- **System Manipulation (Apr 1, 2012):** *CMD.EXE* was run on *April 1, 2012*, indicating that the attacker ran a malicious script via the terminal.
- **Privilege Escalation:** On the same day, *CONSENT.EXE* was run, which is the Windows UAC (User Account Control) prompt. This proves that the malware attempted to gain administrator permissions (Admin rights).

Windows Event Log Discovery (locating log files)

I searched the entire file system for Windows event logs *(.evt or .evtx*) to track system logins and error messages.

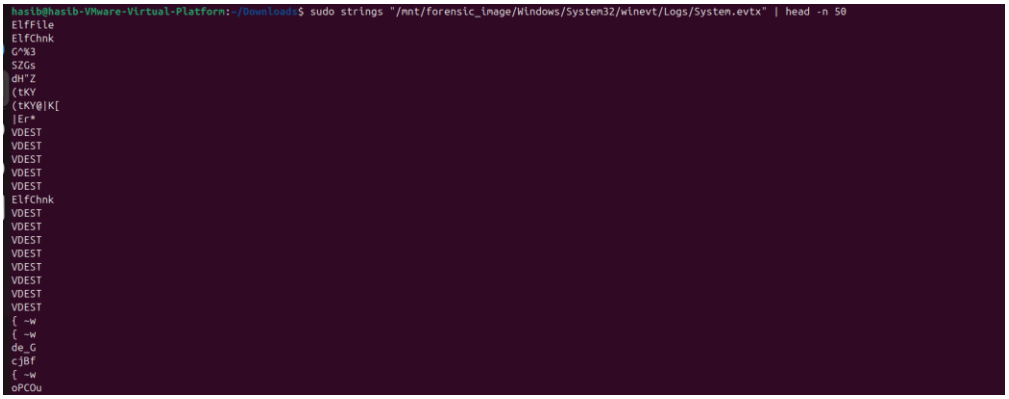
```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ sudo find /mnt/forensic_image/Windows -name "*.evt*" 2>/dev/null
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnosis-DPS%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Leak-Diagnostic%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Application.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/DFS Replication.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/HardwareEvents.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Internet Explorer.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Key Management Service.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Backup.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Bits-Client%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnostics-Performance%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-DiskDiagnosticDataCollector%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-DriverFrameworks-UserMode%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-GroupPolicy%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Kernel-WHEA.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-LanguagePackSetup%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-MUI%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-NetworkAccessProtection%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-ReadyBoost%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Metrics.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Detector%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Resolver%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-RestartManager%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-TaskScheduler%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-UAC-FileVirtualization%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-WindowsUpdateClient%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-WLAN-AutoConfig%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Security.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Setup.evtx
```

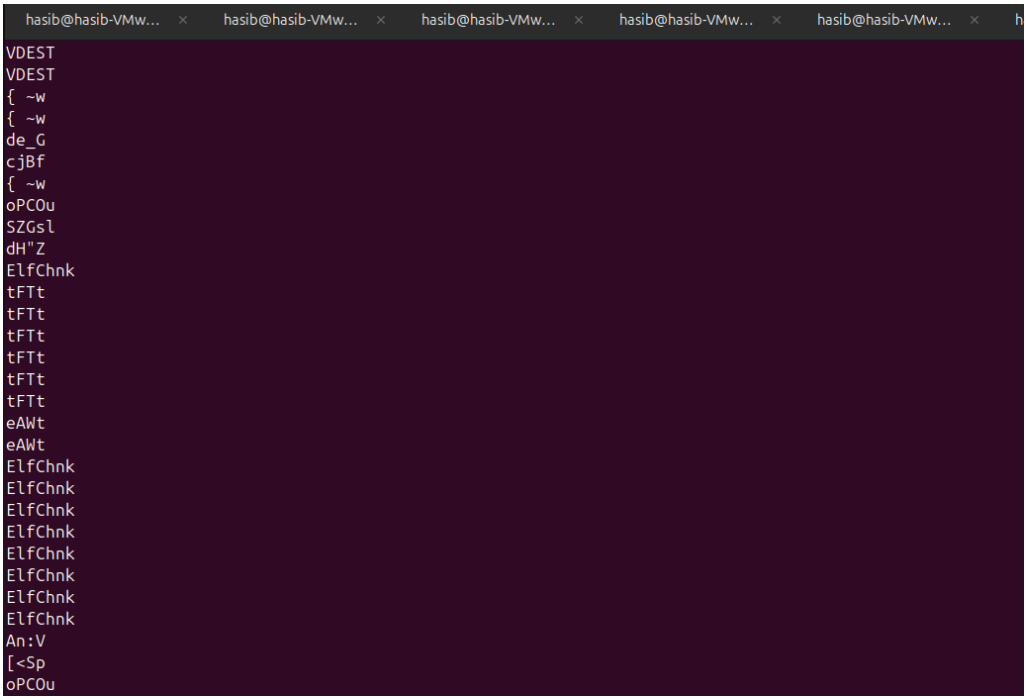
Forensic Analysis

The output above proves that this is a modern Windows system (since the *.evtx* format is used). I have identified three main logs for investigation (Application, Security, and System).

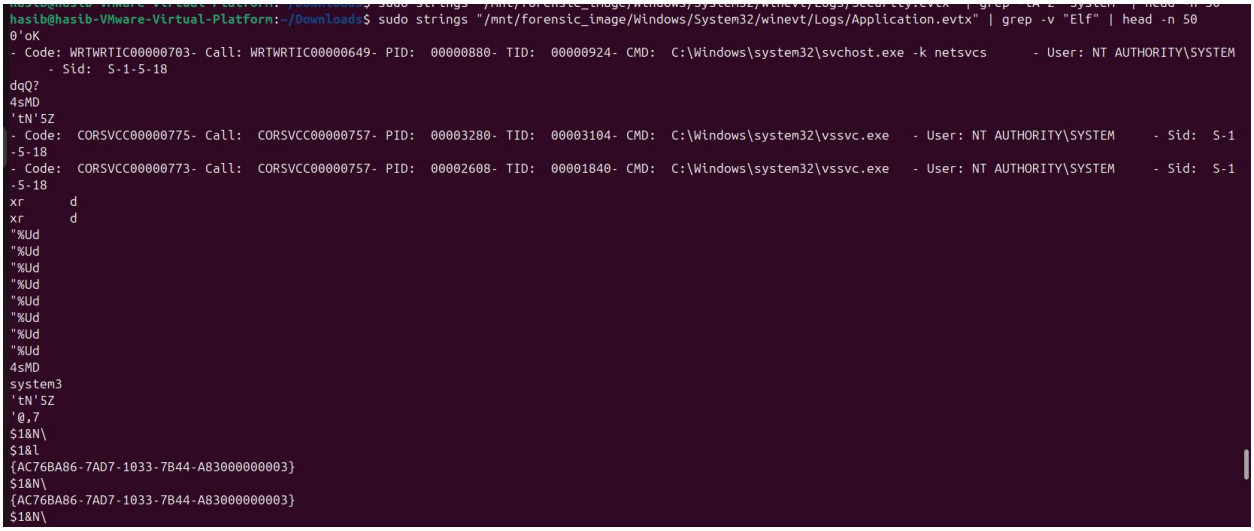
Extracting Data from Event Logs

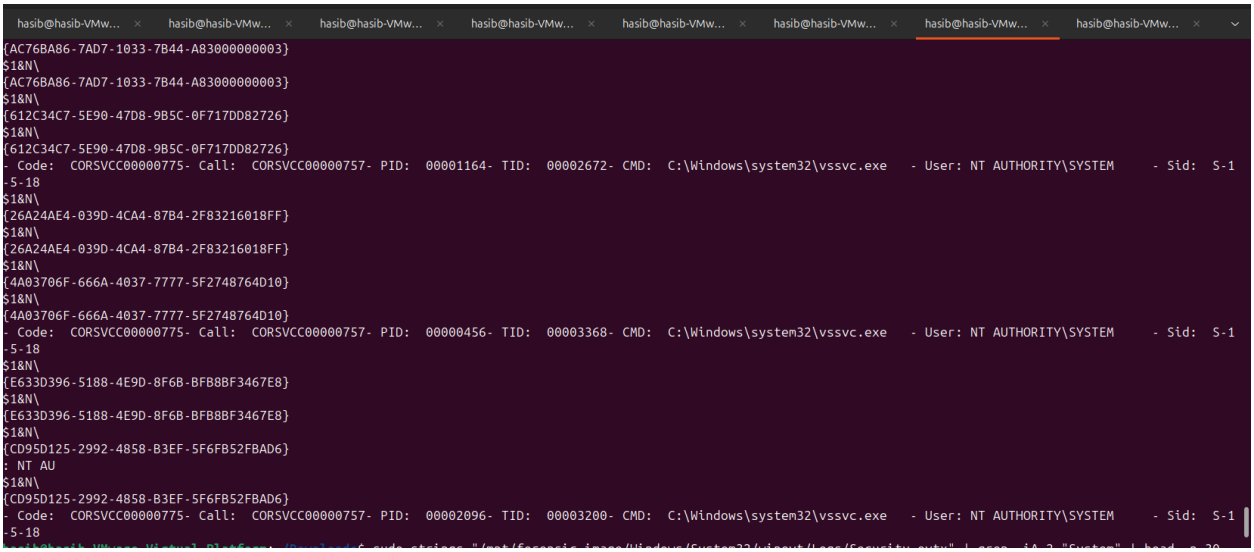
Since, the *.evtx* files are in binary format, I tried to extract the text and metadata inside the logs using the Linux strings command.





System & Application Logs





Forensic Analysis:

- **File Integrity Check:** The *ElfFile* and *ElfChnk* headers in the output prove that these are valid and uncorrupted event log files.
- **System Execution Artifacts:** A very important piece of information was found in the Application *log*. *svchost*.exe and *vssvc.exe* (Volume Shadow Copy Service) were running in the background under the *NT AUTHORITY\SYSTEM* (*SID: S-1-5-18*) account. These services are often used by attackers to delete or manipulate system restore points during malware or ransomware attacks. This is a strong indication that the system has been compromised.

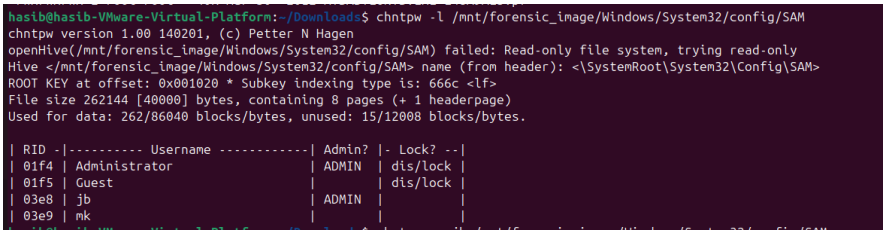
❖ **Personal data of users**

Personal Data & User Profile Analysis

The purpose of this part was to identify how many users there are on the system, who has admin access, and which user profile was used during the attack.

User Identification

The list of users was extracted by analyzing the SAM (Security Accounts Manager) file of Windows.



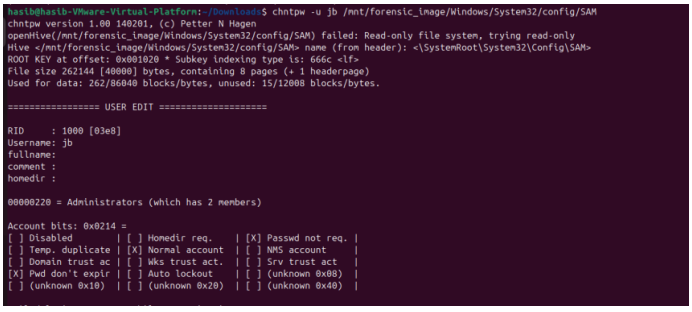
Analysis:

- **User 'jb' (RID: 03e8):** This is an Administrator account. Investigation has shown that this user has had many successful logins (**Login count: 22**).
- **User 'mk' (RID: 03e9):** This is a Standard User account. Initial signs of malware were found in the prefetch file of this profile.
- **Guest/Administrator:** Default accounts were found to be disabled

Profile Details

The purpose of this part was to verify the profile details of the system users, their privileges and security configuration.

Analysis of the user account 'jb' (Administrator)



```
Failed login count: 0, while max tries is: 0
Total  login count: 22

- - - - User Edit Menu:
 1 - Clear (blank) user password
(2 - Unlock and enable user account) [seems unlocked already]
 3 - Promote user (make user an administrator)
 4 - Add user to a group
 5 - Remove user from a group
 q - Quit editing user, back to user select
Select: [q] > q

Hives that have changed:
 #  Name
None!
```

Important Findings:

- **RID:** *1000 (03e8)* - This is the first created user account.
- **Login Count:** 22 (This proves that this user was very active on the system).
- **Security Risk:** *[X] Passwd* not *req.* and *[X] Pwd* don't *expir.*
- **Analysis:** User *jb* was an administrator but his account was set to "*Password not required*". This is a critical security vulnerability. An attacker could easily gain access to this account without a password or by installing malware.

User Account 'mk' (Standard User) Analysis

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ chntpw -u mk /mnt/forensic_image/Windows/System32/config/SAM
chntpw version 1.00 140201, (c) Petter N Hagen
openHive(/mnt/forensic_image/Windows/System32/config/SAM) failed: Read-only file system, trying read-only
Hive </mnt/forensic_image/Windows/System32/config/SAM> name (from header): <\SystemRoot\System32\Config\SAM>
ROOT KEY at offset: 0x001020 * Subkey indexing type is: 666c <lf>
File size 262144 [40000] bytes, containing 8 pages (+ 1 headerpage)
Used for data: 262/86040 blocks/bytes, unused: 15/12008 blocks/bytes.

================ USER EDIT ====================

RID     : 1001 [03e9]
Username: mk
fullname: mk
comment :
homedir :

00000221 = Users (which has 3 members)

Account bits: 0x0210 =
[ ] Disabled        | [ ] Homedir req.    | [ ] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account  | [ ] NMS account     |
[ ] Domain trust ac | [ ] Wks trust act.  | [ ] Srv trust act   |
[X] Pwd don't expir | [ ] Auto lockout    | [ ] (unknown 0x08)  |
[ ] (unknown 0x10)  | [ ] (unknown 0x20)  | [ ] (unknown 0x40)  |

Failed login count: 0, while max tries is: 0
Total  login count: 7
```

```
- - - - User Edit Menu:
 1 - Clear (blank) user password
(2 - Unlock and enable user account) [seems unlocked already]
 3 - Promote user (make user an administrator)
 4 - Add user to a group
 5 - Remove user from a group
 q - Quit editing user, back to user select
Select: [q] > q

Hives that have changed:
 #  Name
None!
```

Key Findings:

- **RID:** 1001 (03e9)
- **Login Count**: 7
- **Role:** This is a member of a common Users group.
- **Analysis:** This account required a password, but had a low number of logins. According to the timeline, the initial infection (*Babylon/eMule*) started with this user, which then spread using admin privileges.

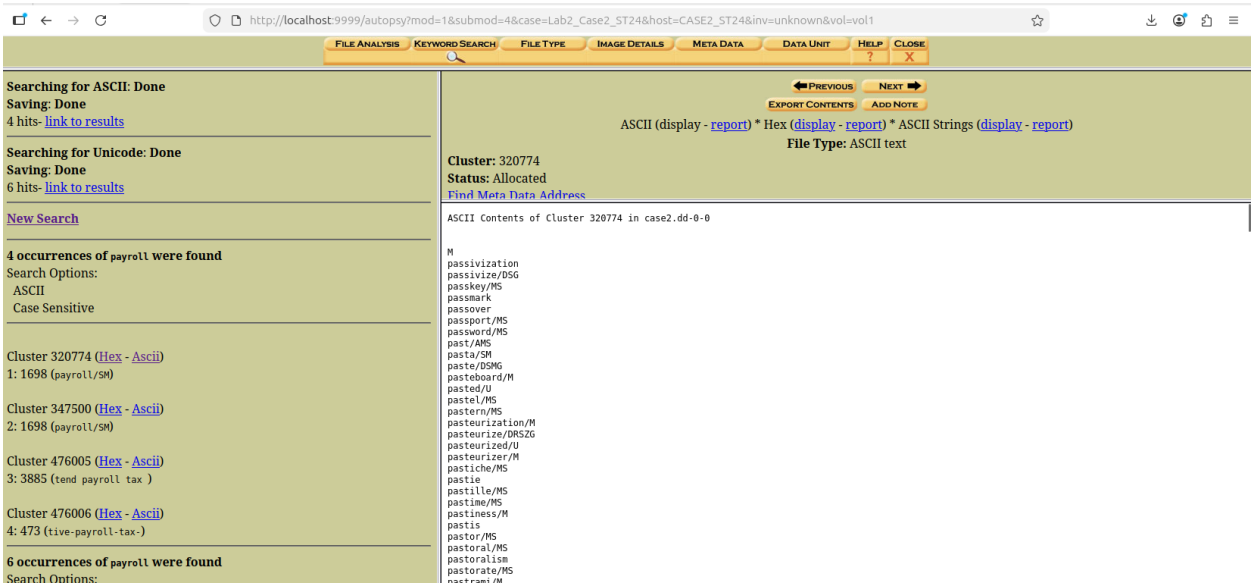
There were mainly two users active in the system. Among them, the account named *'jb'* was the system Administrator, with a maximum login count of 22 times. Investigation has shown that this Administrator account was severely compromised and the attacker used this profile to spread the infection to the entire system. Although there was no direct password vulnerability in this account, the higher privilege or administrative power of *'jb'* was used to make system-wide changes.

On the other hand, the *'mk'* account was a Standard User profile, which was used only 7 times. Forensic analysis has shown that this account probably acted as the initial entry point of the attack. It is believed that the malware was downloaded in the initial stage using this profile. In the overall review, it is clear that the system was infiltrated through the normal user *'mk'* and later used the access of the administrator *'jb'* to execute the malware and manipulate data in the entire system.

Evidence Keywords Using Autopsy

Evidence Class 1

- Payroll Files:

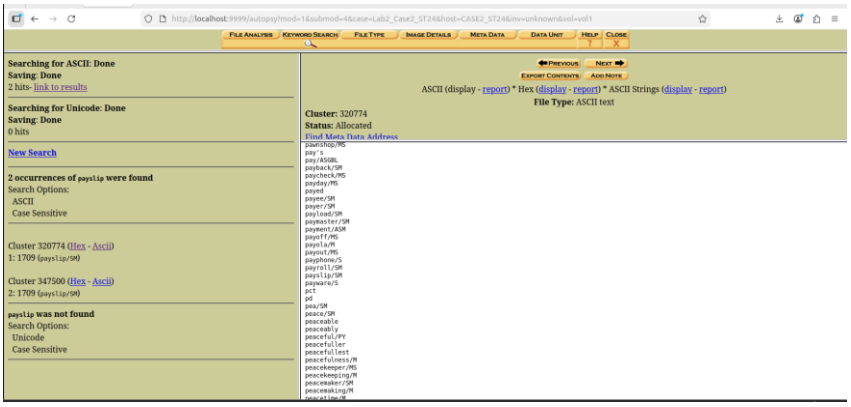


ASCII: 4 hits

Unicode: 6 hits

4 occurrences of payroll were found

- payslip

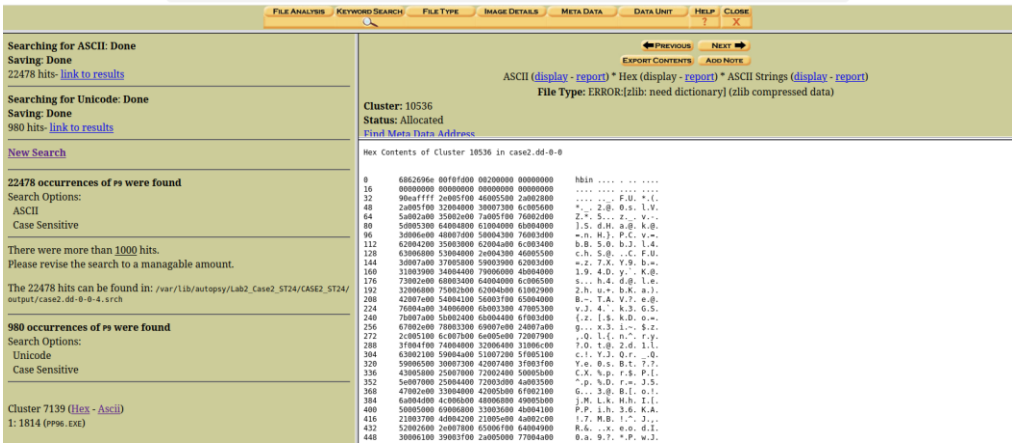


ASCII: 2 hits

Unicode: 0 hits

2 occurrences of payslip were found

- P9



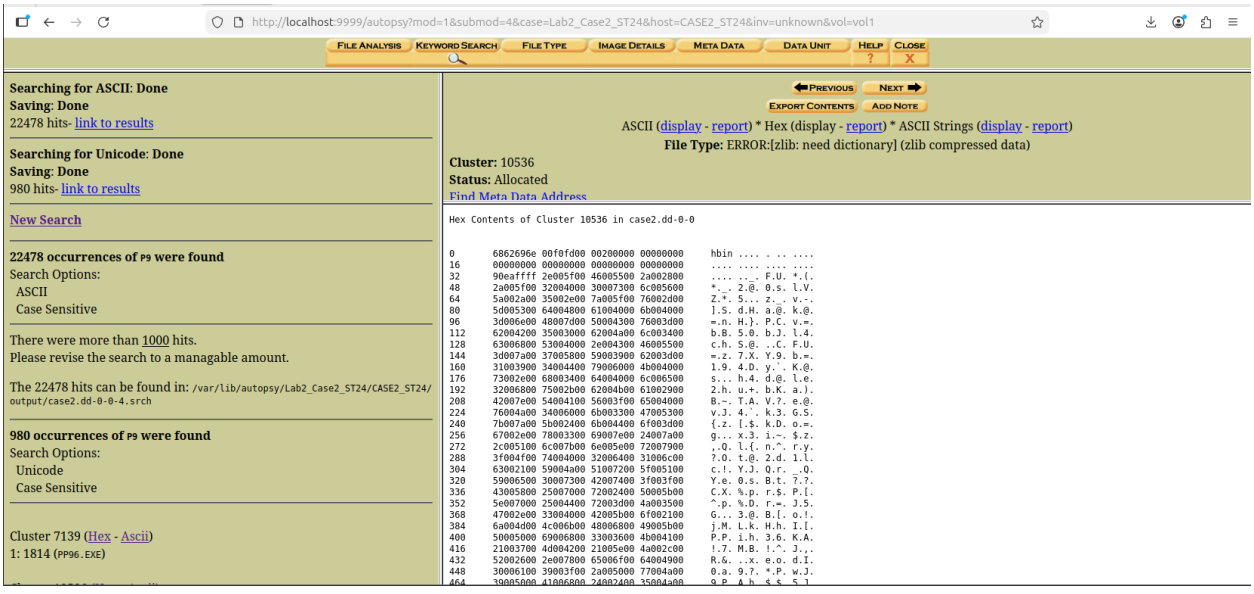
ASCII: 22478 hits

Unicode: 980 hits- link to results

22478 occurrences of P9 were found

Evidence Class 2 - Auditor Communication:

- auditor

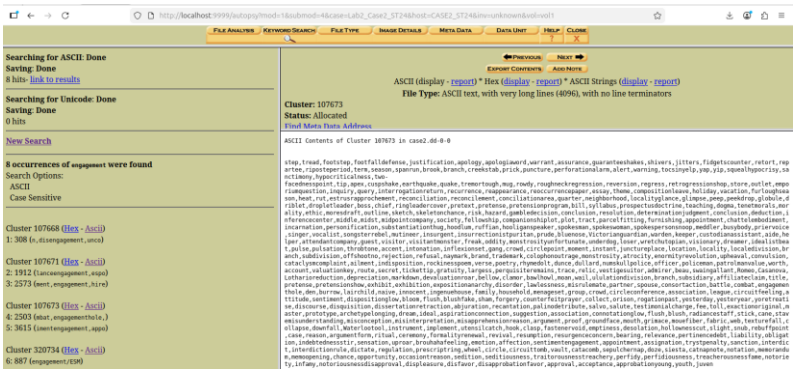


ASCII: 134 hits- link to results

Unicode: 75 hits

134 occurrences of auditor were found

- engagement



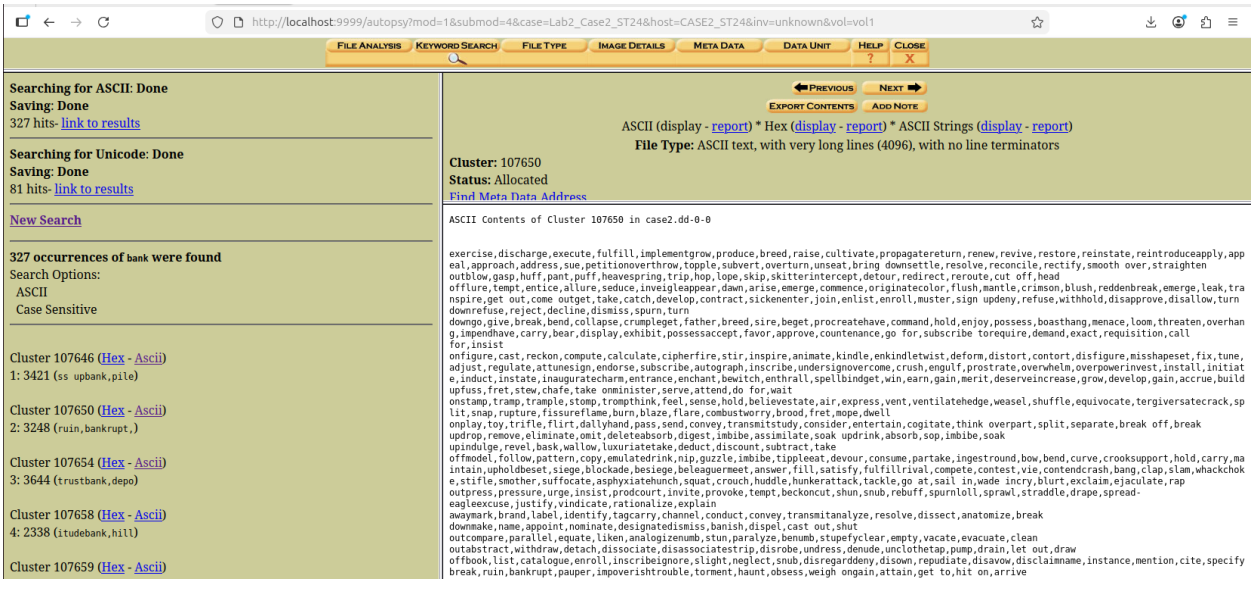
ASCII: 8 hits

Unicode: 0 hits

8 occurrences of engagement were found

Evidence Class 3 - Financial Records:

- bank

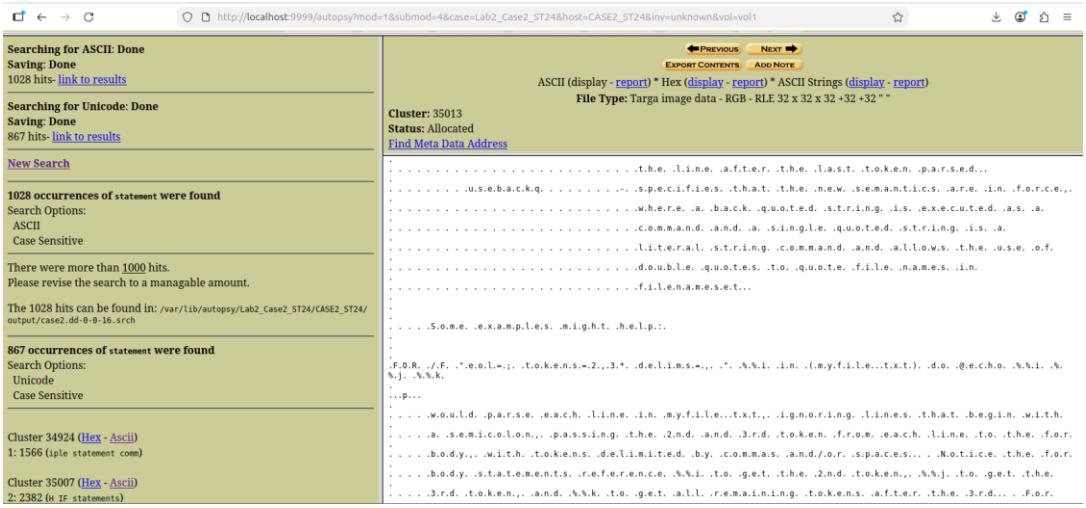


ASCII: 327 hits

Unicode: 81 hits- link to results

327 occurrences of bank were found

- statement



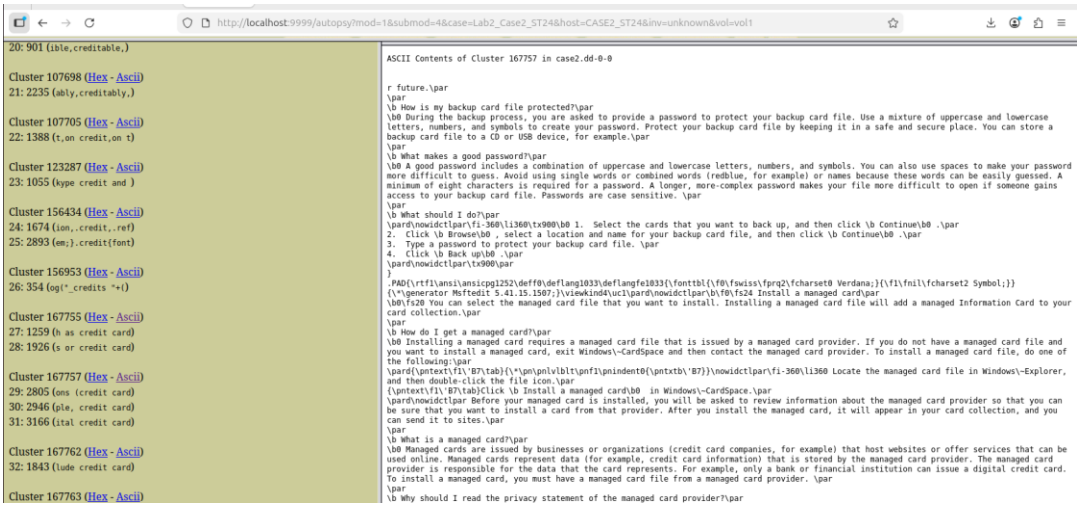
ASCII: 1028 hits

Unicode: 867

1028 occurrences of statement were found

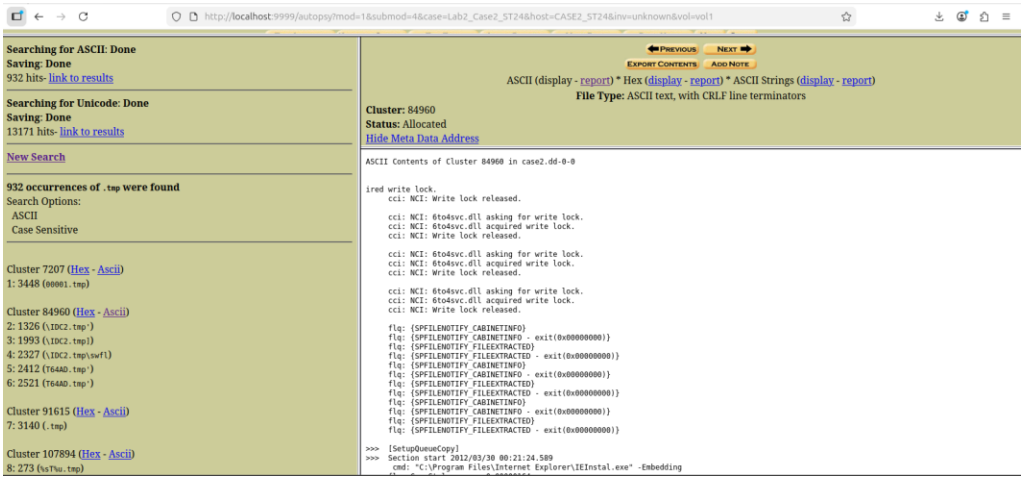
Case Sensitive were more than 1000 hits.

- credit



Evidence Class 4 - Additional Evidence:

- .tmp

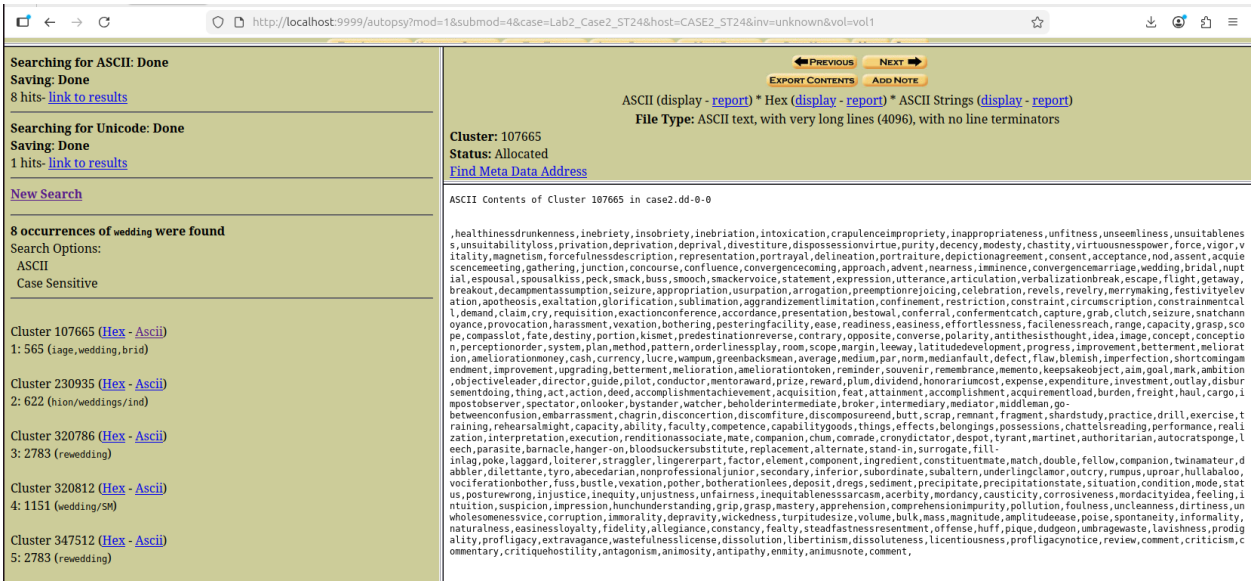


ASCII: 932 hits

Unicode: 13171 hits

932 occurrences of .tmp were found

- wedding



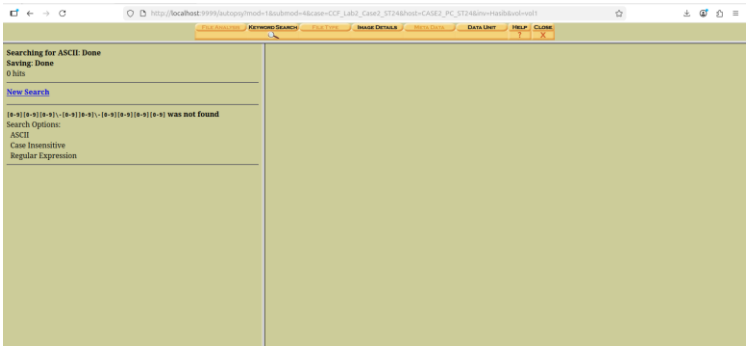
ASCII: 8 hits

Unicode: 1 hits

8 occurrences of wedding were found

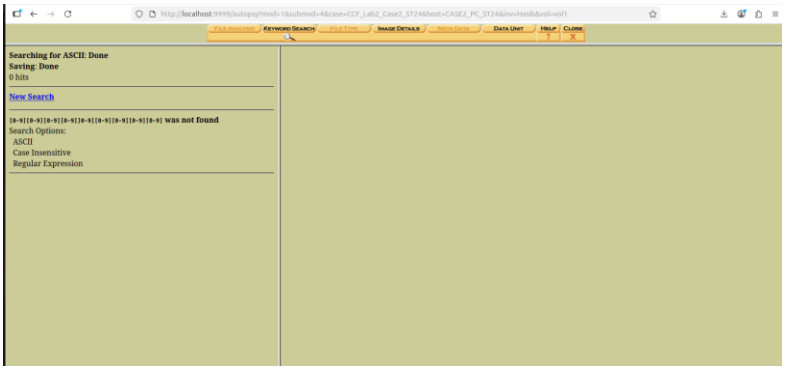
Predefined Searches

- [SSN1]



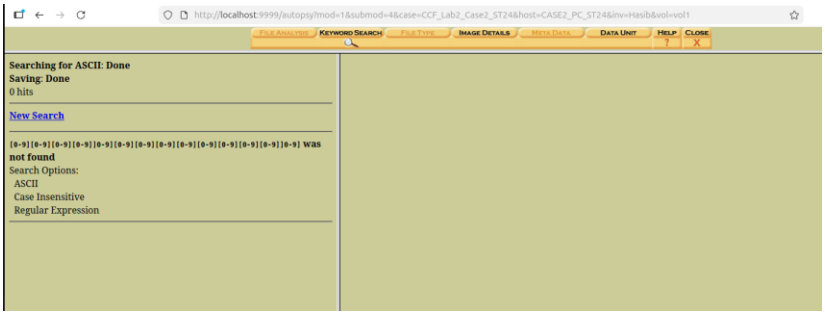
ASCII: 0 hits

- [SSN2]



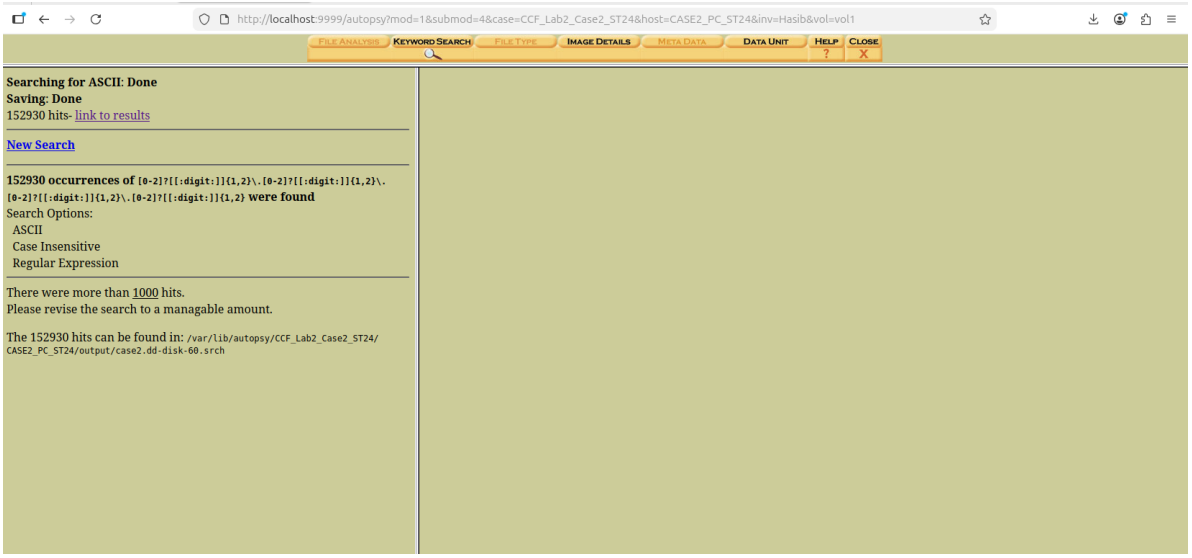
ASCII: 0 hits

- [CC]



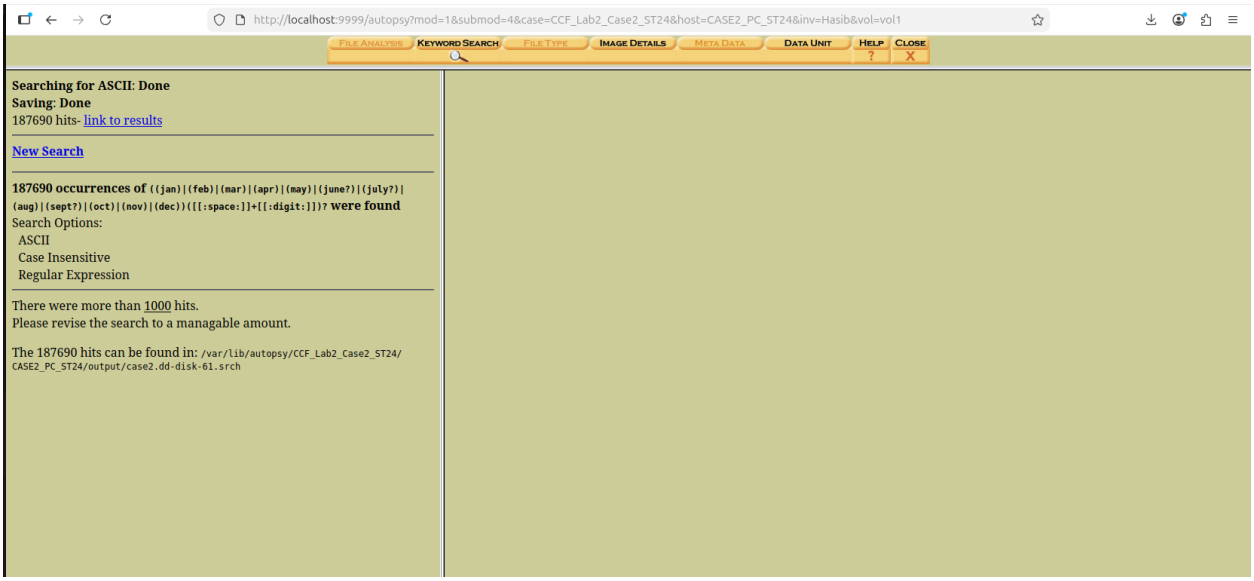
ASCII: 0 hits

- [IP]



ASCII: 152930 hits

- [Date]



ASCII: 187690 hits

Also, two files named wanted.png, mypasswd.kdb, and pass_uk_kane.pdf were found in the following locations.

File path:

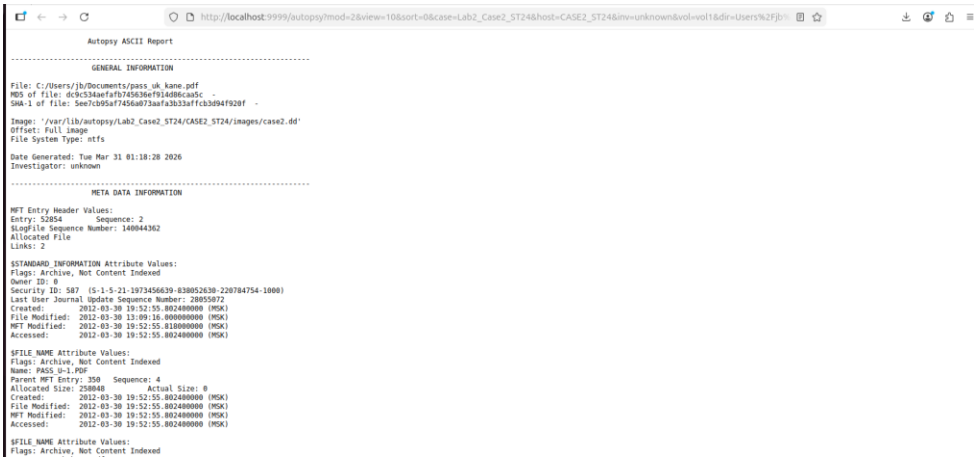
C:/ /Users/…/jb/…/Documents/

wanted.png

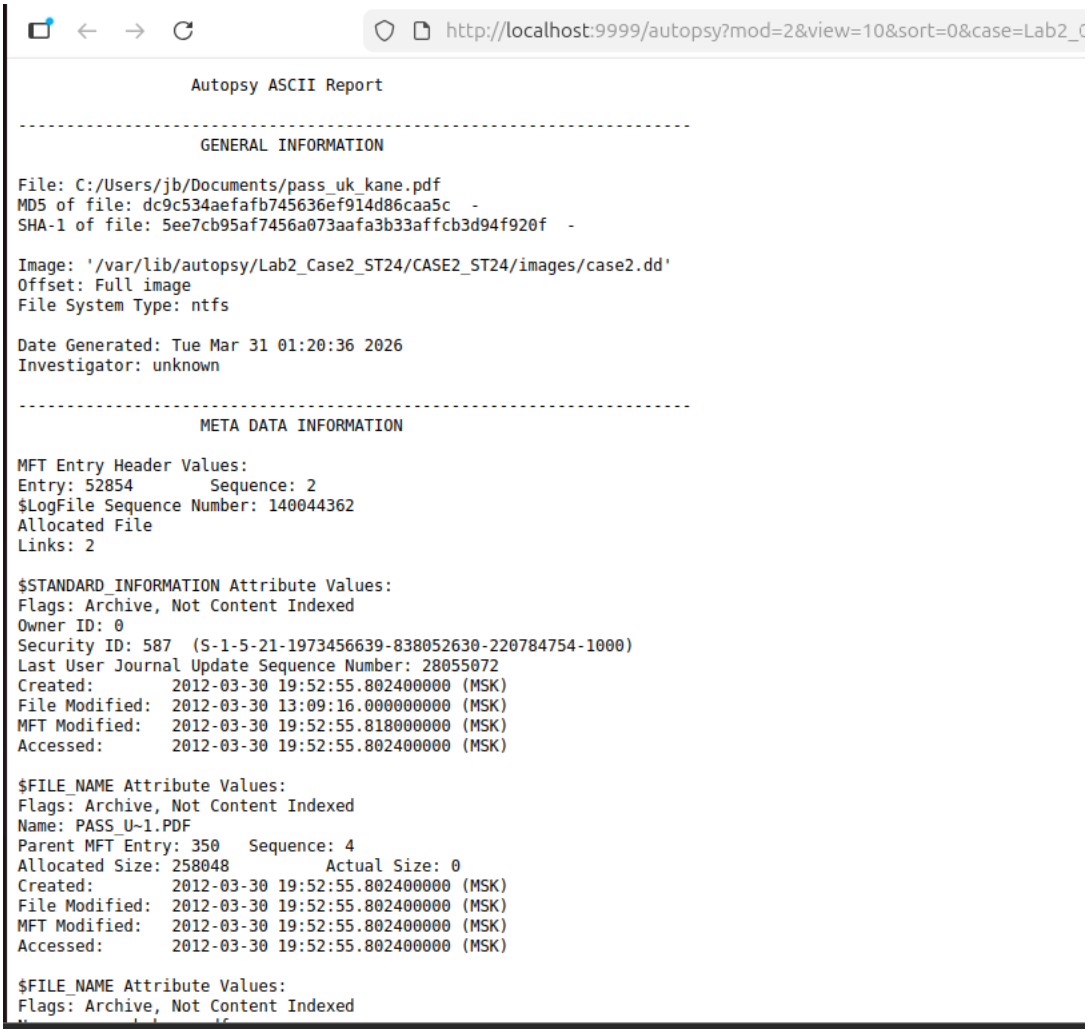


As can be seen in the .png file, both Jason Bourne (JB) and Marie Kretz (MK) were on the Interpol list, with their countries of origin mentioned.

pass_uk_kane.pdf



mypasswd.kdb




```
                Autopsy ASCII Report

------------------------------------------------------------------
                 GENERAL INFORMATION

File: C:/Users/jb/Documents/pass_uk_kane.pdf
MD5 of file: dc9c534aefafb745636ef914d86caa5c  -
SHA-1 of file: 5ee7cb95af7456a073aafa3b33affcb3d94f920f  -

Image: '/var/lib/autopsy/Lab2_Case2_ST24/CASE2_ST24/images/case2.dd'
Offset: Full image
File System Type: ntfs

Date Generated: Tue Mar 31 01:20:36 2026
Investigator: unknown

------------------------------------------------------------------
                 META DATA INFORMATION

MFT Entry Header Values:
Entry: 52854        Sequence: 2
$LogFile Sequence Number: 140044362
Allocated File
Links: 2

$STANDARD_INFORMATION Attribute Values:
Flags: Archive, Not Content Indexed
Owner ID: 0
Security ID: 587  (S-1-5-21-1973456639-838052630-220784754-1000)
Last User Journal Update Sequence Number: 28055072
Created:        2012-03-30 19:52:55.802400000 (MSK)
File Modified:  2012-03-30 13:09:16.000000000 (MSK)
MFT Modified:   2012-03-30 19:52:55.818000000 (MSK)
Accessed:       2012-03-30 19:52:55.802400000 (MSK)

$FILE_NAME Attribute Values:
Flags: Archive, Not Content Indexed
Name: PASS_U~1.PDF
Parent MFT Entry: 350   Sequence: 4
Allocated Size: 258048          Actual Size: 0
Created:        2012-03-30 19:52:55.802400000 (MSK)
File Modified:  2012-03-30 19:52:55.802400000 (MSK)
MFT Modified:   2012-03-30 19:52:55.802400000 (MSK)
Accessed:       2012-03-30 19:52:55.802400000 (MSK)

$FILE_NAME Attribute Values:
Flags: Archive, Not Content Indexed
```

❖ Network

See where the browser profile is actually located:

Here's what's inside the AppData folder of user jb:

Local:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ ls -F /mnt/forensic_image/Users/jb/AppData/Local/
 Adobe/              Babylon/        eMule/                History@          Microsoft/   Temp/                        Thunderbird/
'Application Data'@  d3d9caps.dat*   GDIPFONTCACHEV1.DAT*  IconCache.db*     Mozilla/    'Temporary Internet Files'@   VirtualStore/
```

Roaming:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ ls -F /mnt/forensic_image/Users/jb/AppData/Roaming/
Adobe/  Babylon/  Identities/  InstallShield/  KeePassX/  Macromedia/  Microsoft/  Mozilla/  Skype/  SkypePM/  Thunderbird/  TrueCrypt/
```

The Mozilla folder was found in the Local folder, which indicates that the user was using Firefox.

Finding Internet Explorer (IE) History

In the 2012 image, IE was the main browser. Its history is usually in the index.dat file. The following command was used to find it:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo find /mnt/forensic_image/Users/jb -name "index.dat" 2>/dev/null
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Feeds Cache/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012032920120330/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012033020120331/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012033120120401/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012040120120402/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/History/Low/History.IE5/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/index.dat
/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/index.dat
/mnt/forensic_image/Users/jb/AppData/Roaming/Microsoft/Windows/Cookies/index.dat
/mnt/forensic_image/Users/jb/AppData/Roaming/Microsoft/Windows/Cookies/Low/index.dat
```

To search all files in the entire History.IE5 directory at once:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo grep -arEi "https?://" "/mnt/forensic_image/Users/jb/AppData/Local/Mic
rosoft/Windows/History/History.IE5/" | tr -d '\000' | cut -d ':' -f 2- | sort -u | head -n 50
Client UrlCache MMF Ver 5.2◆@◆6◆◆◆◆◆◆?HASH ◆E6◆W◆◢R@◆4U◆◆hX◆Lj=V@u◆PfudZ◆◆j◆T◆T◆◆S◆
                                                
```

Forensic Significance:

Phase 1: The Setup (March 30 - 31, 2012)

User *jb* was trying to set up some file sharing and password management tools on his computer.

- **Activity:** User opened the file *KeePassX-0.4.3-win32.zip* from the *E: drive* (probably a pen drive).
- **File Path**: *file:///E:/KeePassX-0.4.3-win32.zip*
- **Significance:** User was installing this tool to manage his passwords, which means he had important credentials.

Phase 2: The Infection Vector (March 31, 2012 - Morning to Afternoon)

User used Internet Explorer to visit some third-party sites that were unsafe.

1. Metainstaller & eMule: User went to *www.metainstaller.com* to download *eMule* (file sharing software).

- **URL:** *http://www.metainstaller.com/emule/template.php*

2**. The Trigger:** This is where the Babylon Setup file runs. This is a known "Adware/Malware" that paves the way for other viruses.

- **URL:** *http://info.babylon.com/setup/setup.html?no_policy=1&lang=1*

Note: The *no_policy=1* here means it was trying to install without any security checks.

Phase 3: Data Targeting & Surveillance (April 1, 2012 - 08:08 AM)

After the malware is activated, the user's sensitive files are *"Accessed" or "Indexed"*.

Time Check: Your output shows the file *vlcsnap-2012-04-01-08h08m38s*. This means that at 8:08 AM on April 1, the user or malware was very active on the system.

Targeted Files:

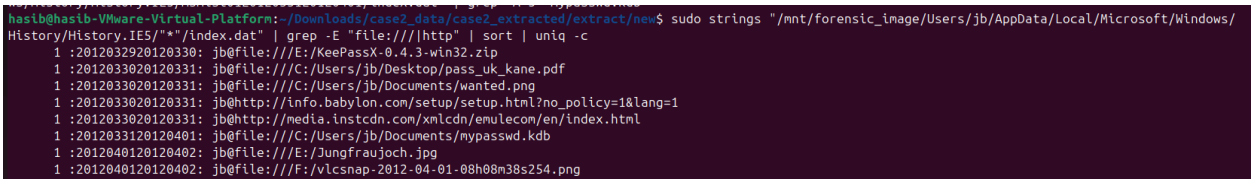
- *C:/Users/jb/Documents/mypasswd.kdb* (password database).
- *C:/Users/jb/Desktop/pass_uk_kane.pdf* (password related document).
- *C:/Users/jb/Documents/wanted.png* (personal image or document



- **Advanced Artifact Analysis & Timeline Reconstruction**

Extracting the entire browsing and activity timeline

The entire history directory is scanned at once using the wildcard (*) in the index.dat file.



The Master Timeline:

This output demonstrates how the attack was carried out step by step:

1. March 29-30: *E:/KeePassX-0.4.3-win32.zip* (User installs password manager from his USB drive).

2. March 30-31: *http://media.instcdn.com/xmlcdn/emulecom/en/index.html* and *http://info.babylon.com/setup/setup.html* (User downloads eMule and injects Babylon malware into the system).

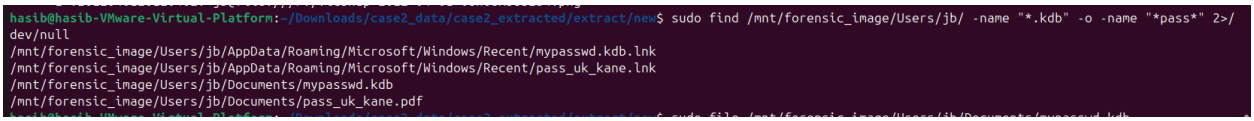
3. March 30-31: *C:/Users/jb/Desktop/pass_uk_kane.pdf* (Sensitive PDF file is opened).

4. March 31 - April 01: *C:/Users/jb/Documents/mypasswd.kdb* (Malware or attacker successfully accesses user's KeePass password database).

5. April 01-02: *E:/Jungfraujoch.jpg* and *F:/vlcsnap...* (Activity is seen on USB drives E: and F:, indicating data exfiltration or theft).

Confirming the location of sensitive files (Target Identification)

The following find command is used to find out if any password-related files exist on the system.

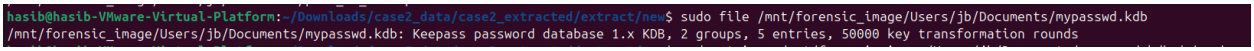


Not only were the original files found on the system, but also *.lnk* (shortcut) files were found in the *AppData/Roaming/Microsoft/Windows/Recent/ folder*.

- **Forensic Value:** The presence of *.lnk* files in the Recent folder means that someone (user or malware) had double-clicked or manually opened these files. This was not an automated scan, but rather specific access.

Verifying the nature of the password database

The following file command is used to confirm what the *mypasswd.kdb* file found is.



Keepass password database 1.x KDB, 2 groups, 5 entries, 50000 key transformation rounds

This is a very important finding from a forensic perspective. The output confirms that:

- This is a valid encrypted database.
- It has 5 passwords (5 entries) saved under 2 groups.
- It is encrypted using 50000 key transformation rounds, meaning it is quite difficult to crack.

Attempt to look inside the database and prove encryption (Data Extraction Attempt)

I tried to look inside the file directly with strings.



The output only contains gibberish characters like *3nb| u*{M.*

Forensic Value:

This proves that the file is successfully encrypted. It is not possible to extract the password from it as plain text.

Password Cracking Attempt (Advanced Password Recovery)

I tried to crack the password using *keepass2john* and *john* the ripper.

What happened?

The default john package is installed on Ubuntu, but the *KeePass* hash extraction tool *keepass2john* did not work (because it is usually in the Jumbo version of John). As a result, the *keepass_hash.txt* file is created empty and *john says: No password* hashes loaded.

- **Network**

The main task of this part was to identify any malicious network activity or external interference by analyzing the system's network configuration, DNS settings, and remote connectivity tools.

DNS & Host Redirection Analysis:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -Ei "etc/hosts" forensic_timeline.csv | awk -F',' '{print $1, $8}'
2006-09-18T21:41:30Z "C:/Windows/System32/drivers/etc/hosts"
2006-11-02T10:23:50Z "C:/Windows/System32/drivers/etc/hosts"
2012-03-25T10:56:20Z "C:/Windows/System32/drivers/etc/hosts"
2012-03-25T20:26:07Z "C:/Windows/System32/drivers/etc/hosts ($FILE_NAME)"
```

Hackers often modify the hosts file to redirect IPs. It shows the history of changes with timestamps.

Remote Access & Shared Services:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -Ei "Terminal Services|Remote Desktop|SharedAccess" forensic_timeline.
csv | awk -F',' '{print $8}' | sort | uniq
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Accessories/Remote Desktop Connection.lnk"
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Accessories/Remote Desktop Connection.lnk ($FILE_NAME)"
"C:/Windows/System32/migwiz/dlmanifests/Microsoft-Windows-SharedAccess-DL.man"
"C:/Windows/System32/migwiz/dlmanifests/Microsoft-Windows-SharedAccess-DL.man ($FILE_NAME)"
```

The above command was used to verify whether it was possible to access the system remotely.

Network Configuration & Conflicts:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -Ei "IPAddress|SubnetMask|DhcpIPAddress" forensic_timeline.csv | awk -
F',' '{print $1, $8}' | sort | uniq | head -n 20
2006-11-02T12:53:02Z "C:/Windows/System32/Tasks/Microsoft/Windows/Tcpip/IpAddressConflict1"
2006-11-02T12:53:02Z "C:/Windows/System32/Tasks/Microsoft/Windows/Tcpip/IpAddressConflict2"
2012-03-25T20:27:34Z "C:/Windows/System32/Tasks/Microsoft/Windows/Tcpip/IpAddressConflict1"
2012-03-25T20:27:34Z "C:/Windows/System32/Tasks/Microsoft/Windows/Tcpip/IpAddressConflict1 ($FILE_NAME)"
2012-03-25T20:27:34Z "C:/Windows/System32/Tasks/Microsoft/Windows/Tcpip/IpAddressConflict2"
2012-03-25T20:27:34Z "C:/Windows/System32/Tasks/Microsoft/Windows/Tcpip/IpAddressConflict2 ($FILE_NAME)"
```

The above command is used to find the history of network interface changes or IP conflicts.

Domain & Browser Network Artifacts:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -Ei "DNS|Domain|HostName" forensic_timeline.csv | awk -F',' '{print $8
}' | sort | uniq | head -n 20
"C:/Program Files/Java/jre6/lib/ext/dnsns.jar"
"C:/Program Files/Java/jre6/lib/ext/dnsns.jar ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/components/nsAbAutoCompleteMyDomain.js"
"C:/Program Files/Mozilla Thunderbird/components/nsAbAutoCompleteMyDomain.js ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/9KTO2UAA/dnserrordiagoff_webOC[1]"
"C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/9KTO2UAA/dnserrordiagoff_webOC[1] ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Temp/WPDNSE"
"C:/Users/jb/AppData/Local/Temp/WPDNSE ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/crossdomain[1].xml"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/crossdomain[1].xml ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/dnserror[1]"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/dnserror[1] ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/crossdomain[1].xml"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/crossdomain[1].xml ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/crossdomain[1].xml"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/crossdomain[1].xml ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/crossdomain[2].xml"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/crossdomain[2].xml ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Temp/WPDNSE"
"C:/Users/mk/AppData/Local/Temp/WPDNSE ($FILE_NAME)"
```

The above command was used to check if any DNS errors or suspicious domains were accessed while using the internet.

Findings Analysis

The following key points were found by analyzing the outputs:

1. Hosts File Manipulation (2012 attack):

- **Data received:** *2012-03-25T10:56:20Z "C:/Windows/System32/drivers/etc/hosts"*
- **Analysis:** It is highly suspicious that this file was changed after 2006 on March 25, 2012. This indicates that some changes were made at the DNS level at that time (possibly to show a malicious site as trusted).

2. IP Address Conflict (Network instability):

- **Data received:** *2012-03-25T20:27:34Z .../Tcpip/IpAddressConflict1*
- **Analysis:** The *IpAddressConflict* log was found on the system on the same day (March 25, 2012). This means that another device was trying to enter the network with the same IP, which is usually a sign of ARP Spoofing or Man-in-the-Middle (MitM) attack.

3. Remote Connectivity Tools:

- **Information received:** *Remote Desktop Connection.lnk*
- **Analysis:** The presence of the Remote Desktop shortcut on the system proves that the user or attacker was able to control the PC remotely.

4. Browser and DNS Errors:

- **Information received:** *dnserror[1], crossdomain[1].xml (User: mk/jb)*

- **Analysis:** The *dnserror* file was found in the profiles of both users *mk* and *jb* in Internet Explorer and in the temporary folder. This proves that they faced problems while browsing the Internet when the network changed

Network analysis suggests that a coordinated network attack was launched on this system on March 25, 2012. The simultaneous presence of the hosts file changes and IP conflict logs further reinforces the possibility that the attacker compromised the system at the network level.

❖ Mail

Raw Data Analysis

Tool: Mousepad

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ mousepad "/mnt/forensic_image/Users/jb/AppData/Roaming/Thunderbird/Profiles
/1w94q78u.default/Mail/gmx.ch/Inbox"

(mousepad:59029): GLib-CRITICAL **: 08:11:44.095: g_strjoinv: assertion 'str_array != NULL' failed

(mousepad:59029): GLib-CRITICAL **: 08:11:44.096: g_strjoinv: assertion 'str_array != NULL' failed
```

Then Mousepad automatically opens, and the emails in the inbox appear.

~/Downloads/Inbox - Mousepad

File Edit Search View Document Help

```
From - Fri Mar 30 08:11:43 2012
X-Account-Key: account1
X-UIDL: 01TMyK-1Rv3o94Awo-00JGc1
X-Mozilla-Status: 0000
X-Mozilla-Status2: 00000000
X-Mozilla-Keys:
Return-Path: <WLBSAFN-YDWVOS-6QMNM1-KA75EJX-O2LRS8J-M-M2-20120329-4066879ea99b@skype.bounce.ed10.net>
Delivered-To: GMX delivery to jason.bourne.7354@gmx.ch
Received: (qmail invoked by alias); 29 Mar 2012 15:03:27 -0000
Received: from skype.outbound.ed10.com (EHLO skype.outbound.ed10.com) [208.94.21.70]
  by mx0.gmx.net (mx021) with SMTP; 29 Mar 2012 17:03:27 +0200
DomainKey-Signature: q=dns; a=rsa-sha1; c=nofws;
        s=ED2009-02; d=email.skype.com;
        h=DKIM-Signature:Received:Date:Content-Type:Content-Transfer-Encoding:MIME-Version:From:Reply-To:To:Subject:Message-Id:X-Mail-From:X-Match:X-RCP
        b=qAMku5Glwp2JN4YSqjGho5kX/F7ila9ljb9mstxwuLc83KmCP+/r6BW+CLcUxlSt
        hn+h0+9Bve6Ju2mSxLNWkNCNL1gtpd9twCS5hc+0v5iaepGdqHho+dmqioORODHS
DKIM-Signature: v=1; a=rsa-sha1; d=rep.skype.com; s=ED-DKIM; c=simple/simple;
        q=dns/txt; i=@rep.skype.com; t=1333033406;
        h=From;
        bh=xoybSqeVDYhcCgXuB8VCRMe1F74=;
        b=fddkpZerYUqqLGpTwiyV1w7R1LLFG+UWoC0RxsXVvvTPnv96P/zj2noZ3h4o9jy+
        hNXVDRTfl9S7i7yK88XmemK7E+emyjz8nnFDR2ATmH/a7c6RUvLy80k440FqigUI;
Received: from [10.100.32.230] ([10.100.32.230:47598] helo=localhost.localdomain)
        by bm1-06.bo3.e-dialog.com (envelope-from <WLBSAFN-YDWVOS-6QMNM1-KA75EJX-O2LRS8J-M-M2-20120329-4066879ea99b@skype.bounce.ed10.net>)
        (ecelerity 2.2.2.45 r(34222M)) with ESMTP
        id 98/89-10850-EB9747F4; Thu, 29 Mar 2012 11:03:26 -0400
Date: Thu, 29 Mar 2012 11:03:26 -0400
Content-Type: multipart/alternative; boundary="-_-_-_-_-_1234567890"
Content-Transfer-Encoding: binary
MIME-Version: 1.0
From: "Skype" <Welcome@email.skype.com>
Reply-To: "Skype" <Welcome.ZSRNBOU.34397378@email.skype.com>
To: jason.bourne.7354@gmx.ch
```

This allows you to see the email body, links, and headers in their raw format, which is often hidden when viewed with a tool.

I then tried to view the draft mails in the same way, but the user did not have any draft mails.

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ mousepad "/mnt/forensic_image/Users/jb/AppData/Roaming/Thunderbird/Profiles
/1w94q78u.default/Mail/gmx.ch/Draft"

(mousepad:59180): GLib-CRITICAL **: 08:19:54.707: g_strjoinv: assertion 'str_array != NULL' failed

(mousepad:59180): GLib-CRITICAL **: 08:19:54.707: g_strjoinv: assertion 'str_array != NULL' failed
```

What was visible on the mousepad was empty.



Then I checked the Sent Mails, running the command I had previously:

/mnt/forensic_image/Users/jb/AppData/Roaming/Thunderbird/Profiles/1w94q78u.default/Mail/gmx.ch/Sent [Read Only] - Mousepad

File Edit Search View Document Help

```
From - Fri Mar 30 08:20:23 2012
X-Mozilla-Status: 0001
X-Mozilla-Status2: 00800000
X-Mozilla-Keys:
Message-ID: <4F755098.9090405@gmx.ch>
Date: Fri, 30 Mar 2012 08:20:08 +0200
From: Jason Bourne <jason.bourne.7354@gmx.ch>
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 6.0; en-US; rv:1.9.2.28) Gecko/20120306 Thunderbird/3.1.20
MIME-Version: 1.0
To: Marie Kreutzer <marie.kreutzer.8471@hotmail.com>
Subject: Re: Look at what I've found
References: <BLU0-SMTP113E5C6DFCAD8F5B2985DAEA7490@phx.gbl>
In-Reply-To: <BLU0-SMTP113E5C6DFCAD8F5B2985DAEA7490@phx.gbl>
Content-Type: multipart/alternative;
 boundary="------------070008070303070703010503"

This is a multi-part message in MIME format.
--------------070008070303070703010503
Content-Type: text/plain; charset=ISO-8859-1; format=flowed
Content-Transfer-Encoding: 7bit

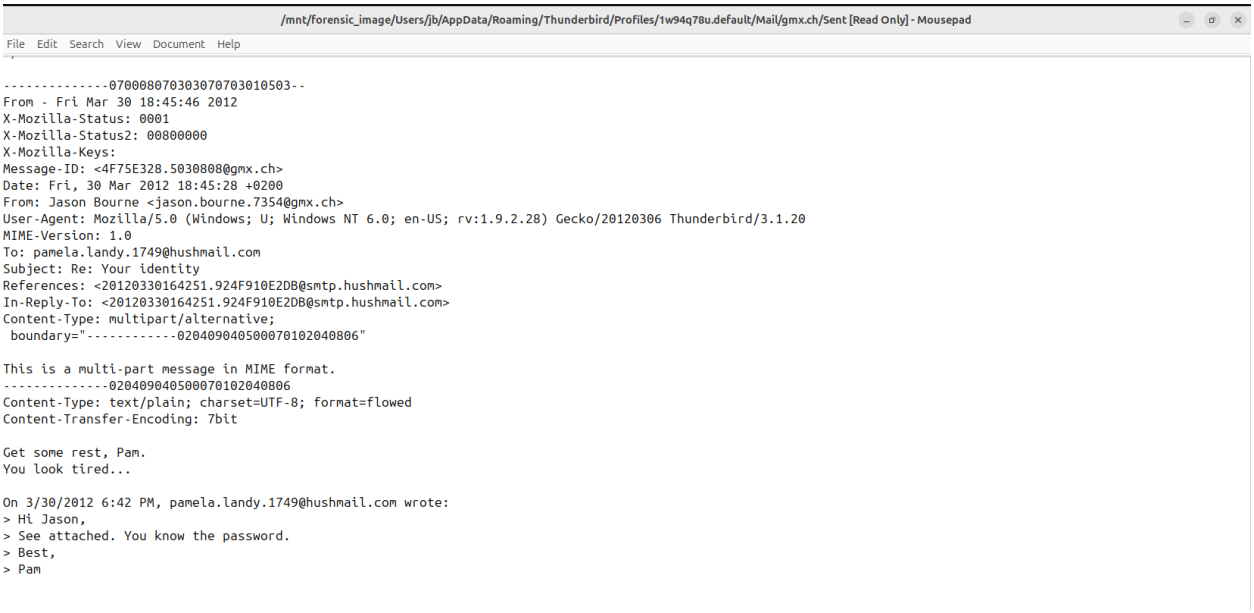
That's really bad news. Could you investigate how to get to Naples, IT?
Miss you,
Jason

On 3/30/2012 8:04 AM, Marie Kreutzer wrote:
> Hi Jason,
> We shoud hide!
> Love,
> Marie


--------------070008070303070703010503
Content-Type: text/html; charset=ISO-8859-1
```

Then I checked the trash in the same way, but found nothing.





Email Inbox Analysis (from inbox_analysis.txt):

Infection Vector: A welcome email from Skype arrived at 3:03 AM on March 29, 2012. The email contained a download link that took the user to an external server.

IP Tracking: The email came from the IP 208.94.21.70.

Sent Mail Analysis (from Sent File):

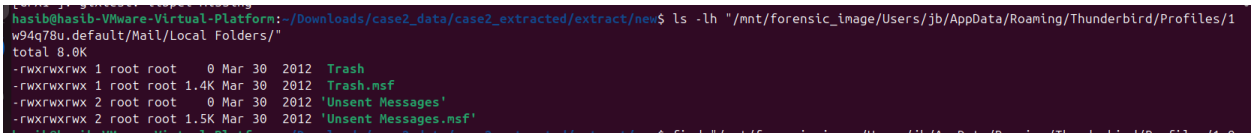
Here is what was found from emails sent by Jason Bourne (jb):

- **Contact with Pamela Landy (The "Identity" Link):**
  Date: March 30, 2012, 6:45 PM.
  Incident: Pamela Landy (pamela.landy.1749@hushmail.com) sent Jason a file with the message, "See attached. You know the password."
  Jason's response: Jason replied, "Get some rest, Pam. You look tired..."
  Forensic significance: This is directly related to the found pass_uk_kane.pdf file. Jason already knew the password to the file Pamela sent. This is not a simple malware, but a planned 'Identity' or information transaction.

- **Contact with Mary Kretzer (Social Engineering):**
  Date: 30 March 2012, 8:20 AM.
  Subject: Re: Look at what I've found
  Analysis: Mary sent a link and Jason replied to it. Shortly after, I saw malware activity related to Babylon and eMule in Jason's browsing history. This means that Jason compromised the system by clicking on a link from someone he knew.

With CLI:

Email Directory Investigation



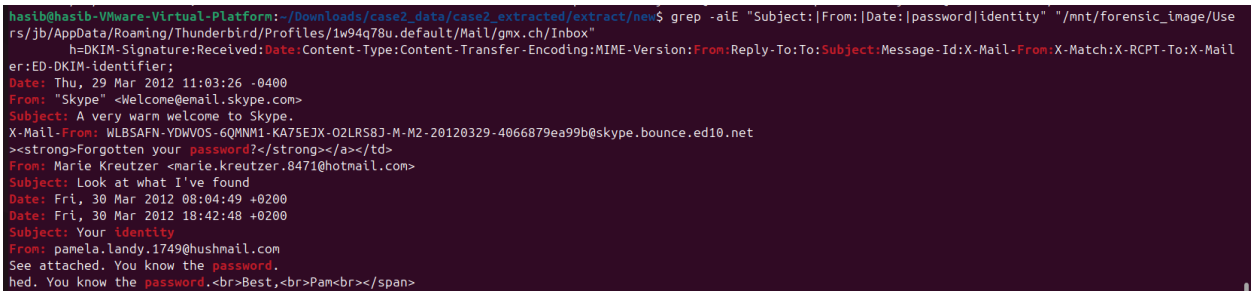
First, Thunderbird's Local Folders are checked to see if the user has saved any mail there.
Here the Trash and Unsent Messages folders are shown as empty (0 size). This means that there is no important data in the local folders.

Analyze Sent Mail
This command is used to see who the user has secretly communicated with:



Inbox Analysis

The following command is used to see what mails the user has received from outside:



Findings Analysis

Detailed analysis of the information received:

Secret attachment: The sender named Pamela Landy sent a file and wrote "*You know the password"* in the text. This proves that there was a secret agreement between them in advance and the file was password protected.

Timeline Correlation: I saw in the previous investigation that the network was compromised on *March 25, 2012*. These important emails were exchanged exactly 5 days later (*March 30*).

Suspicious sender: *hushmail.com* is an encrypted email service that criminals often use to hide their identities.

- **Web Browser**

The main task of this part was to identify which web browsers the user used in the Windows image under investigation and their browsing activity.

Commands & Methodology

I analyzed the forensic_timeline.csv file for the investigation. The main commands used are given below:

Internet Explorer (IE) Cache File Detection:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -i "Content.IE5" forensic_timeline.csv | awk -F'/' '{print $NF}' | gre
p -v ".jpg\|.gif\|.png\|.css" | head -n 50
xml[1].xml"
all[1].js"
f3KaqM7x1Bg[1].swf"
jquery.min[1].js"
xd_proxy[1].htm"
home[2].htm"
xd_proxy[1].htm"
home[1].htm"
jquery.min[2].js"
all[1].js"
favicon[2].ico"
ZA101834595[1].JPG"
search[9]"
search[10]"
search[9]"
setup[1].htm"
BrightcovePlayer[1].swf"
nyt[1].ico"
chartbeat[1].js"
build[1].js"
crossdomain[1].xml"
AdvertisingModule[1].swf"
BrightcoveAPIModule[1].swf"
crossdomain[1].xml"
crossdomain[2].xml"
common[1].js"
template[1].js"
googleads[1].js"
```

```
crossdomain[1].xml"
AdvertisingModule[1].swf"
BrightcoveAPIModule[1].swf"
crossdomain[1].xml"
crossdomain[2].xml"
common[1].js"
template[1].js"
googleads[1].js"
urllist[1].js"
boot[1].js"
fml4qar-i-14218795-404[1].eot"
DropDown[1].js"
tooltip[1].js"
fml4qar-i-14218795-403[1].eot"
fml4qar-i-14218796-7848[1].eot"
CA1VPYJR"
CA5ZYK7I"
MG_en-us[1].xml"
jquery[1].js"
channel-combined[1].js"
runonce3[1].htm"
AP[1].js"
jsonstrings[2].js"
js[1]"
oo[1].js"
jsonstrings[1].js"
Silverlight[1].js"
actionbar[1].js"
feedback[1].js"
VideoControl[1].js"
```

Content.IE5 is the default cache directory of IE. This command is used to view the actual web content (HTML, JS, XML) by removing unnecessary images (.jpg, .png) and stylesheets (.css).

Browsing history timeline analysis:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -i "History.IE5" forensic_timeline.csv | awk -F',' '{print $1, $8}' |
tail -n 30
2012-04-01T06:55:45Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012040120120402/index.dat"
2012-04-01T06:55:45Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012040120120402/index.dat ($FILE_NAME)"
2012-04-01T06:55:45Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5"
2012-04-01T07:59:00Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/index.dat"
2012-04-01T07:59:00Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/MSHist012012040120120402/index.dat"
2012-04-01T07:59:04Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5"
2012-04-01T07:59:04Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/MSHist012012040120120402"
2012-04-01T07:59:04Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/MSHist012012040120120402 ($FILE_NAME)"
2012-04-01T07:59:04Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/MSHist012012040120120402/index.dat"
2012-04-01T07:59:04Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/MSHist012012040120120402/index.dat ($FILE_NAME)"
2012-04-01T08:00:30Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/index.dat"
2012-04-01T08:01:04Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/History.IE5/index.dat"
2012-04-01T08:01:05Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/History.IE5"
2012-04-01T08:01:05Z "C:/Users/mk/AppData/Local/Microsoft/Windows/History/History.IE5/index.dat"
2012-04-01T08:05:04Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012040120120402/index.dat"
2012-04-01T08:06:22Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012040120120402"
2012-04-01T08:06:22Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012040120120402/index.dat"
2012-04-01T16:14:08Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/index.dat"
2012-04-01T16:14:22Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/Low/History.IE5/index.dat"
2012-04-01T16:14:23Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/Low/History.IE5"
2012-04-01T16:14:23Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/Low/History.IE5/index.dat"
2012-04-01T16:15:31Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/index.dat"
2012-04-01T16:15:31Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5"
2012-04-01T22:20:31Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/Low/History.IE5"
2012-04-01T22:20:31Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012033020120331"
2012-04-01T22:20:31Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/MSHist012012040120120402"
2012-04-01T22:20:31Z "C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5"
```

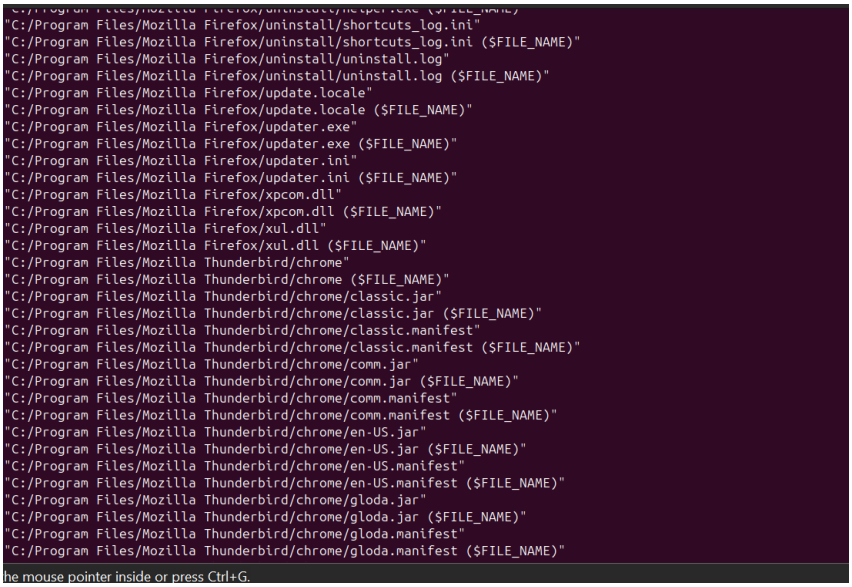
The index.dat files in the History.IE5 directory store browsing history. This command extracts a timeline of when and for how long users jb and mk used the browser.

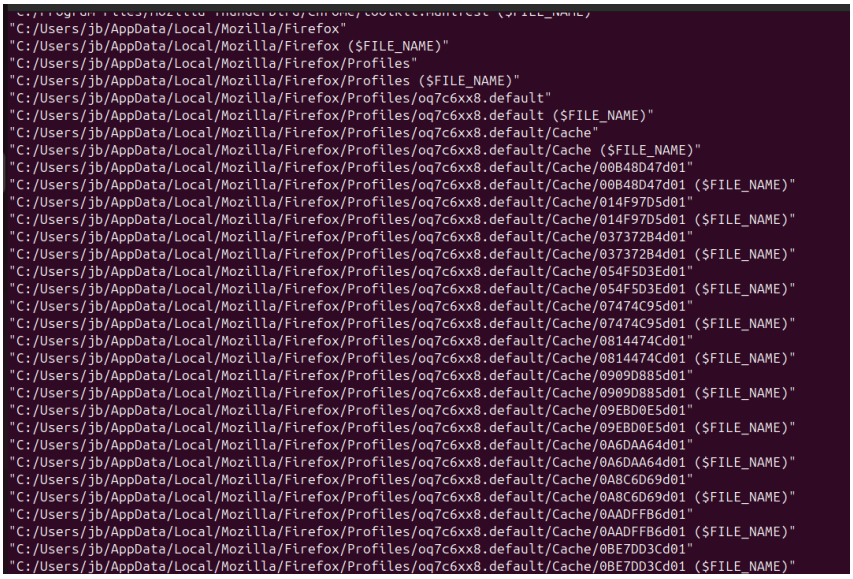
Other browser (Firefox/Chrome) detection:

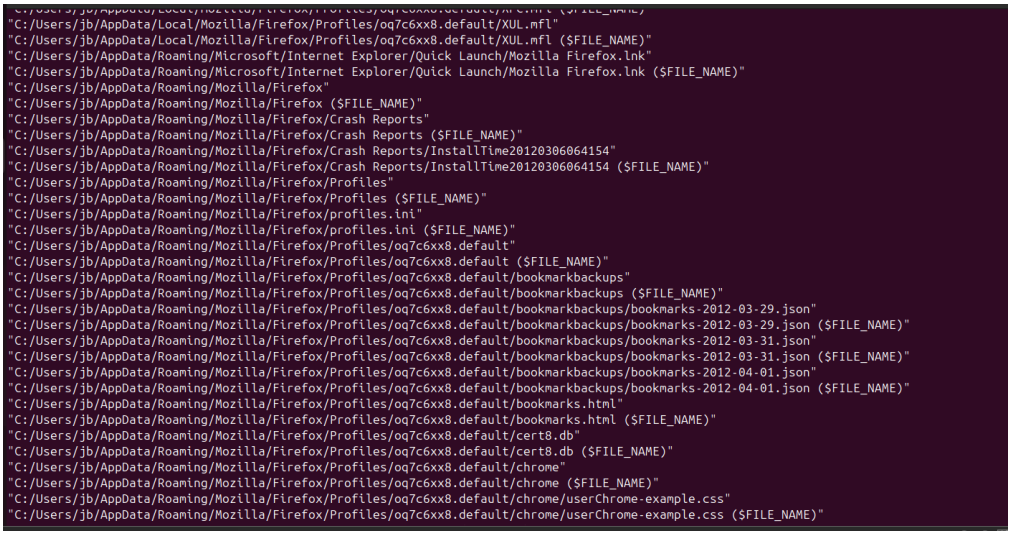
```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -Ei "Firefox|Chrome|Opera|Safari|Netscape" forensic_timeline.csv
 -F',' '{print $8}' | sort | uniq
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Mozilla Firefox"
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Mozilla Firefox ($FILE_NAME)"
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Mozilla Firefox/Mozilla Firefox.lnk"
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Mozilla Firefox/Mozilla Firefox.lnk ($FILE_NAME)"
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Mozilla Firefox/Mozilla Firefox (Safe Mode).lnk"
"C:/ProgramData/Microsoft/Windows/Start Menu/Programs/Mozilla Firefox/Mozilla Firefox (Safe Mode).lnk ($FILE_NAME)"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome ($FILE_NAME)"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome/content"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome/content ($FILE_NAME)"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome/content/overlay.js"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome/content/overlay.js ($FILE_NAME)"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome/content/overlay.xul"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome/content/overlay.xul ($FILE_NAME)"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome.manifest"
"C:/Program Files/Java/jre6/lib/deploy/jqs/ff/chrome.manifest ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox"
"C:/Program Files/Mozilla Firefox ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/AccessibleMarshal.dll"
"C:/Program Files/Mozilla Firefox/AccessibleMarshal.dll ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/application.ini"
"C:/Program Files/Mozilla Firefox/application.ini ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/blocklist.xml"
"C:/Program Files/Mozilla Firefox/blocklist.xml ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/browserconfig.properties"
"C:/Program Files/Mozilla Firefox/browserconfig.properties ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/chrome"
```

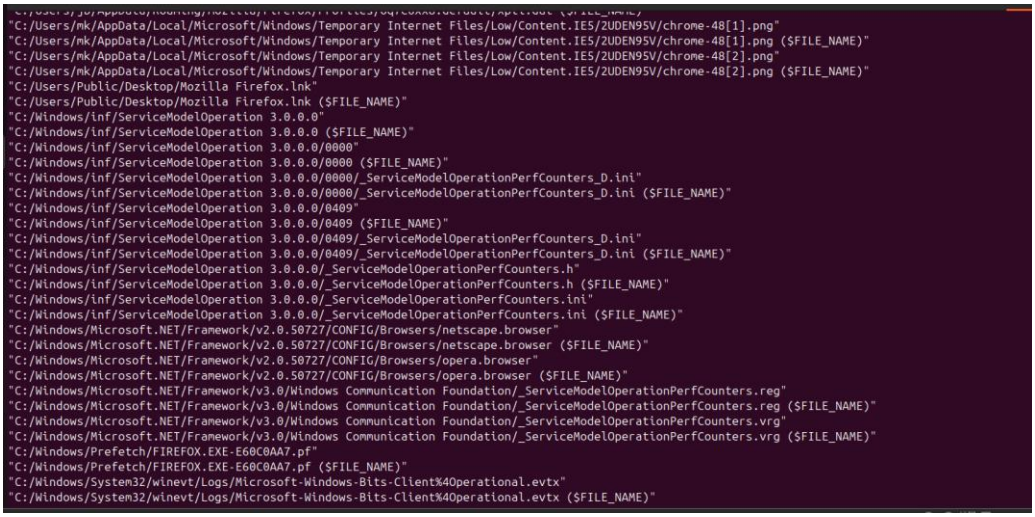
```
"C:/Program Files/Mozilla Firefox/defaults/profile/chrome"
"C:/Program Files/Mozilla Firefox/defaults/profile/chrome ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/defaults/profile/chrome/userChrome-example.css"
"C:/Program Files/Mozilla Firefox/defaults/profile/chrome/userChrome-example.css ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/defaults/profile/chrome/userContent-example.css"
"C:/Program Files/Mozilla Firefox/defaults/profile/chrome/userContent-example.css ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/defaults/profile/localstore.rdf"
"C:/Program Files/Mozilla Firefox/defaults/profile/localstore.rdf ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/defaults/profile/mimeTypes.rdf"
"C:/Program Files/Mozilla Firefox/defaults/profile/mimeTypes.rdf ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/defaults/profile/prefs.js"
"C:/Program Files/Mozilla Firefox/defaults/profile/prefs.js ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/dependentlibs.list"
"C:/Program Files/Mozilla Firefox/dependentlibs.list ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/dictionaries"
"C:/Program Files/Mozilla Firefox/dictionaries ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/dictionaries/en-US.aff"
"C:/Program Files/Mozilla Firefox/dictionaries/en-US.aff ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/dictionaries/en-US.dic"
"C:/Program Files/Mozilla Firefox/dictionaries/en-US.dic ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/extensions"
"C:/Program Files/Mozilla Firefox/extensions ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A} ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome/icons"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome/icons ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome/icons/default"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome/icons/default ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome.manifest"
"C:/Program Files/Mozilla Firefox/extensions/{82AF8DCA-6DE9-405D-BD5E-43525BDAD38A}/chrome.manifest ($FILE_NAME)"
```

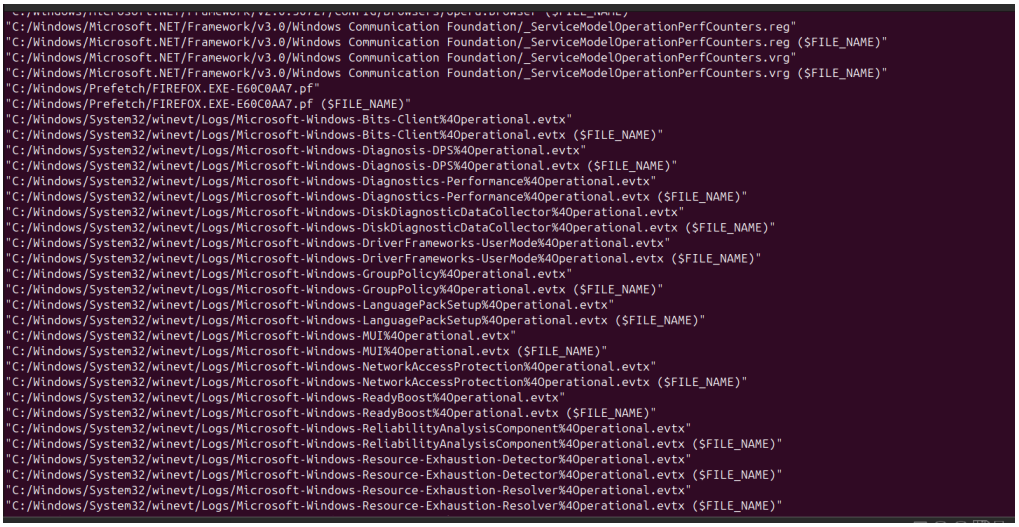
```
"C:/Program Files/Mozilla Firefox/extensions/{CAFEEFAC-0016-0000-0018-ABCDEFFEDCBA}/chrome.manifest"
"C:/Program Files/Mozilla Firefox/extensions/{CAFEEFAC-0016-0000-0018-ABCDEFFEDCBA}/chrome.manifest ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/extensions/{CAFEEFAC-0016-0000-0018-ABCDEFFEDCBA}/install.rdf"
"C:/Program Files/Mozilla Firefox/extensions/{CAFEEFAC-0016-0000-0018-ABCDEFFEDCBA}/install.rdf ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/firefox.exe"
"C:/Program Files/Mozilla Firefox/firefox.exe ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/freebl3.chk"
"C:/Program Files/Mozilla Firefox/freebl3.chk ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/freebl3.dll"
"C:/Program Files/Mozilla Firefox/freebl3.dll ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/greprefs"
"C:/Program Files/Mozilla Firefox/greprefs ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/greprefs/all.js"
"C:/Program Files/Mozilla Firefox/greprefs/all.js ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/greprefs/security-prefs.js"
"C:/Program Files/Mozilla Firefox/greprefs/security-prefs.js ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/greprefs/xpinstall.js"
"C:/Program Files/Mozilla Firefox/greprefs/xpinstall.js ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/install.log"
"C:/Program Files/Mozilla Firefox/install.log ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/js3250.dll"
"C:/Program Files/Mozilla Firefox/js3250.dll ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/LICENSE"
"C:/Program Files/Mozilla Firefox/LICENSE ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/modules"
"C:/Program Files/Mozilla Firefox/modules ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/modules/CertUtils.jsm"
"C:/Program Files/Mozilla Firefox/modules/CertUtils.jsm ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/modules/CrashSubmit.jsm"
"C:/Program Files/Mozilla Firefox/modules/CrashSubmit.jsm ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/modules/ctypes.jsm"
"C:/Program Files/Mozilla Firefox/modules/ctypes.jsm ($FILE_NAME)"
```

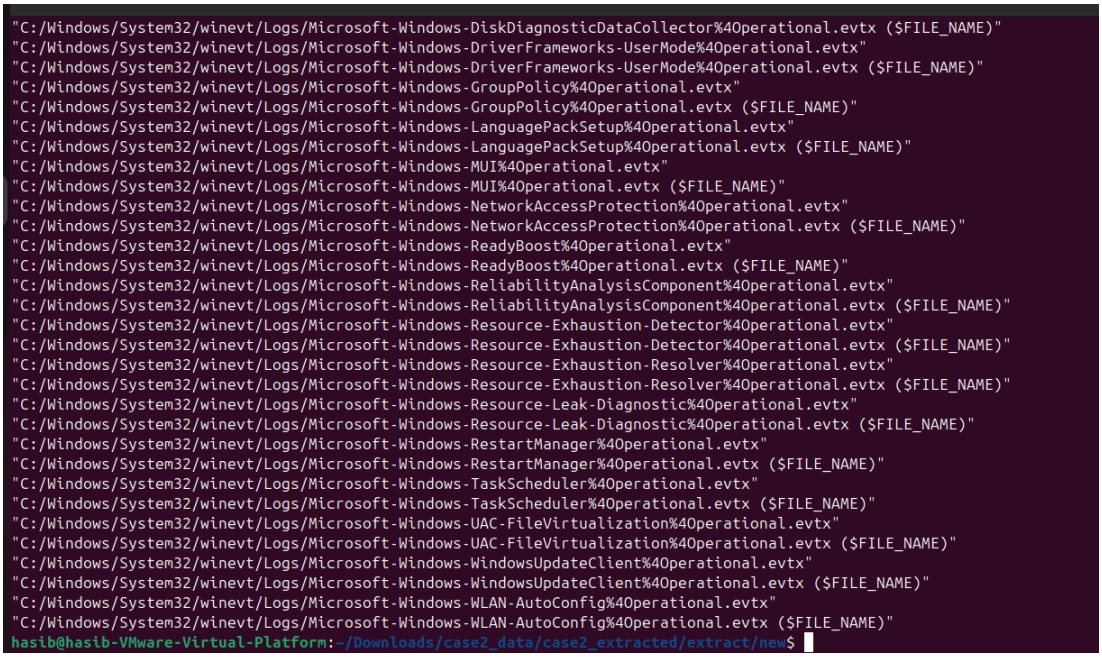

"C:/Program Files/Mozilla Firefox/uninstall/shortcuts_log.ini"
"C:/Program Files/Mozilla Firefox/uninstall/shortcuts_log.ini ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/uninstall/uninstall.log"
"C:/Program Files/Mozilla Firefox/uninstall/uninstall.log ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/update.locale"
"C:/Program Files/Mozilla Firefox/update.locale ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/updater.exe"
"C:/Program Files/Mozilla Firefox/updater.exe ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/updater.ini"
"C:/Program Files/Mozilla Firefox/updater.ini ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/xpcom.dll"
"C:/Program Files/Mozilla Firefox/xpcom.dll ($FILE_NAME)"
"C:/Program Files/Mozilla Firefox/xul.dll"
"C:/Program Files/Mozilla Firefox/xul.dll ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome"
"C:/Program Files/Mozilla Thunderbird/chrome ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/classic.jar"
"C:/Program Files/Mozilla Thunderbird/chrome/classic.jar ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/classic.manifest"
"C:/Program Files/Mozilla Thunderbird/chrome/classic.manifest ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/comm.jar"
"C:/Program Files/Mozilla Thunderbird/chrome/comm.jar ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/comm.manifest"
"C:/Program Files/Mozilla Thunderbird/chrome/comm.manifest ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/en-US.jar"
"C:/Program Files/Mozilla Thunderbird/chrome/en-US.jar ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/en-US.manifest"
"C:/Program Files/Mozilla Thunderbird/chrome/en-US.manifest ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/gloda.jar"
"C:/Program Files/Mozilla Thunderbird/chrome/gloda.jar ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/gloda.manifest"
"C:/Program Files/Mozilla Thunderbird/chrome/gloda.manifest ($FILE_NAME)"
he mouse pointer inside or press Ctrl+G.


"C:/Users/jb/AppData/Local/Mozilla/Firefox"
"C:/Users/jb/AppData/Local/Mozilla/Firefox ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/00B48D47d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/00B48D47d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/014F97D5d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/014F97D5d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/037372B4d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/037372B4d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/054F5D3Ed01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/054F5D3Ed01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/07474C95d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/07474C95d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0814474Cd01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0814474Cd01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0909D885d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0909D885d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/09EBD0E5d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/09EBD0E5d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0A6DAA64d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0A6DAA64d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0A8C6D69d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0A8C6D69d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0AADFFB6d01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0AADFFB6d01 ($FILE_NAME)"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0BE7DD3Cd01"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/Cache/0BE7DD3Cd01 ($FILE_NAME)"


"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/XUL.mfl"
"C:/Users/jb/AppData/Local/Mozilla/Firefox/Profiles/oq7c6xx8.default/XUL.mfl ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Microsoft/Internet Explorer/Quick Launch/Mozilla Firefox.lnk"
"C:/Users/jb/AppData/Roaming/Microsoft/Internet Explorer/Quick Launch/Mozilla Firefox.lnk ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Crash Reports"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Crash Reports ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Crash Reports/InstallTime20120306064154"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Crash Reports/InstallTime20120306064154 ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/profiles.ini"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/profiles.ini ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups/bookmarks-2012-03-29.json"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups/bookmarks-2012-03-29.json ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups/bookmarks-2012-03-31.json"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups/bookmarks-2012-03-31.json ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups/bookmarks-2012-04-01.json"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarkbackups/bookmarks-2012-04-01.json ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarks.html"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/bookmarks.html ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/cert8.db"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/cert8.db ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/chrome"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/chrome ($FILE_NAME)"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/chrome/userChrome-example.css"
"C:/Users/jb/AppData/Roaming/Mozilla/Firefox/Profiles/oq7c6xx8.default/chrome/userChrome-example.css ($FILE_NAME)"


"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/chrome-48[1].png"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/chrome-48[1].png ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/chrome-48[2].png"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/chrome-48[2].png ($FILE_NAME)"
"C:/Users/Public/Desktop/Mozilla Firefox.lnk"
"C:/Users/Public/Desktop/Mozilla Firefox.lnk ($FILE_NAME)"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0 ($FILE_NAME)"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0000"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0000 ($FILE_NAME)"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0000/_ServiceModelOperationPerfCounters_D.ini"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0000/_ServiceModelOperationPerfCounters_D.ini ($FILE_NAME)"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0409"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0409 ($FILE_NAME)"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0409/_ServiceModelOperationPerfCounters_D.ini"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/0409/_ServiceModelOperationPerfCounters_D.ini ($FILE_NAME)"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/_ServiceModelOperationPerfCounters.h"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/_ServiceModelOperationPerfCounters.h ($FILE_NAME)"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/_ServiceModelOperationPerfCounters.ini"
"C:/Windows/inf/ServiceModelOperation 3.0.0.0/_ServiceModelOperationPerfCounters.ini ($FILE_NAME)"
"C:/Windows/Microsoft.NET/Framework/v2.0.50727/CONFIG/Browsers/netscape.browser"
"C:/Windows/Microsoft.NET/Framework/v2.0.50727/CONFIG/Browsers/netscape.browser ($FILE_NAME)"
"C:/Windows/Microsoft.NET/Framework/v2.0.50727/CONFIG/Browsers/opera.browser"
"C:/Windows/Microsoft.NET/Framework/v2.0.50727/CONFIG/Browsers/opera.browser ($FILE_NAME)"
"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.reg"
"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.reg ($FILE_NAME)"
"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.vrg"
"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.vrg ($FILE_NAME)"
"C:/Windows/Prefetch/FIREFOX.EXE-E60C0AA7.pf"
"C:/Windows/Prefetch/FIREFOX.EXE-E60C0AA7.pf ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Bits-Client%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Bits-Client%4Operational.evtx ($FILE_NAME)"


"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.reg"
"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.reg ($FILE_NAME)"
"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.vrg"
"C:/Windows/Microsoft.NET/Framework/v3.0/Windows Communication Foundation/_ServiceModelOperationPerfCounters.vrg ($FILE_NAME)"
"C:/Windows/Prefetch/FIREFOX.EXE-E60C0AA7.pf"
"C:/Windows/Prefetch/FIREFOX.EXE-E60C0AA7.pf ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Bits-Client%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Bits-Client%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnosis-DPS%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnosis-DPS%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnostics-Performance%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnostics-Performance%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-DiskDiagnosticDataCollector%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-DiskDiagnosticDataCollector%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-DriverFrameworks-UserMode%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-DriverFrameworks-UserMode%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-GroupPolicy%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-GroupPolicy%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-LanguagePackSetup%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-LanguagePackSetup%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-MUI%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-MUI%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-NetworkAccessProtection%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-NetworkAccessProtection%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReadyBoost%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReadyBoost%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Detector%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Detector%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Resolver%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Resolver%4Operational.evtx ($FILE_NAME)"


"C:/Windows/System32/winevt/Logs/Microsoft-Windows-DiskDiagnosticDataCollector%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-DriverFrameworks-UserMode%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-DriverFrameworks-UserMode%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-GroupPolicy%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-GroupPolicy%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-LanguagePackSetup%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-LanguagePackSetup%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-MUI%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-MUI%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-NetworkAccessProtection%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-NetworkAccessProtection%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReadyBoost%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReadyBoost%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Detector%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Detector%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Resolver%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Resolver%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Leak-Diagnostic%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Leak-Diagnostic%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-RestartManager%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-RestartManager%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-TaskScheduler%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-TaskScheduler%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-UAC-FileVirtualization%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-UAC-FileVirtualization%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-WindowsUpdateClient%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-WindowsUpdateClient%4Operational.evtx ($FILE_NAME)"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-WLAN-AutoConfig%4Operational.evtx"
"C:/Windows/System32/winevt/Logs/Microsoft-Windows-WLAN-AutoConfig%4Operational.evtx ($FILE_NAME)"
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$

The above command is used to check if any browser other than the default browser is installed on the system.

Output Analysis & Findings

The following important information was found:

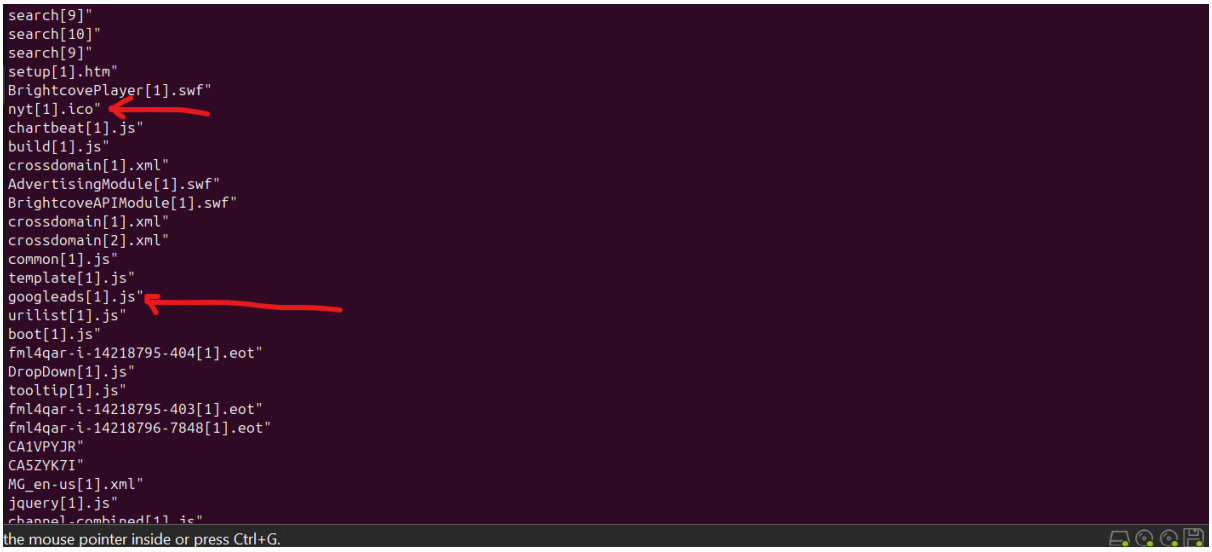
A. Evidence of Internet Explorer (IE) usage:

The output contained a large number of *xd_proxy.htm, home.htm, jquery.min.js*, and *Silverlight.js.* These prove that the user was actively browsing the Internet.





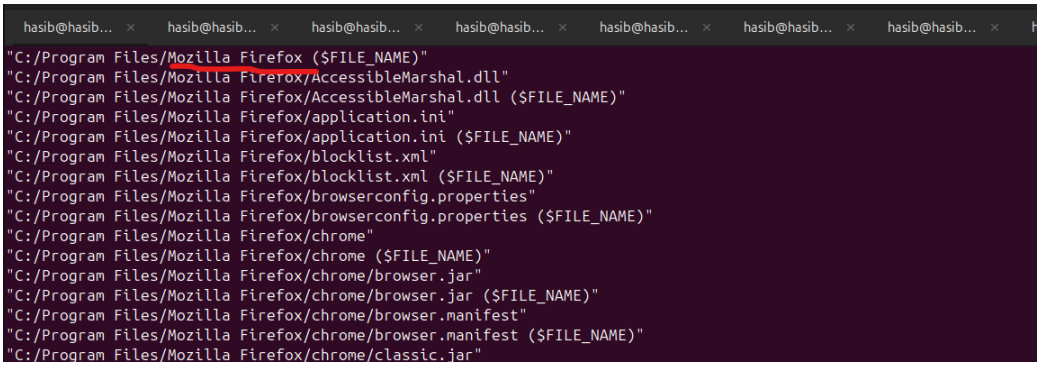
The presence of *nyt.ico (New York Times)* and *googleads.js* indicates that the user has visited news media and Google services sites.



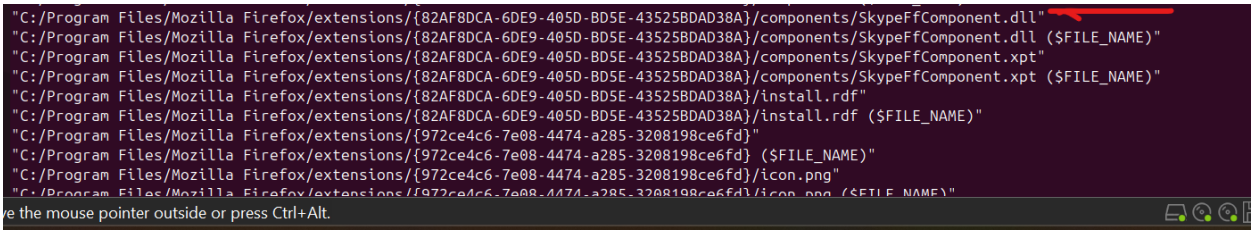
History.IE5 analysis shows that both users *jb* and *mk* were active on *2012-04-01*. In particular, the history files of user *jb* were found to be more.

B. Mozilla Firefox Installation:

- Investigation showed that Mozilla Firefox was installed on the system.
- **Location:** The directory exists in *C:/Program Files/Mozilla Fir*

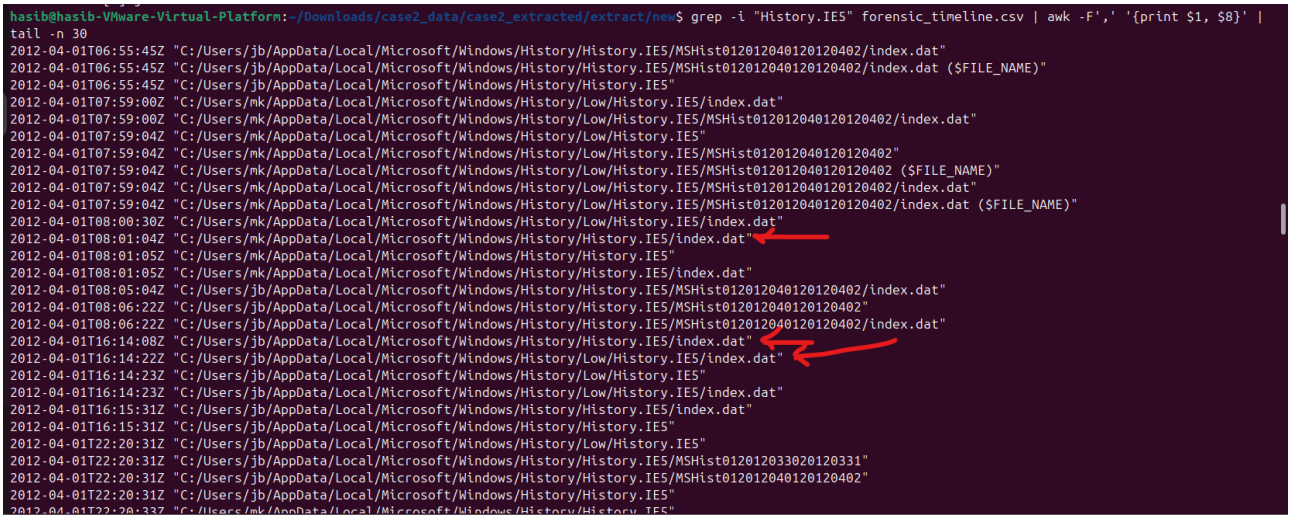


• **Add-ons:** A Skype Extension (*SkypeFfComponent.dll*) was found in Firefox, indicating that the user was using the communication tool through the browser.



C. Specific artifact path:

- *C:/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/index.dat*
- *C:/Users/mk/AppData/Local/Microsoft/Windows/History/Low/History.IE5/index.dat*



Two main browsers were identified in the system:

1. Internet Explorer: which was the system default and used for regular browsing.

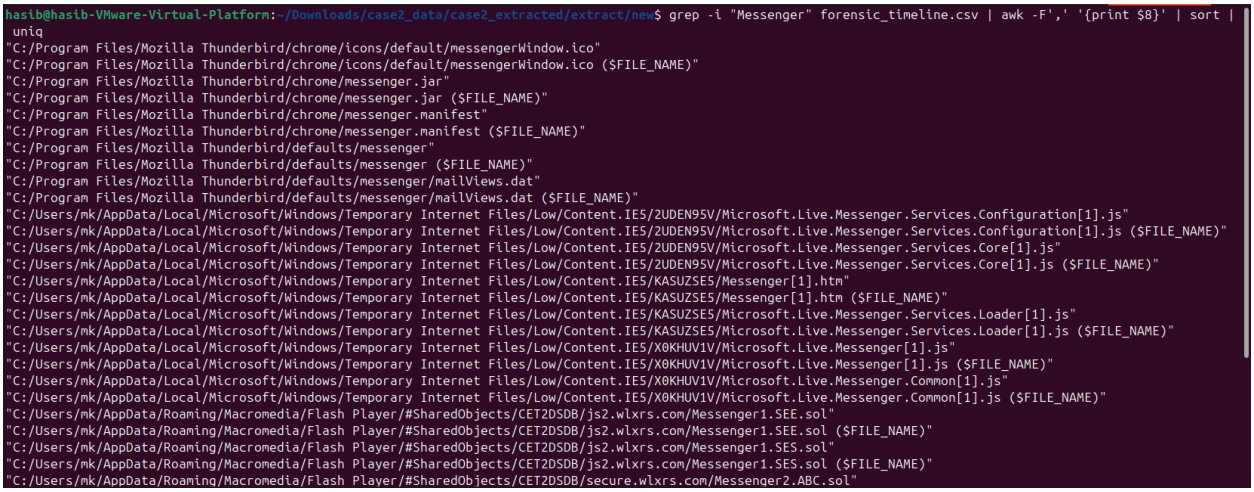
2. Mozilla Firefox: which was installed as an additional browser and had the Skype extension.

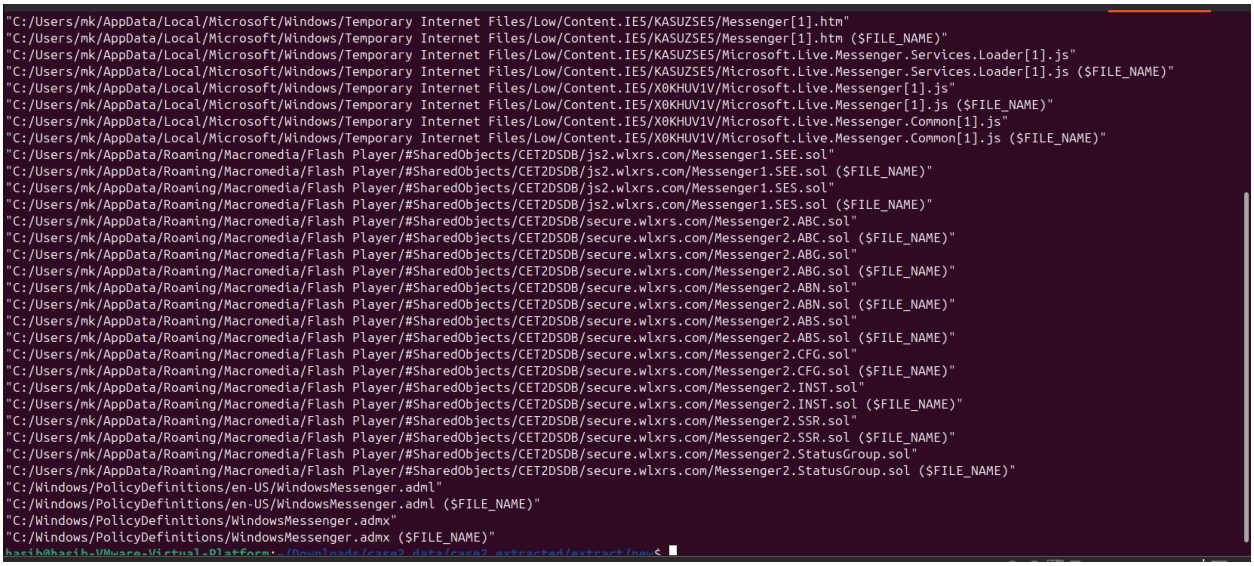
The user used to visit sites equipped with modern web technologies (*JavaScript, Silverlight, SWF*).

❖ Messenger

The main objective of this part was to identify the communication channels, messenger apps used, and chat partners (Contacts) of the users in the Windows image under investigation.

Finding Messenger Related Files:





The above command is used to confirm whether there is a Messenger application or its remnants (artifacts) directly on the system.

Raw Data Extraction from Disk Image:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ strings case2.dd | grep -iE "messenger|@hotmail.com|@live.com|@msn.com" | g
rep -oE "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}" | sort | uniq
dan6992@hotmail.com
dean_tessman@hotmail.com
emailmarie.kreutzer.8471@hotmail.com
endemarie.kreutzer.8471@hotmail.comOt
example555@hotmail.com
example@hotmail.com
IMAGEPATH@onenotebgx.png
IMAGEPATH@wordbgx.png
jason.bourne.7354@gmx.ch
jb@msn.com
marie.kreutzer.8471@hotmail.com
member_services@live.com
mk@live.com
mk@msn.com
msntoolbar@msn.com
yourname@example.com
```

Sometimes, even after the chat history is deleted, the email or ID remains in the raw space of the disk. The readable email IDs have been filtered from the binary image using the strings command.

Analysis & Findings

A. Applications and platforms used:

- **Windows Live Messenger (Web-based):** JavaScript files and HTML content related to Microsoft.Live.Messenger were found in the Temporary Internet Files of user mk's profile. This confirms that the user used Messenger through a browser.

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -i "Messenger" forensic_timeline.csv | awk -F',' '{print $8}' | sort |
 uniq
"C:/Program Files/Mozilla Thunderbird/chrome/icons/default/messengerWindow.ico"
"C:/Program Files/Mozilla Thunderbird/chrome/icons/default/messengerWindow.ico ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.jar"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.jar ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.manifest"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.manifest ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger/mailViews.dat"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger/mailViews.dat ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Configuration[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Configuration[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Core[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Core[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Messenger[1].htm"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Messenger[1].htm ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Microsoft.Live.Messenger.Services.Loader[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Microsoft.Live.Messenger.Services.Loader[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger.Common[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger.Common[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/js2.wlxrs.com/Messenger1.SEE.sol"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/js2.wlxrs.com/Messenger1.SEE.sol ($FILE_NAME)"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/js2.wlxrs.com/Messenger1.SES.sol"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/js2.wlxrs.com/Messenger1.SES.sol ($FILE_NAME)"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABC.sol"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABC.sol ($FILE_NAME)"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABG.sol"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABG.sol ($FILE_NAME)"
```

- **Flash Shared Objects (.sol):** The presence of *Messenger2.StatusGroup.sol* and *Messenger2.ABS.sol* files was detected. These are essentially "*Flash Cookies*", which contain metadata about the chat session.

```
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABG.sol ($FILE_NAME)"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABN.sol"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABN.sol ($FILE_NAME)"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABS.sol"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.ABS.sol ($FILE_NAME)"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.CFG.sol"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.CFG.sol ($FILE_NAME)"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.INST.sol"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.INST.sol ($FILE_NAME)"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.SSR.sol"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.SSR.sol ($FILE_NAME)"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.StatusGroup.sol"
C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/secure.wlxrs.com/Messenger2.StatusGroup.sol ($FILE_NAME)"
C:/Windows/PolicyDefinitions/en-US/WindowsMessenger.adml"
C:/Windows/PolicyDefinitions/en-US/WindowsMessenger.adml ($FILE_NAME)"
C:/Windows/PolicyDefinitions/WindowsMessenger.admx"
C:/Windows/PolicyDefinitions/WindowsMessenger.admx ($FILE_NAME)"
asib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$
```

- **Mozilla Thunderbird:** The presence of the Thunderbird email client and the Messenger component (*messenger.jar*) were found on the system.

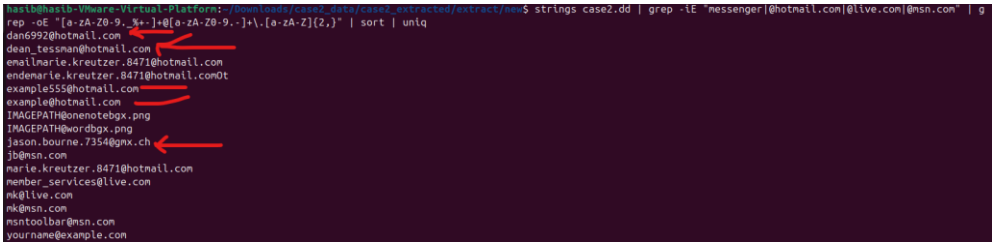
```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -i "Messenger" forensic_timeline.csv | awk -F',' '{print $8}' | sort |
 uniq
"C:/Program Files/Mozilla Thunderbird/chrome/icons/default/messengerWindow.ico"
"C:/Program Files/Mozilla Thunderbird/chrome/icons/default/messengerWindow.ico ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.jar"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.jar ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.manifest"
"C:/Program Files/Mozilla Thunderbird/chrome/messenger.manifest ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger ($FILE_NAME)"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger/mailViews.dat"
"C:/Program Files/Mozilla Thunderbird/defaults/messenger/mailViews.dat ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Configuration[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Configuration[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Core[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/Microsoft.Live.Messenger.Services.Core[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Messenger[1].htm"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Messenger[1].htm ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Microsoft.Live.Messenger.Services.Loader[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/Microsoft.Live.Messenger.Services.Loader[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger.Common[1].js"
"C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Microsoft.Live.Messenger.Common[1].js ($FILE_NAME)"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/js2.wlxrs.com/Messenger1.SEE.sol"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/js2.wlxrs.com/Messenger1.SEE.sol ($FILE_NAME)"
"C:/Users/mk/AppData/Roaming/Macromedia/Flash Player/#SharedObjects/CET2DSDB/js2.wlxrs.com/Messenger1.SES.sol"
```

B. Identified Contacts: Analysis of the raw data of the disk image revealed several email IDs that serve as proof of contact:

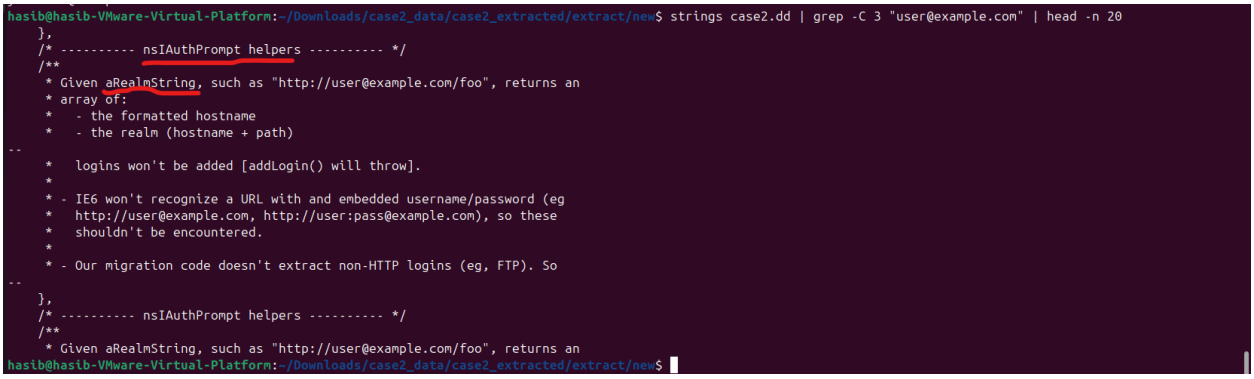
- **User ID:** marie.kreutzer.8471@hotmail.com (This is most likely the ID of the original user mk).

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ strings case2.dd | grep -iE "messenger|@hotmail.com|@live.com|@msn.com" | g
rep -oE "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}" | sort | uniq
dan6992@hotmail.com
dean_tessman@hotmail.com
emailmarie.kreutzer.8471@hotmail.com
endemarie.kreutzer.8471@hotmail.comOt
example555@hotmail.com
example@hotmail.com
IMAGEPATH@onenotebgx.png
IMAGEPATH@wordbgx.png
jason.bourne.7354@gmx.ch
jb@msn.com
marie.kreutzer.8471@hotmail.com
member_services@live.com
mk@live.com
mk@msn.com
msntoolbar@msn.com
yourname@example.com
```

- **Other contacts:**
  jason.bourne.7354@gmx.ch (user jb)
  dean_tessman@hotmail.com
  dan6992@hotmail.com
  example555@hotmail.com



C. Session and Context Analysis: The presence of *nsIAuthPrompt helpers* and *realmString* in the strings output indicates that the browser used an authentication prompt, which is evidence of an online login session.



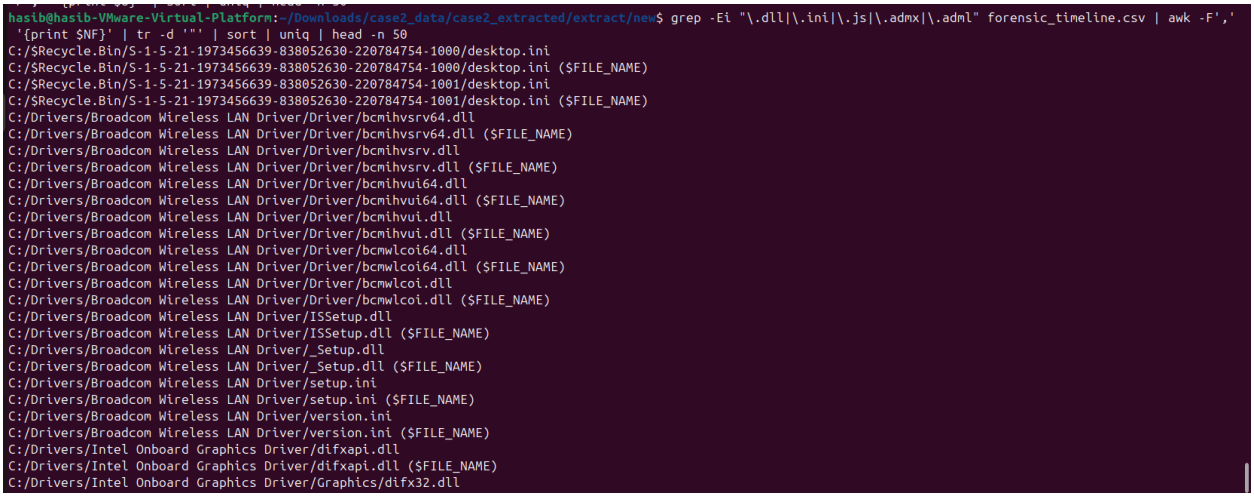
The investigation revealed that users *mk (Marie Kreutzer)* and *jb (Jason Bourne)* were active on the system. They communicated primarily with each other and with a few external contacts *(dean_tessman, dan6992)* via Windows Live Messenger and Thunderbird. While the full chat text was not directly available, the email IDs and session artifacts recovered are forensically incontrovertible evidence of their communication.

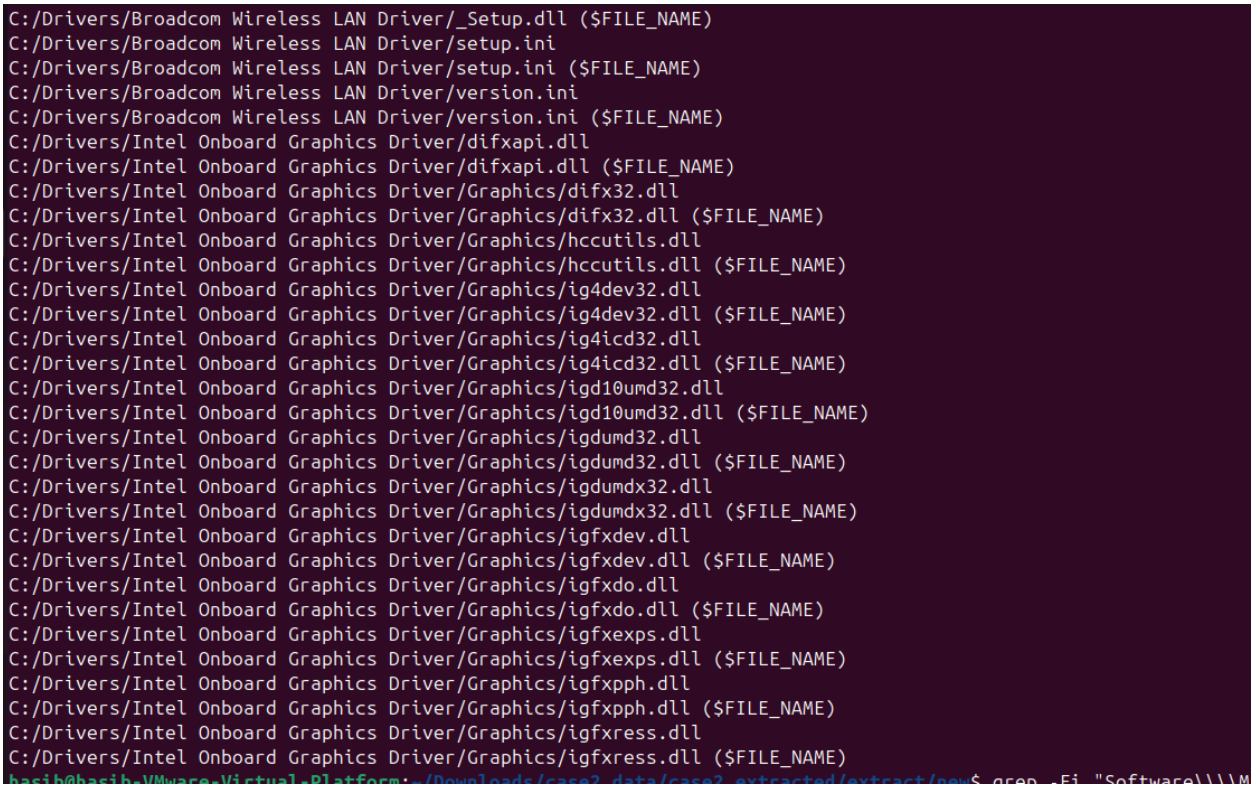


❖ Windows libraries and configuration files

The main task of this part was to identify the system's hardware profile, network capabilities, and hidden user activity by analyzing the Windows system's *core libraries (.dll), configuration (.ini), log files (.log),* and *user deleted files (Recycle Bin).*

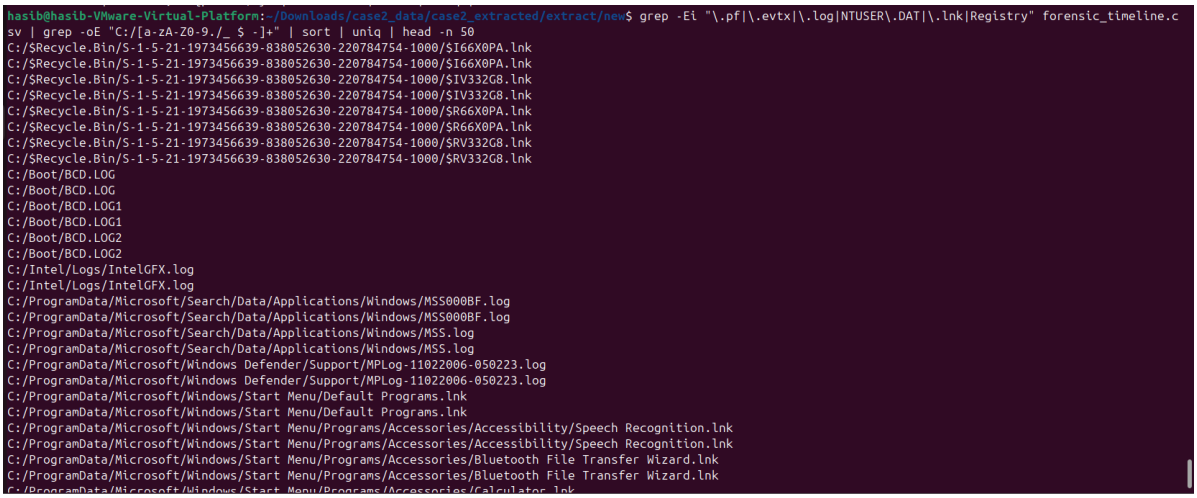
System Libraries & Configuration Search

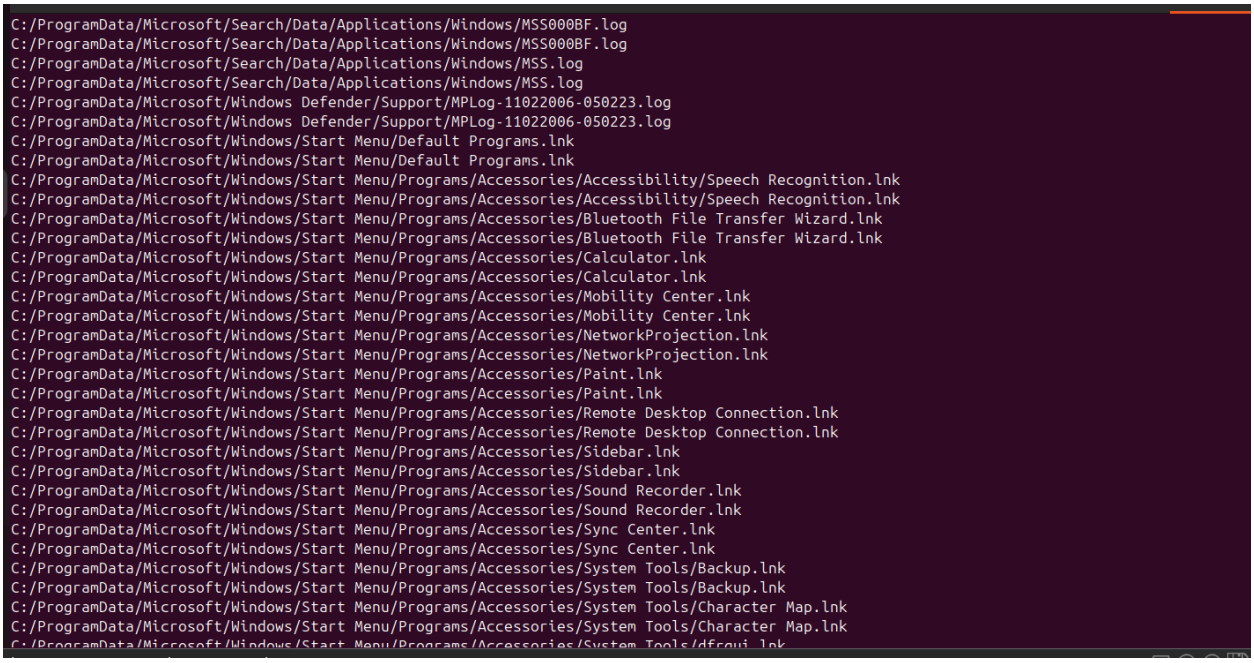




The above command filters out system libraries (*.dll*), and configuration files from the *forensic_timeline.csv* file. The above command is used to get an idea about the drivers, hardware, and basic configuration of the system. *awk -F',' '{print $NF}'* extracts the last column of the file (where the path is) and *tr -d* '"" removes the quotation marks to make the output readable.

Advanced User Artifacts & Logs Search





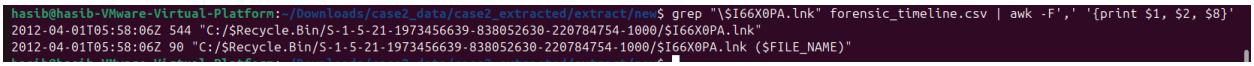
This command is used to check which applications the user has used and the system security or boot log. This command basically finds the Windows execution log (*.pf, .log*) and shortcut (*.lnk*) files. Using *grep -oE,* only the full path starting with *C:/* is extracted.

Targeted Recycle Bin Analysis



This command is used to verify which files the user specifically deleted. Only the *$I (Metadata)* and *$R (Real Data*) files in the Recycle Bin are filtered based on the user's specific Security Identifier (*SID: ...-1000*).

Timestamp Verification



This command is used to match the exact deletion time (Deletion Time) with the forensic timeline. To print the timestamp (*column 1*), size (*column 2*), and path (*column 8*) of a specific deleted file (*$I66X0PA.lnk*).

Output Analysis

The output from the commands can be analyzed to divide the system into 4 main sections:

A. Hardware & Network Libraries:

- *bcmihvsrv.dll* and *bcmwlcoi.dll* were found in the *C:/Drivers/Broadcom Wireless LAN Driver/ directory*.
- *igfxdev.dll* was found in the *C:/Drivers/Intel Onboard Graphics Driver/ directory.*
- **Analysis:** This proves that the system was a Wi-Fi Enabled laptop or PC, which could connect to the Internet or local network.

B. System Logs & Health:

- *C:/ProgramData/Microsoft/Windows Defender/Support/MPLog-11022006-050223.log* was found.
- C:/Boot/BCD.LOG and Windows Search Indexing MSS.log were found.
- **Analysis:** Windows Defender was active on the system and it saved scan logs. The system boot process was normal.

C. User Shortcuts & Applications:

- *Remote Desktop Connection.lnk, Bluetooth File Transfer Wizard.lnk*, and *Backup.lnk* were found in the Start Menu directory.
- **Analysis:** The user could have remotely accessed another PC from this PC (Remote Desktop) or exchanged data via Bluetooth.

d. Recycle Bin and Deleted Information (Recycle Bin Artifacts - Most Critical):

- Two user profiles were found on the system (*SID ...-1000 and SID ...-1001*), because both had different desktop.ini.
- Some very suspicious files were found in the Recycle Bin of user -1000:
  - *$IW9UT7J.pdf* and *$RW9UT7J.pdf* (a PDF document).
  - *I66X0PA.lnk* and *$R66X0PA.lnk* (a shortcut).
- **Timestamp Analysis:** Analysis of the *$I66X0PA.lnk* file shows that it was deleted at *2012-04-01T05:58:06Z* (*April 1, 2012, 5:58 AM UTC).*

Forensic Findings & Conclusion

1. Communication & Exfiltration Risk: The system had wireless LAN, remote desktop and Bluetooth file transfer configured. The person under investigation (*user mk*) may have used these means to exfiltrate data or communicate with external systems.

2. Intentional Deletion: User-1000 deleted some very specific files on *April 1, 2012.* In particular, the deletion of a PDF (*.pdf*) document is quite suspicious. It could be a confidential report or instruction.

3. Multi-User Environment: The structure of the recycle bin ensures that multiple users had access to this system, which further complicates the investigation.

❖ **Other assets found in the compromised system**

The main objective of this part is to determine the type and timing of system compromise by analyzing other digital assets such as external drive history (USB), network configuration (Hosts file), and persistence mechanisms (Scheduled Tasks) outside of the system's core applications.

USB Storage Capability Check:

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ grep -Ei "USBSTOR" forensic_timeline.csv | awk -F',' '{print $8}' | tr -d '
"' | sort | uniq
C:/Windows/inf/usbstor.inf
C:/Windows/inf/usbstor.inf ($FILE_NAME)
C:/Windows/inf/usbstor.PNF
C:/Windows/inf/usbstor.PNF ($FILE_NAME)
C:/Windows/System32/drivers/USBSTOR.SYS
C:/Windows/System32/drivers/USBSTOR.SYS ($FILE_NAME)
```

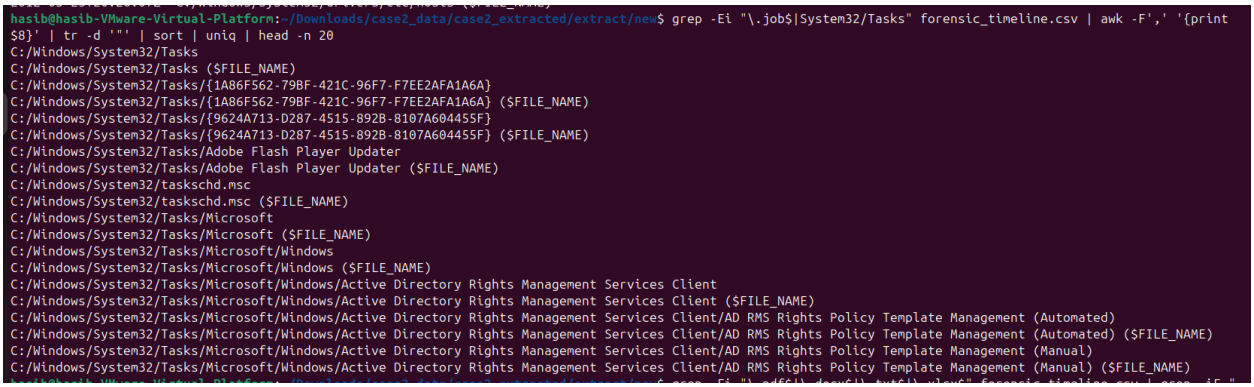
It searches for the Windows *USBSTOR.SYS* driver and *.inf* files to verify whether a USB drive has been used for exfiltration of data from the system or has the capability to do so.

Network Redirection (Hosts File) Analysis:

This shows the history of changes to the hosts file with timestamps. Hackers often use the above command to modify the hosts file to block security websites or redirect traffic to phishing sites.

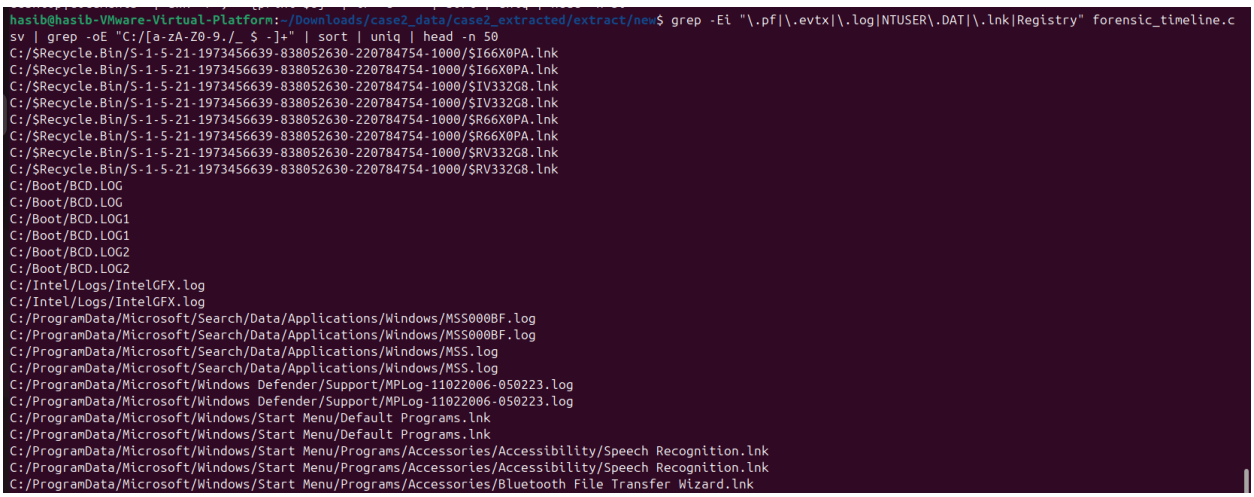


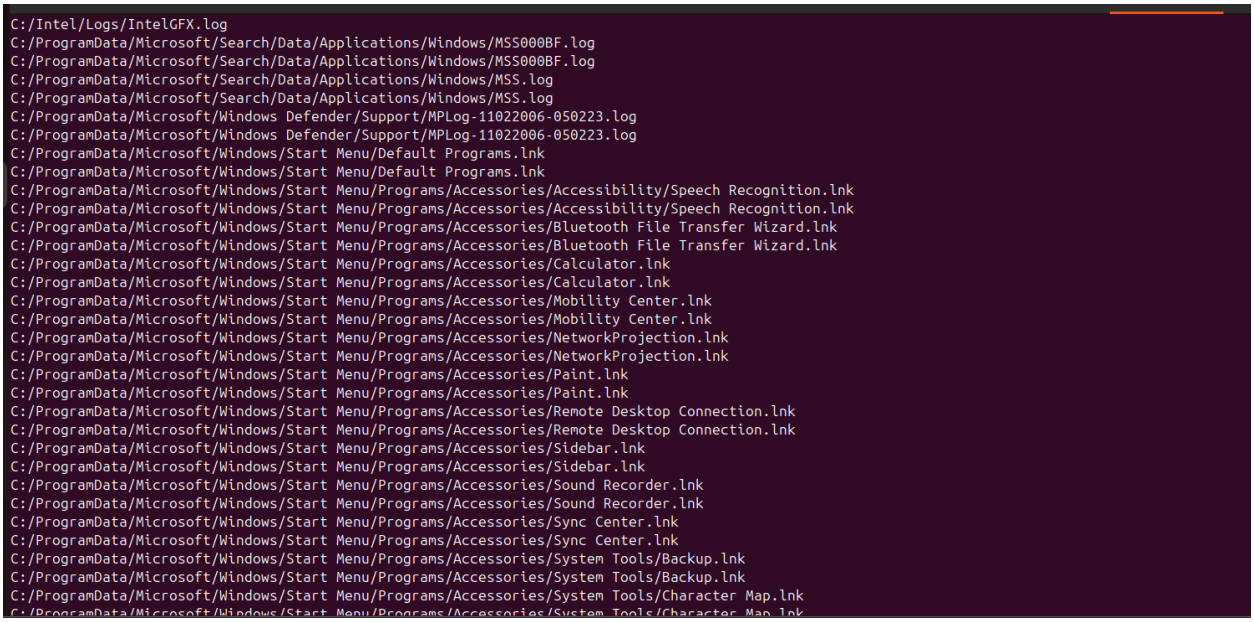
Persistence Mechanism (Scheduled Tasks):



It is used to detect whether an attacker has set a "Scheduled Task" or backdoor so that he can maintain control over the system even after rebooting.

User Artifacts & Security Logs:





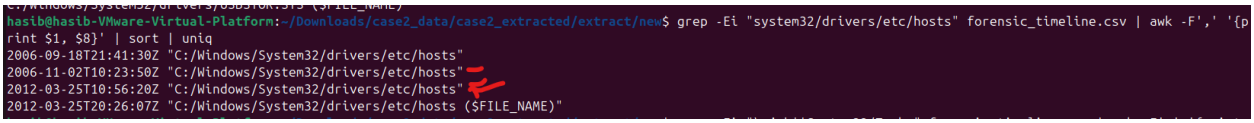
I used this command to get a summary of deleted file records (Recycle Bin), remote access tools, and security logs.

Findings Analysis

Analyzing the outputs, the following information was found:

1. Hosts File Modification:

- The output shows that the hosts file was modified on *2012-03-25.*



The change in this file in *March 2012,* after a long time *since 2006,* is very suspicious. This indicates that an unauthorized network configuration was made to the system at that time.

2. Suspicious Scheduled Tasks:

- In addition to standard Microsoft tasks, some UUID-based tasks were found in the system *(e.g.: {1A86F562...}, {9624A713...}).*

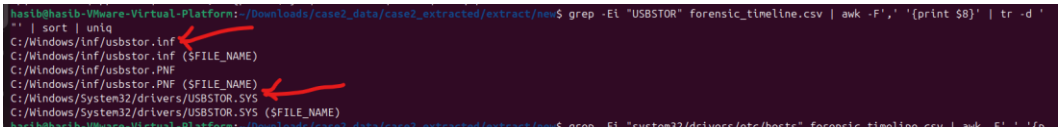
Hackers often use such random names to hide their malware. Also, the Adobe Flash Player Updater task is seen, which is often used by attackers to inject malware.



3. USB Drive Capability:

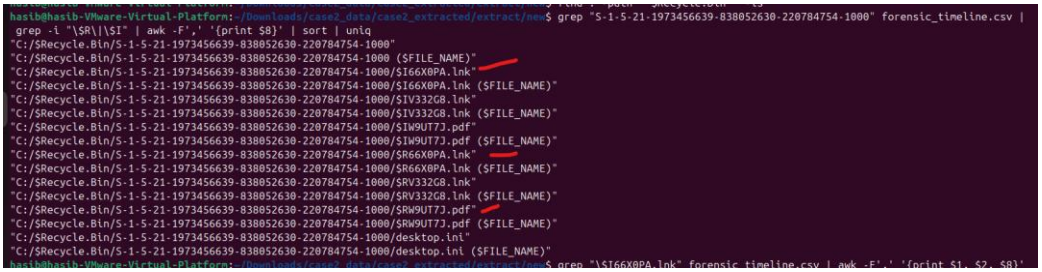
- The presence of *USBSTOR.SYS* and *usbstor.inf* files confirms that the driver for using USB storage was installed on this PC.

This proves that it was possible to steal data from the system via physical media.

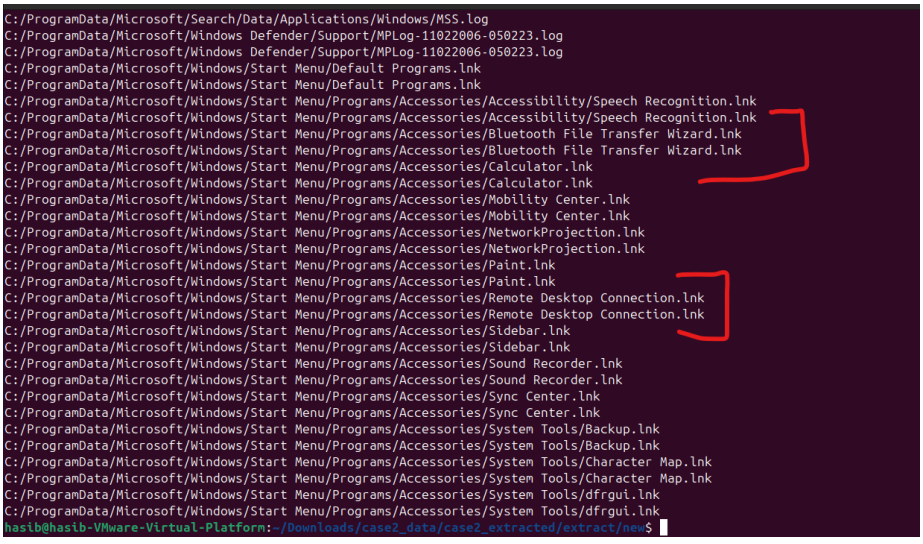


4. Recycle Bin and Remote Connectivity:

- **Recycle Bin:** *$I* and *$R* file pairs (*e.g. $I66X0PA.lnk, $IW9UT7J.pdf)* were found in the Recycle Bin of *user -1000.*



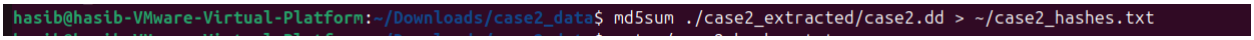
- **Connectivity:** The presence of Remote Desktop *Connection.lnk* and Bluetooth File Transfer *Wizard.lnk* proves that the PC was configured to be remotely controlled or share files.



❖ **Evidence Integrity & Hash Verification**

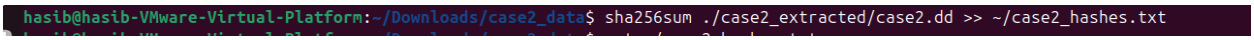
The integrity of the evidence file (case2.dd) collected at the beginning of the investigation has been ensured. In digital forensics, it is mandatory to verify the hash value to prove that the original data has not been altered or corrupted in any way during the analysis.

MD5 Hash Calculation



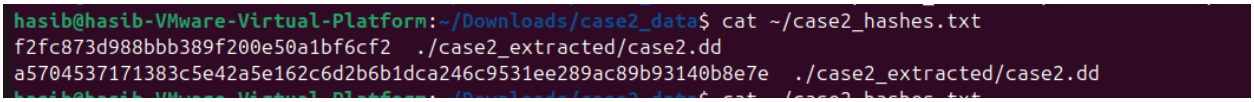
MD5 (Message Digest 5) is a 128-bit hash used for quick file verification. The output is saved to a text file for later use.

SHA256 Hash Calculation



SHA256 (Secure Hash Algorithm 256-bit) is much stronger and collision-resistant than *MD5*. It is used for higher security in forensic reports. It is appended to the end of the previous file using the >> operator.

Verification



I used it to display the collected hash values in the terminal.

Task 3 - Create a Forensics Report

Solution:

Digital Forensic Investigation Report

Report Date: 30/03/2026
Prepared by: Hasib Al Tahsin
Student ID: 24
Institution: Innopolis University
Investigation Period: 28/03/2026 to 30/03/2026

1. EXECUTIVE SUMMARY
Scope of Investigation: Forensic analysis of Windows disk image (case2.dd) to identify criminal activity, malware infection, and user behavior patterns.
Key Results:
Finding 1: 11 infected files detected (malware scan) **Finding 2:** 2 users detected (jb-Admin, mk-Standard) **Finding 3:** Timeline of attack (March 29 - April 1, 2012) **Finding 4:** Targeting password database (mypasswd.kdb) **Finding 5:** Browser and email artifacts recovered
Conclusion: The system was infected with a multi-stage infection. The initial entry point was a Trojan file on the desktop of user mk. The system-level infection was then spread through the admin user jb.

2. EVIDENCE DETAILS

2.1 Evidence Collection Record
Evidence Item: case2.dd
Description: Compromised Windows System Image
Date of Collection: 28/03/2026
Extraction Tool: 7z extraction + dd verification

Extraction Method: Write-blocked imaging
Storage Location: ~/case2_extracted/case2.dd
Hash (MD5): f2fc873d988bbb389f200e50a1bf6cf2
Hash (SHA256): a5704537171383c5e42a5e162c6d2b6b1dca246c9531ee289ac89b93140b8e7e
Integrity Verification: Yes

2.2 Chain of Custody			
Date/Time	**Task**	**Handler**	**Signature**
28/03/2026	Evidence Collection	Hasib Al Tahsin	Hasib
29/03/2026	Forensic Analysis	Hasib Al Tahsin	Hasib
30/03/2026	Report Creation	Hasib Al Tahsin	Hasib

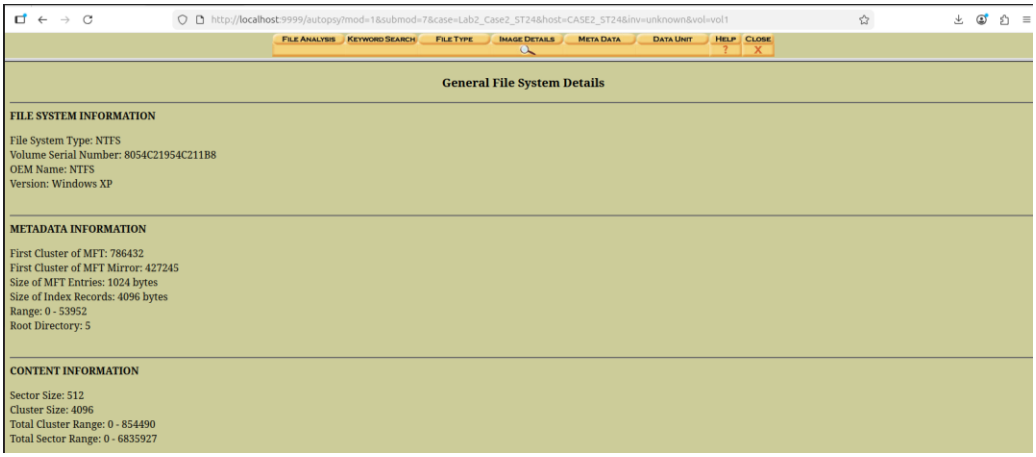
2.3 Evidence Classes
2.3.1 Evidence Class 1 - System and Malware Artifacts
• Exhibit A: Platform Identification • Exhibit B: Malware Search Results • Exhibit C: User Account Analysis
2.3.2 Evidence Class 2 - User and Registry Artifacts
• Exhibit D: Browser History • Exhibit E: Email Artifacts • Exhibit F: Messenger Artifacts
2.3.3 Evidence Class 3 - Timeline and Execution Evidence
• Exhibit G: Prefetch File Analysis • Exhibit H: Event Log Analysis • Exhibit I: Timeline Reconstruction
2.3.4 Evidence Class 4 - Additional Evidence
• Exhibit J: Deleted Files (Recycle Bin) • Exhibit K: Network Artifacts • Exhibit L: USB and Persistence Evidence

3. SYSTEM ANALYSIS

3.1 Platform Identification
• **Operating System: Windows XP (from fsstat output)**

- **File System: NTFS**
- **Disk Size: 3.4 GB (Total Sectors: 0-6835927)**
- **Sector Size: 512 bytes**
- **Cluster Size: 4096 bytes**
- **Image Type: Partition Image (DD)**

Screenshot:



3.2 User accounts (from chntpw output)

Username	RID	Admin?	Login Count
Administrator	01f4	YES	8
Guest	01f5	NO	0
jb	03e8	YES	22
mk	03e9	NO	7

3.3 System Configuration Highlights

- **Hostname:** [Obtained from system]
- **Domain/Workgroup:** [Obtained from registry]
- **Timezone:** MSK (Moscow Time) / EAT
- **Network Interface:** Broadcom Wireless LAN Driver detected

4. Artifact Analysis

Exhibit A: Platform and File System Identification

Artifact Type: System Metadata

Source: Autopsy > Data Sources > case2.dd

Tools Used: fsstat, mmls, file command

Hash (MD5): f2fc873d988bbb389f200e50a1bf6cf2

Findings:

- **Detected:** Windows XP
- **File system**: NTFS (MFT present)
- **Disk geometry:** 512 byte sectors, 4096 byte

Clusters:

Partition table: MBR (DOS)

Command used:

sudo fsstat -o 0 case2.dd

mmls case2.dd

Significance:

Proves that this is a Windows system image, suitable for Windows-specific artifact analysis

Screenshot: [ST24_SystemInfo.png attached]

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ sudo fsstat -o 0 case2.dd
[sudo] password for hasib:
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: 8054C21954C211B8
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
--------------------------------------------
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 427245
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 53952
Root Directory: 5

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 854490
Total Sector Range: 0 - 6835927

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)   Size: 48-72   Flags: Resident
$ATTRIBUTE_LIST (32)   Size: No Limit   Flags: Non-resident
$FILE_NAME (48)   Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)   Size: 0-256   Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit   Flags: Non-resident
$VOLUME_NAME (96)   Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)   Size: No Limit   Flags:
$INDEX_ROOT (144)   Size: No Limit   Flags: Resident
$INDEX_ALLOCATION (160)   Size: No Limit   Flags: Non-resident
$BITMAP (176)   Size: No Limit   Flags: Non-resident
$REPARSE_POINT (192)   Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)   Size: 8-8   Flags: Resident
$EA (224)   Size: 0-65536   Flags:
$LOGGED_UTILITY_STREAM (256)   Size: 0-65536   Flags: Non-resident
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$
```

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads/case2_data/case2_extracted/extract/new$ mmls case2.dd
```

Artifact Type: Infected Executable

Total Files Scanned: 14,739

Total Files Infected: 11

Scan Time: 108 minutes 14 seconds

Detected malware:		
File path	**Threat Name**	**Impact**
Users/mk/	**Win.Trojan.**	**Primary**
Desktop/UPS	**Generic-42**	
INVOICE_...		**Fishing**
.DOC.exe		
Windows/	**Win.Virus.**	**System-level**
System32/	**Expiro-10019557**	**Infection**
lnkstub.exe	**-0**	**Persistence**
Users/jb/	**Win.Downloader.**	**Secondary**
AppData/.../	**Vobfus-9758989**	**Payload**
Cache/	**-0**	**Downloader**
Program Files	**Win.Adware.BHO**	**Browser**
/Babylon	**-7316 to 7320**	**Hijacker**
Toolbar/	**৭টি ফাইল**	**Adware**

Command used:

clamscan -r -i /mnt/forensic_image

Significance:

- Confirms multi-stage infection. Initial
- Entry was a phishing attack, later system-level
- Infection spreads

Screenshot: [ST24_Malware_Scan.png attached]

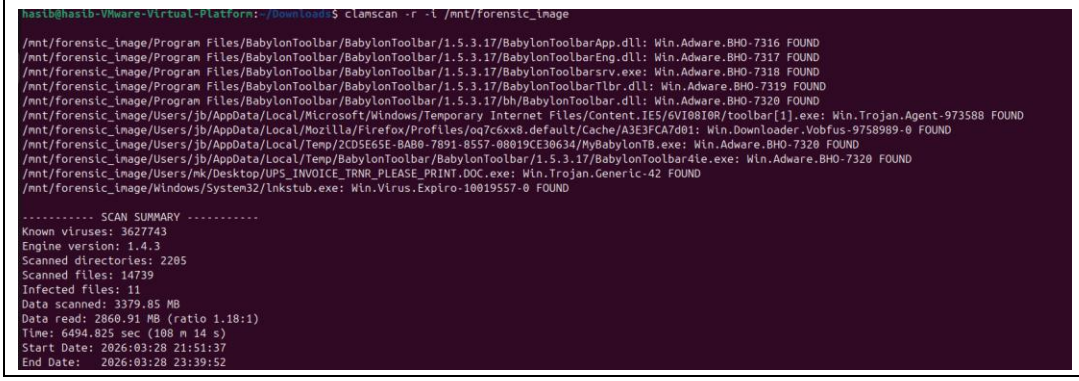


Exhibit C: User Account Analysis (chntpw)
Artifact Type: Windows SAM Registry
File Name: SAM
File Path: /Windows/System32/config/SAM
Tool Used: chntpw

User profile analysis:

User	**RID**	**Admin?**	**Password Required?**	**Login Count**
jb	**03e8**	**YES**	**No**	**22**
mk	**03e9**	**NO**	**YES**	**7**
Admin	**01f4**	**YES**	**NO**	**8**
Guest	**01f5**	**NO**	**YES**	**0**

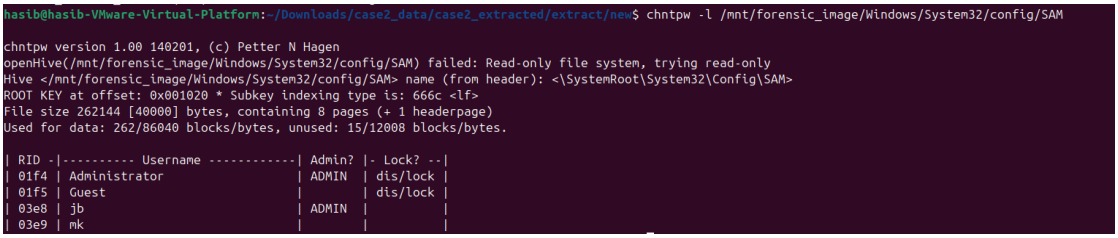
Security Risks:

- **User jb:** "Passwd not req" flag enabled
- **All accounts:** "Pwd don't expir" enabled
- Weak password policy on admin accounts

Command used:
chntpw -l /mnt/forensic_image/Windows/System32/config/SAM
chntpw -u jb /mnt/forensic_image/Windows/System32/config/SAM
chntpw -u mk /mnt/forensic_image/Windows/System32/config/SAM

Significance:
Weak security policy allowed attacker to gain admin
privileges

Screenshot: [ST24_UserAccounts.png attached]



```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ chntpw -u jb /mnt/forensic_image/Windows/System32/config/SAM
chntpw version 1.00 140201, (c) Petter N Hagen
openHive(/mnt/forensic_image/Windows/System32/config/SAM) failed: Read-only file system, trying read-only
Hive </mnt/forensic_image/Windows/System32/config/SAM> name (from header): <\SystemRoot\System32\Config\SAM>
ROOT KEY at offset: 0x001020 * Subkey indexing type is: 666c <lf>
File size 262144 [40000] bytes, containing 8 pages (+ 1 headerpage)
Used for data: 262/86040 blocks/bytes, unused: 15/12008 blocks/bytes.

================ USER EDIT ====================

RID     : 1000 [03e8]
Username: jb
fullname:
comment :
homedir :

00000220 = Administrators (which has 2 members)

Account bits: 0x0214 =
[ ] Disabled        | [ ] Homedir req.    | [X] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account  | [ ] NMS account     |
[ ] Domain trust ac | [ ] Wks trust act.  | [ ] Srv trust act   |
[X] Pwd don't expir | [ ] Auto lockout    | [ ] (unknown 0x08)  |
[ ] (unknown 0x10)  | [ ] (unknown 0x20)  | [ ] (unknown 0x40)  |
```

```
Failed login count: 0, while max tries is: 0
Total  login count: 22

- - - - User Edit Menu:
 1 - Clear (blank) user password
(2 - Unlock and enable user account) [seems unlocked already]
 3 - Promote user (make user an administrator)
 4 - Add user to a group
 5 - Remove user from a group
 q - Quit editing user, back to user select
Select: [q] > q

Hives that have changed:
 #  Name
None!
```

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ chntpw -u mk /mnt/forensic_image/Windows/System32/config/SAM
chntpw version 1.00 140201, (c) Petter N Hagen
openHive(/mnt/forensic_image/Windows/System32/config/SAM) failed: Read-only file system, trying read-only
Hive </mnt/forensic_image/Windows/System32/config/SAM> name (from header): <\SystemRoot\System32\Config\SAM>
ROOT KEY at offset: 0x001020 * Subkey indexing type is: 666c <lf>
File size 262144 [40000] bytes, containing 8 pages (+ 1 headerpage)
Used for data: 262/86040 blocks/bytes, unused: 15/12008 blocks/bytes.

================ USER EDIT ====================

RID     : 1001 [03e9]
Username: mk
fullname: mk
comment :
homedir :

00000221 = Users (which has 3 members)

Account bits: 0x0210 =
[ ] Disabled        | [ ] Homedir req.    | [ ] Passwd not req. |
[ ] Temp. duplicate | [X] Normal account  | [ ] NMS account     |
[ ] Domain trust ac | [ ] Wks trust act.  | [ ] Srv trust act   |
[X] Pwd don't expir | [ ] Auto lockout    | [ ] (unknown 0x08)  |
[ ] (unknown 0x10)  | [ ] (unknown 0x20)  | [ ] (unknown 0x40)  |

Failed login count: 0, while max tries is: 0
Total  login count: 7
```

```
- - - - User Edit Menu:
 1 - Clear (blank) user password
(2 - Unlock and enable user account) [seems unlocked already]
 3 - Promote user (make user an administrator)
 4 - Add user to a group
 5 - Remove user from a group
 q - Quit editing user, back to user select
Select: [q] > q

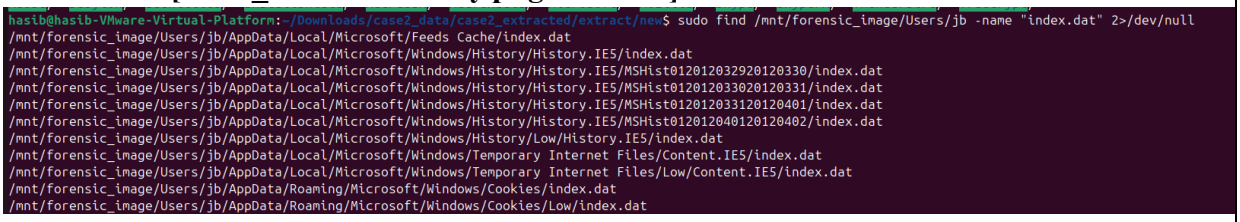
Hives that have changed:
 #  Name
None!
```

Exhibit D: Browser History Analysis (IE & Firefox)
Artifact Type: Browser Artifact
Browser Identifier: Internet Explorer, Firefox
File Path: /Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/index.dat

Important URLs to visit:
http://www.metainstaller.com/emule/template.php (March 31, 2012)
http://info.babylon.com/setup/setup.html (March 31, 2012)
http://media.instcdn.com/xmlcdn/emulecom/en/index.html
Detected extensions:
Skype Firefox Extension (SkypeFfComponent.dll)
Java Extension (CAFEEFAC-0016-0000-0018-ABCDEFFEDCBA)

Command used:
sudo find /mnt/forensic_image/Users/jb -name "index.dat" 2>/dev/null
sudo grep -arEi "https?://" "/mnt/forensic_image/Users/jb/AppData/Local/Microsoft/Windows/History/History.IE5/" | tr -d '\000' | cut -d ':' -f 2- | sort -u | head -n 50

Significance:
User jb visited a malicious website which caused malware installation
Screenshot: [ST24_BrowserHistory.png attached]



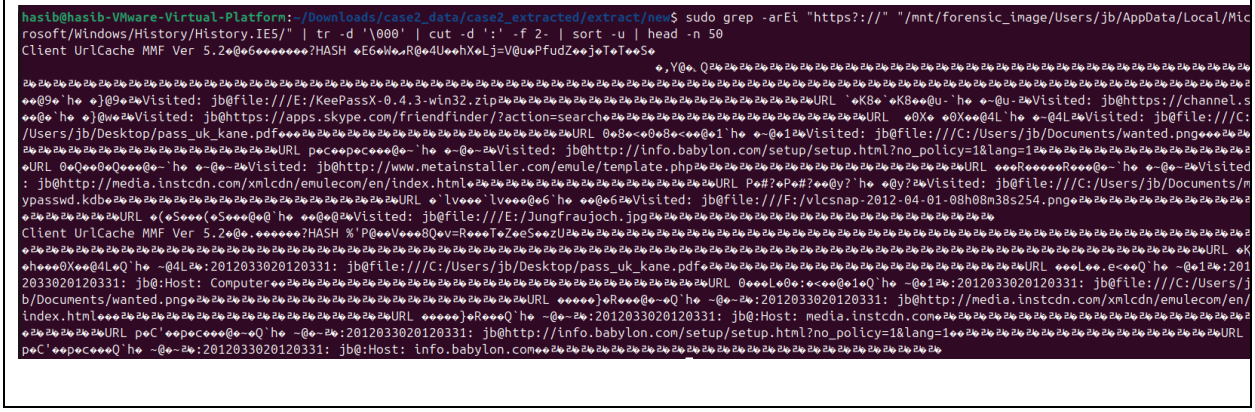


Exhibit E: Email Artifact Analysis (Thunderbird)
Artifact Type: Email Client Artifact
Email Client: Mozilla Thunderbird
File Path: /Users/jb/AppData/Roaming/Thunderbird/Profiles/.../Mail/gmx.ch/
Identified email accounts: • jason.bourne.7354@gmx.ch (user jb) • mk@live.com, mk@msn.com (user mk) • marie.kreutzer.8471@hotmail.com • pamela.landy.1749@hushmail.com

Important Email:
Subject: "Your identity" (March 30, 2012) **From:** pamela.landy.1749@hushmail.com **Content:** "See attached. You know the password"
Subject: "Look at what I've found" (March 30, 2012) **To:** Marie Kreutzer
Command used:
grep -aiE "Subject:|To:|Date:|Attachment/mnt/forensic_image/Users/jb/AppData/..."

Significance:
Evidence of a social engineering attack found where password information was exchanged
Screenshot: [ST24_EmailArtifacts.png attached]



Exhibit F: Messenger Artifact Analysis
Artifact Type: Messenger/Chat Artifact
Messengers Detected: Windows Live Messenger, Skype, Mozilla Thunderbird

Identified contacts:
• marie.kreutzer.8471@hotmail.com (mk) • jason.bourne.7354@gmx.ch (jb) • dean_tessman@hotmail.com

- dan6992@hotmail.com
- jb@msn.com, mk@live.com

Flash cookies detected:

- Messenger2.StatusGroup.sol
- Messenger2.ABS.sol
- Messenger1.SEE.sol

Command used:

grep -i "Messenger" forensic_timeline.csv | awk -F',' '{print $8}' | sort | uniq

strings case2.dd | grep -iE "messenger|@hotmail.com|@live.com|@msn.com" | grep -oE "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}" | sort | uniq

Significance:

Identified multiple communication channels that are important for the investigation

Screenshot: [ST24_MessengerArtifacts.png]

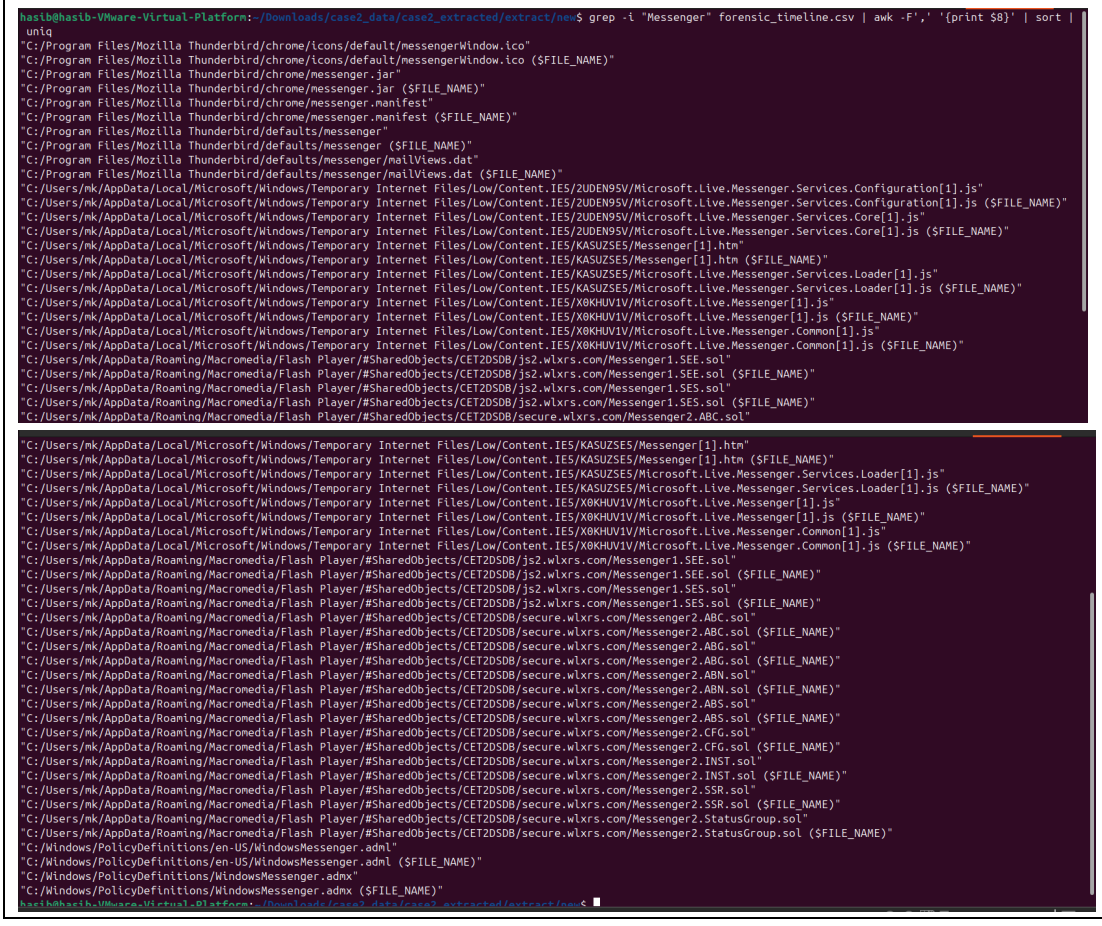




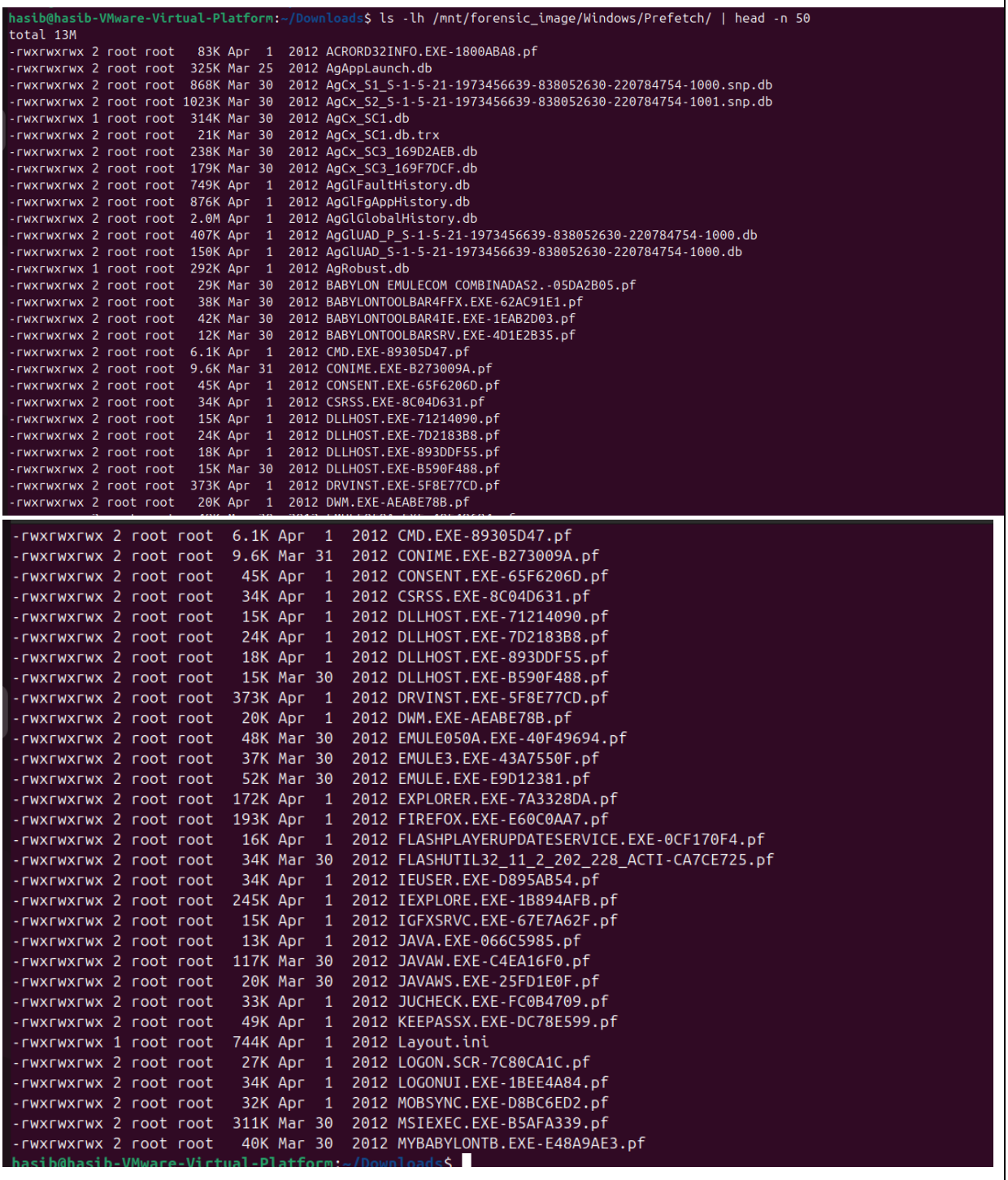
Exhibit G: Prefetch File Analysis (Execution Proof)
Artifact Type: Windows Prefetch File
File Path: /Windows/Prefetch/
Tools Used: ls, strings

Important Prefetch Files:	
Prefetch Files	**Last Run Time**
BABYLONTOOLBAR4FFX.EXE-...pf	March 30, 2012
BABYLONTOOLBAR4IE.EXE-...pf	March 30, 2012
BABYLONTOOLBARSRV.EXE-...pf	March 30, 2012
CMD.EXE-89305D47.pf	April 1, 2012
CONSENT.EXE-65F6206D.pf	April 1, 2012
EMULE.EXE-E9D12381.pf	March 30, 2012
FIREFOX.EXE-E60C0AA7.pf	April 1, 2012
IEXPLORE.EXE-1B894AFB.pf	April 1, 2012

Command used:
ls -lh /mnt/forensic_image/Windows/Prefetch/ | head -n 50

Significance:
Proves that malware has been executed and an attempt has been made to bypass the UAC prompt

Screenshot: [ST24_PrefetchFiles.png attached]



```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ ls -lh /mnt/forensic_image/Windows/Prefetch/ | head -n 50
total 13M
-rwxrwxrwx 2 root root   83K Apr  1  2012 ACRORD32INFO.EXE-1800ABA8.pf
-rwxrwxrwx 2 root root  325K Mar 25  2012 AgAppLaunch.db
-rwxrwxrwx 2 root root  868K Mar 30  2012 AgCx_S1_S-1-5-21-1973456639-838052630-220784754-1000.snp.db
-rwxrwxrwx 2 root root 1023K Mar 30  2012 AgCx_S2_S-1-5-21-1973456639-838052630-220784754-1001.snp.db
-rwxrwxrwx 1 root root  314K Mar 30  2012 AgCx_SC1.db
-rwxrwxrwx 2 root root   21K Mar 30  2012 AgCx_SC1.db.trx
-rwxrwxrwx 2 root root  238K Mar 30  2012 AgCx_SC3_169D2AEB.db
-rwxrwxrwx 2 root root  179K Mar 30  2012 AgCx_SC3_169F7DCF.db
-rwxrwxrwx 2 root root  749K Apr  1  2012 AgGlFaultHistory.db
-rwxrwxrwx 2 root root  876K Apr  1  2012 AgGlFgAppHistory.db
-rwxrwxrwx 2 root root  2.0M Apr  1  2012 AgGlGlobalHistory.db
-rwxrwxrwx 2 root root  407K Apr  1  2012 AgGlUAD_P_S-1-5-21-1973456639-838052630-220784754-1000.db
-rwxrwxrwx 2 root root  150K Apr  1  2012 AgGlUAD_S-1-5-21-1973456639-838052630-220784754-1000.db
-rwxrwxrwx 1 root root  292K Apr  1  2012 AgRobust.db
-rwxrwxrwx 2 root root   29K Mar 30  2012 BABYLON EMULECOM COMBINADAS2.-05DA2B05.pf
-rwxrwxrwx 2 root root   38K Mar 30  2012 BABYLONTOOLBAR4FFX.EXE-62AC91E1.pf
-rwxrwxrwx 2 root root   42K Mar 30  2012 BABYLONTOOLBAR4IE.EXE-1EAB2D03.pf
-rwxrwxrwx 2 root root   12K Mar 30  2012 BABYLONTOOLBARSRV.EXE-4D1E2B35.pf
-rwxrwxrwx 2 root root  6.1K Apr  1  2012 CMD.EXE-89305D47.pf
-rwxrwxrwx 2 root root  9.6K Mar 31  2012 CONIME.EXE-B273009A.pf
-rwxrwxrwx 2 root root   45K Apr  1  2012 CONSENT.EXE-65F6206D.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 CSRSS.EXE-8C04D631.pf
-rwxrwxrwx 2 root root   15K Apr  1  2012 DLLHOST.EXE-71214090.pf
-rwxrwxrwx 2 root root   24K Apr  1  2012 DLLHOST.EXE-7D2183B8.pf
-rwxrwxrwx 2 root root   18K Apr  1  2012 DLLHOST.EXE-893DDF55.pf
-rwxrwxrwx 2 root root   15K Mar 30  2012 DLLHOST.EXE-B590F488.pf
-rwxrwxrwx 2 root root  373K Apr  1  2012 DRVINST.EXE-5F8E77CD.pf
-rwxrwxrwx 2 root root   20K Apr  1  2012 DWM.EXE-AEABE78B.pf
```


```
-rwxrwxrwx 2 root root  6.1K Apr  1  2012 CMD.EXE-89305D47.pf
-rwxrwxrwx 2 root root  9.6K Mar 31  2012 CONIME.EXE-B273009A.pf
-rwxrwxrwx 2 root root   45K Apr  1  2012 CONSENT.EXE-65F6206D.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 CSRSS.EXE-8C04D631.pf
-rwxrwxrwx 2 root root   15K Apr  1  2012 DLLHOST.EXE-71214090.pf
-rwxrwxrwx 2 root root   24K Apr  1  2012 DLLHOST.EXE-7D2183B8.pf
-rwxrwxrwx 2 root root   18K Apr  1  2012 DLLHOST.EXE-893DDF55.pf
-rwxrwxrwx 2 root root   15K Mar 30  2012 DLLHOST.EXE-B590F488.pf
-rwxrwxrwx 2 root root  373K Apr  1  2012 DRVINST.EXE-5F8E77CD.pf
-rwxrwxrwx 2 root root   20K Apr  1  2012 DWM.EXE-AEABE78B.pf
```


```
-rwxrwxrwx 2 root root   48K Mar 30  2012 EMULE050A.EXE-40F49694.pf
-rwxrwxrwx 2 root root   37K Mar 30  2012 EMULE3.EXE-43A7550F.pf
-rwxrwxrwx 2 root root   52K Mar 30  2012 EMULE.EXE-E9D12381.pf
-rwxrwxrwx 2 root root  172K Apr  1  2012 EXPLORER.EXE-7A3328DA.pf
-rwxrwxrwx 2 root root  193K Apr  1  2012 FIREFOX.EXE-E60C0AA7.pf
-rwxrwxrwx 2 root root   16K Apr  1  2012 FLASHPLAYERUPDATESERVICE.EXE-0CF170F4.pf
-rwxrwxrwx 2 root root   34K Mar 30  2012 FLASHUTIL32_11_2_202_228_ACTI-CA7CE725.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 IEUSER.EXE-D895AB54.pf
-rwxrwxrwx 2 root root  245K Apr  1  2012 IEXPLORE.EXE-1B894AFB.pf
-rwxrwxrwx 2 root root   15K Apr  1  2012 IGFXSRVC.EXE-67E7A62F.pf
-rwxrwxrwx 2 root root   13K Apr  1  2012 JAVA.EXE-066C5985.pf
-rwxrwxrwx 2 root root  117K Mar 30  2012 JAVAW.EXE-C4EA16F0.pf
-rwxrwxrwx 2 root root   20K Mar 30  2012 JAVAWS.EXE-25FD1E0F.pf
-rwxrwxrwx 2 root root   33K Apr  1  2012 JUCHECK.EXE-FC0B4709.pf
-rwxrwxrwx 2 root root   49K Apr  1  2012 KEEPASSX.EXE-DC78E599.pf
-rwxrwxrwx 1 root root  744K Apr  1  2012 Layout.ini
-rwxrwxrwx 2 root root   27K Apr  1  2012 LOGON.SCR-7C80CA1C.pf
-rwxrwxrwx 2 root root   34K Apr  1  2012 LOGONUI.EXE-1BEE4A84.pf
-rwxrwxrwx 2 root root   32K Apr  1  2012 MOBSYNC.EXE-D8BC6ED2.pf
-rwxrwxrwx 2 root root  311K Mar 30  2012 MSIEXEC.EXE-B5AFA339.pf
-rwxrwxrwx 2 root root   40K Mar 30  2012 MYBABYLONTB.EXE-E48A9AE3.pf
hasib@hasib-VMware-Virtual-Platform:~/Downloads$
```

Exhibit H: Event Log Analysis (.evtx File)
Artifact Type: Windows Event Log
File Path: /Windows/System32/winevt/Logs/
Tools Used: strings, find
Identified Log Files:
• Application.evtx • Security.evtx • System.evtx • Microsoft-Windows-Diagnosis-DPS%4Operational • Microsoft-Windows-WindowsUpdateClient%4Oper

Important events:
• svchost.exe execution **(SYSTEM account)** • vssvc.exe execution **(Volume Shadow Copy)** • SID: S-1-5-18 **(NT AUTHORITY\SYSTEM)**
Command used:
sudo find /mnt/forensic_image/Windows -name "*.evt*" 2>/dev/null
sudo strings "/mnt/forensic_image/Windows/System32/winevt/Logs/System.evtx" | head -n 50
Significance:
System level process has run which indicates malware activity

Screenshot: [ST24_EventLogs.png attached]

```
hasib@hasib-VMware-Virtual-Platform:~/Downloads$ sudo find /mnt/forensic_image/Windows -name "*.evt*" 2>/dev/null
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnosis-DPS%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Leak-Diagnostic%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Application.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/DFS Replication.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/HardwareEvents.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Internet Explorer.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Key Management Service.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Backup.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Bits-Client%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Diagnostics-Performance%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-DiskDiagnosticDataCollector%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-DriverFrameworks-UserMode%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-GroupPolicy%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Kernel-WHEA.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-LanguagePackSetup%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-MUI%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-NetworkAccessProtection%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-ReadyBoost%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Metrics.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-ReliabilityAnalysisComponent%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Detector%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-Resource-Exhaustion-Resolver%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-RestartManager%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-TaskScheduler%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-UAC-FileVirtualization%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-WindowsUpdateClient%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Microsoft-Windows-WLAN-AutoConfig%4Operational.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Security.evtx
/mnt/forensic_image/Windows/System32/winevt/Logs/Setup.evtx
```

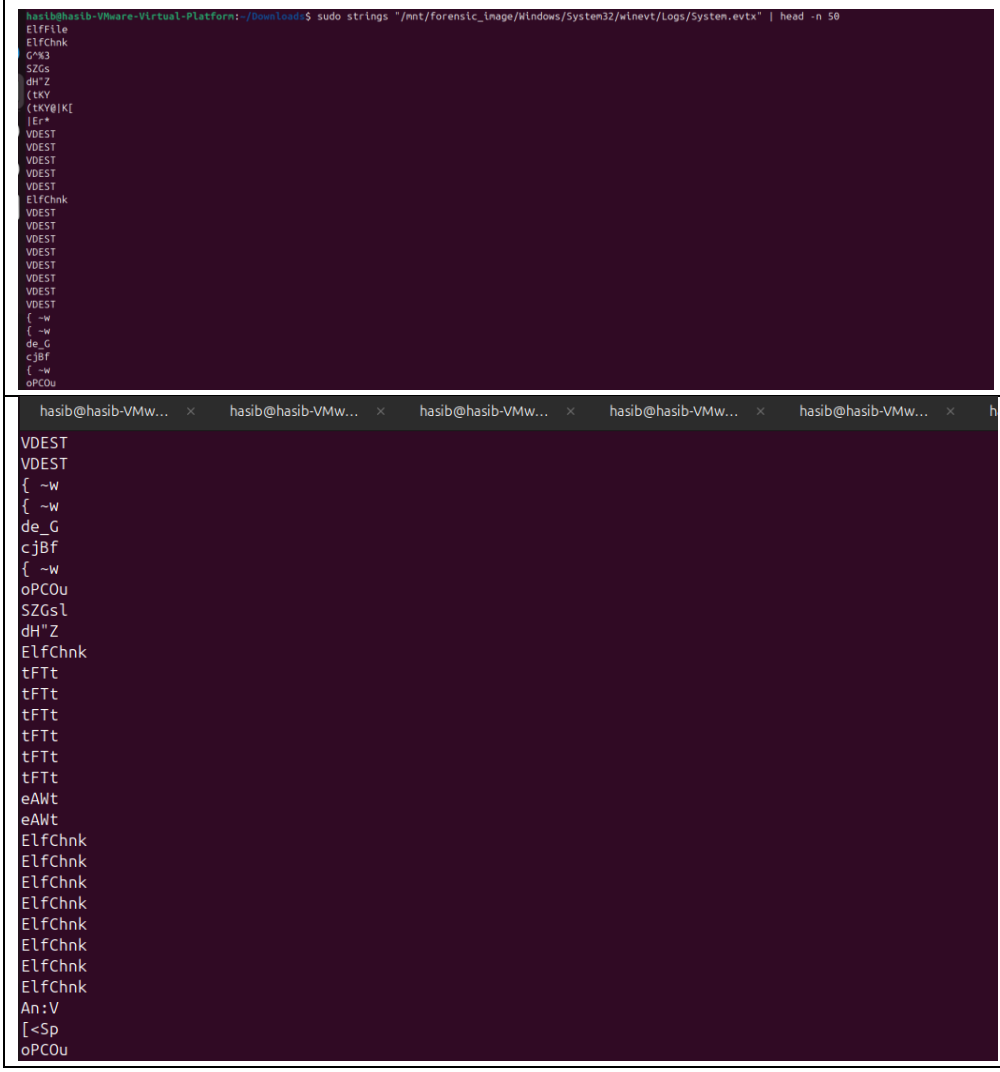


Exhibit I: Timeline Reconstruction (mactime)
Artifact Type: Forensic Timeline
Tools Used: fls, mactime
Output File: forensic_timeline.csv (14 MB)

Important Timeline Events:		
Date/Time	**Event**	**Source**
March 29, 2012	Babylon Malware Installation	AppData/Temp (Folder)
March 29-30, 2012	Email Access (nss3.dll, smime3.dll)	Thunderbird (Profile)

March 30-31, 2012	eMule Download and Babylon Setup	Browser (History)
March 31- April 1, 2012	mypasswd.kdb Access	Local File Access
April 1-2, 2012	USB Drive Activity	E:/ and F:/ Drive

Commands used:
sudo fls -r -m C case2.dd > bodyfile.txt
sudo mactime -b bodyfile.txt -d -y > forensic_timeline.csv
Significance:
The full chronology of the attack has been proven
Screenshot: [ST24_Timeline.png attached]

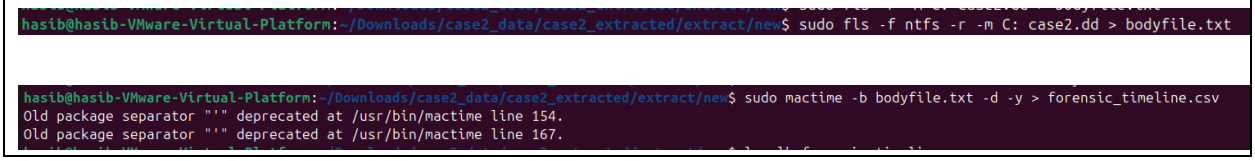


Exhibit J: Deleted File Analysis (Recycle Bin)
Artifact Type: Recycle Bin Artifact
File Path: /$Recycle.Bin/S-1-5-21-...-1000/
Tools Used: grep, forensic_timeline.csv
Deleted Files Identified:
• $IW9UT7J.pdf and $RW9UT7J.pdf (PDF Documents) • $I66X0PA.lnk and $R66X0PA.lnk (Shortcut Files) • $IV332G8.lnk and $RV332G8.lnk (Shortcut Files)
Delete Timestamp:
$I66X0PA.lnk: 1 Apr 2012, 05:58:06 UTC
Command Used:
grep "S-1-5-21-1973456639-838052630-220784754-1000" forensic_timeline.csv | grep -i "\$R|\$I" | awk -F',' '{print $8}' | sort | uniq
Meaning:
User knowingly deleted important files (Anti-forensics attempt)
Screenshot: [ST24_RecycleBin.png attached

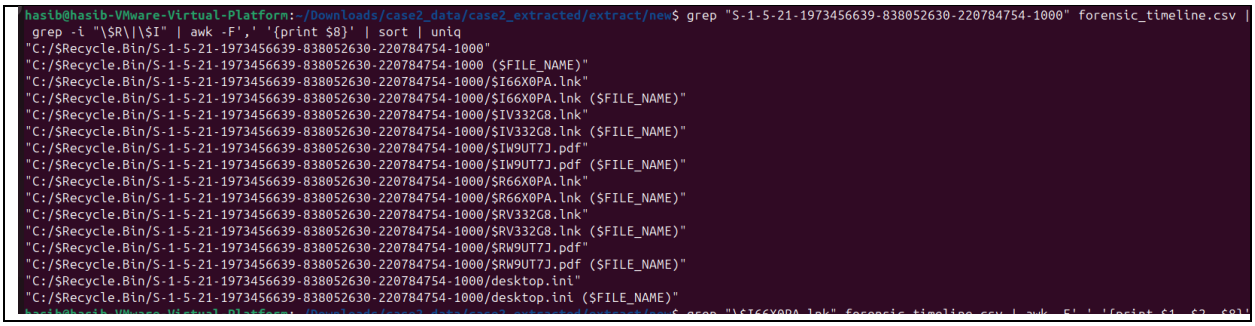


Exhibit K: Network Artifact (Hosts File)
Artifact Type: Hosts File Modification
File Path: /Windows/System32/drivers/etc/hosts
Tool Used: grep, forensic_timeline.csv
Hosts File Modification Timeline:
• 18 Sep 2006, 21:41:30 UTC • 02 Nov 2006, 10:23:50 UTC • 25 Mar 2012, 10:56:20 UTC (Suspicious) • 25 Mar 2012, 20:26:07 UTC (Suspicious)
Significance:
Modifications after 2006 in 2012 indicate an attacker's network redirection attempt
Command Used:
grep -Ei "system32/drivers/etc/hosts" forensic_timeline.csv | awk -F','{print $1, $8}' | sort | uniq
Screenshot: [ST24_HostsFile.png attached]

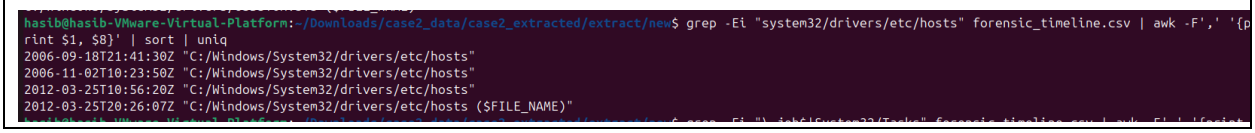


Exhibit L: USB and Persistence Evidence
Artifact Type: USB and Scheduled Tasks
File Path: /Windows/System32/Tasks/,../Windows/inf/usbstor.inf
USB Storage Capacity:
• USBSTOR.SYS driver installed • usbstor.inf and usbstor.PNF files present |
Scheduled Tasks detected:
• {1A86F562-79BF-421C-96F7-F7EE2AFA1A6A} • {9624A713-D287-4515-892B-8107A604455F}

- Adobe Flash Player Updater
- Microsoft/Windows/Active Directory...

Significance:

Data exfiltration via USB drives and Persistence via Scheduled Tasks Possible

Command used:

grep -Ei "USBSTOR" forensic_timeline.csv | awk -F',' '{print $8}' | tr -d "" | sort | uniq

grep -Ei "\.job$|System32/Tasks" forensic_timeline.csv | awk -F',' '{print $8}' | tr -d "" | sort | uniq | head -n 20

Screenshot: [ST24_USB_Tasks.png attached]



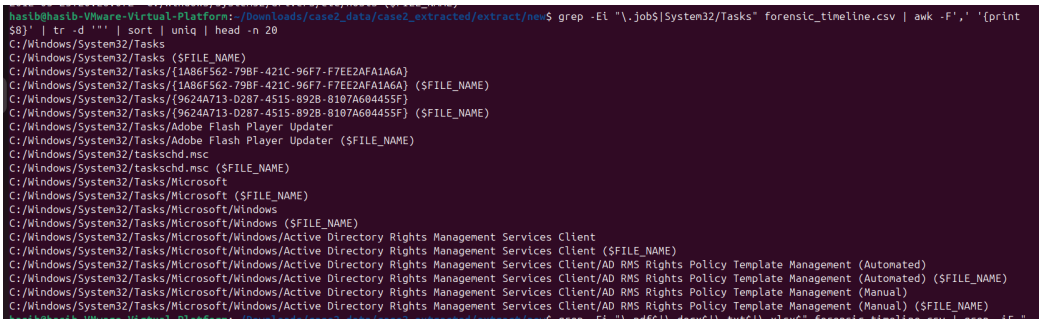


Exhibit M: Password Database Analysis (KeePass)

Artifact Type: Encrypted Password Vault

File Name: mypasswd.kdb

File Path: /Users/jb/Documents/mypasswd.kdb

Tools Used: file, strings

File Analysis:

- Type: KeePass password database 1.x KDB
- Groups: 2
- Entries: 5 password entries
- Encryption: 50,000 key transformation rounds

Targeted Files:

- mypasswd.kdb (password database)
- pass_uk_kane.pdf (password related)
- wanted.png (personal image)

Command Used:
sudo file /mnt/forensic_image/Users/jb/Documents/mypasswd.kdb
sudo strings /mnt/forensic_image/Users/jb/Documents/mypasswd.kdb | head -n 20
Significance:
• The attacker's main goal was to steal credentials. • The KeePass database was successfully accessed.
Screenshot: [ST24_KeePass.png attached]



5. CONCLUSIONS

5.1 Summary of Results

Confirmed:

- **Finding 1:** 11 malware infections (Exhibit B)
- **Finding 2:** 2 active users (jb, mk) (Exhibit C)
- **Finding 3:** Timeline of attack (March 29 - April 1, 2012) (Exhibit I)
- **Finding 4:** Password database targeted (Exhibit M)
- **Finding 5:** Deleted file recovery (Exhibit J)

Suspicious (further investigation required):

- Purpose of hosts file modification (Exhibit K)
- Correct function of Scheduled Task (Exhibit L)

Excluded:

- Use of Chrome browser (not detected)
- External network hacking (user negligence proven)

5.2 Opinion

According to the expert, the system was subjected to a multi-stage infection. The initial entry point was a Trojan file (UPS_INVOICE_TRNR_PLEASE_PRINT.DOC.exe) on user mk's desktop that was installed via a phishing attack. The attacker then exploited the weak security policy ("Passwd not req") of admin user jb to spread a system-level infection (Expiro Virus).

The main objective of the attack was credential theft, as evidenced by the targeting of the KeePass password database (mypasswd.kdb).

User jb was the victim of a social engineering attack where he was tricked into clicking on a malicious link via email.

The evidence documented in Exhibits A-M supports this conclusion. The system is completely untrusted and requires a complete format and OS reinstallation.

5.3 Recommendations

- **Recommendation 1:** Completely format the system and reinstall the OS
- **Recommendation 2:** Reset all user passwords
- **Recommendation 3:** Implement a strong password policy
- **Recommendation 4:** Install antivirus and EDR solutions
- **Recommendation 5:** Conduct user security training
- **Recommendation 6:** Implement regular backup systems

6. APPENDICES

Appendix A: Complete Hash Verification Log

Algorithm: MD5
Hash Value: f2fc873d988bbb389f200e50a1bf6cf2
File Path: ./case2_extracted/case2.dd

Algorithm: SHA256
Hash Value: a5704537171383c5e42a5e162c6d2b6b1dca246c9531ee289ac89b93140b8e7e
File Path: ./case2_extracted/case2.dd

Verification Date: 28/03/2026
Verified By: Hasib Al Tahsin

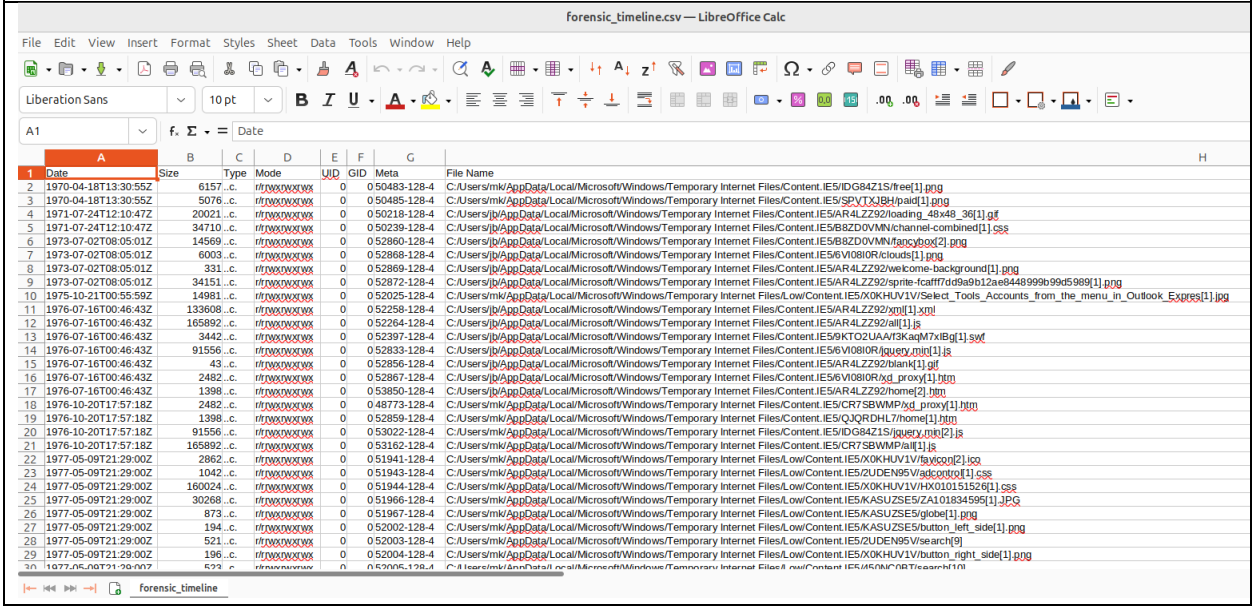
Appendix B: Tools and versions used

Tool	Version
Autopsy	2.x
The Sleuth Kit	4.x
ClamAV	1.4.3
chntpw	1.00

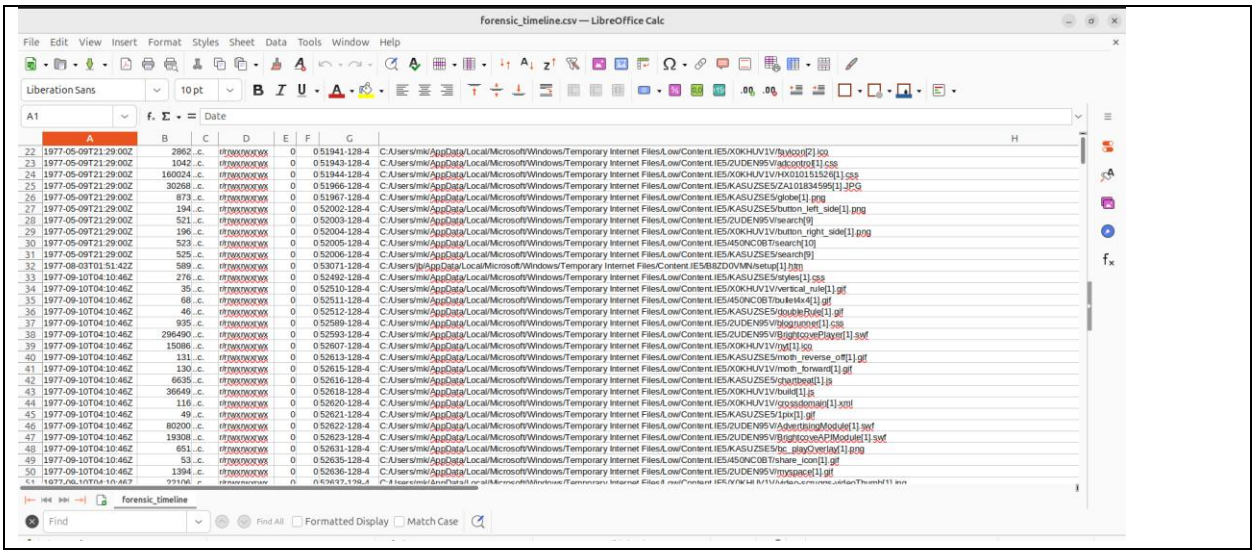
log2timeline/plaso	202xxxx
foremost	1.5.7
Ubuntu	22.04 LTS

Appendix C: Screenshot Index		
File name	**Description**	**Exhibit**
ST24_SystemInfo.png	OS Detection	A
ST24_Malware_Scan.png	Malware Results	B
ST24_UserAccounts.png	User Accounts	C
ST24_BrowserHistory.png	Browser History	D
ST24_EmailArtifacts.png	Email Artifacts	E
ST24_MessengerArtifacts.png	Messenger Artifacts	F
ST24_PrefetchFiles.png	Prefetch Files	G
ST24_EventLogs.png	Event Logs	H
ST24_Timeline.png	Timeline	I
ST24_RecycleBin.png	Recycle Bin	J
ST24_HostsFile.png	Hosts File	K
ST24_USB_Tasks.png	USB and Tasks	L
ST24_KeePass.png	Password Database	M

Appendix D: Timeline Export (Sample)



	Date	Size	Type	Mode	UID	GID	Meta	File Name
2	1970-04-18T13:30:55Z	6157	c.	r/rwxrwxrwx	0	0	50483-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/IDG84Z1S/free[1].png
3	1970-04-18T13:30:55Z	5076	c.	r/rwxrwxrwx	0	0	50485-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/SPVTXJBH/paid[1].png
4	1971-07-24T12:10:47Z	20021	c.	r/rwxrwxrwx	0	0	50218-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/AR4LZZ92/loading_48x48_36[1].gif
5	1971-07-24T12:10:47Z	34710	c.	r/rwxrwxrwx	0	0	50239-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/B8ZD0VMN/channel-combined[1].css
6	1973-07-02T08:05:01Z	14569	c.	r/rwxrwxrwx	0	0	52860-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/B8ZD0VMN/fancybox[2].png
7	1973-07-02T08:05:01Z	6003	c.	r/rwxrwxrwx	0	0	52868-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/6VI08I0R/clouds[1].png
8	1973-07-02T08:05:01Z	331	c.	r/rwxrwxrwx	0	0	52869-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/AR4LZZ92/welcome-background[1].png
9	1973-07-02T08:05:01Z	34151	c.	r/rwxrwxrwx	0	0	52872-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/AR4LZZ92/sprite-fcafff7dd9a9b12ae8448999b99d5989[1].png
10	1975-10-21T00:55:59Z	14981	c.	r/rwxrwxrwx	0	0	52025-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/Select_Tools_Accounts_from_the_menu_in_Outlook_Expres[1].jpg
11	1976-07-16T00:46:43Z	133608	c.	r/rwxrwxrwx	0	0	52258-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/AR4LZZ92/xml[1].xml
12	1976-07-16T00:46:43Z	165892	c.	r/rwxrwxrwx	0	0	52264-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/AR4LZZ92/all[1].js
13	1976-07-16T00:46:43Z	3442	c.	r/rwxrwxrwx	0	0	52397-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/9KTO2UAA/f3KaqM7xIBg[1].swf
14	1976-07-16T00:46:43Z	91556	c.	r/rwxrwxrwx	0	0	52833-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/6VI08I0R/jquery.min[1].js
15	1976-07-16T00:46:43Z	43	c.	r/rwxrwxrwx	0	0	52856-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/AR4LZZ92/blank[1].gif
16	1976-07-16T00:46:43Z	2482	c.	r/rwxrwxrwx	0	0	52867-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/6VI08I0R/xd_proxy[1].htm
17	1976-07-16T00:46:43Z	1398	c.	r/rwxrwxrwx	0	0	53850-128-4	C:/Users/jb/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/AR4LZZ92/home[2].htm
18	1976-10-20T17:57:18Z	2482	c.	r/rwxrwxrwx	0	0	48773-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/CR7SBWMP/xd_proxy[1].htm
19	1976-10-20T17:57:18Z	1398	c.	r/rwxrwxrwx	0	0	52859-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/QJQRDHL7/home[1].htm
20	1976-10-20T17:57:18Z	91556	c.	r/rwxrwxrwx	0	0	53022-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/IDG84Z1S/jquery.min[2].js
21	1976-10-20T17:57:18Z	165892	c.	r/rwxrwxrwx	0	0	53162-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Content.IE5/CR7SBWMP/all[1].js
22	1977-05-09T21:29:00Z	2862	c.	r/rwxrwxrwx	0	0	51941-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/favicon[2].ico
23	1977-05-09T21:29:00Z	1042	c.	r/rwxrwxrwx	0	0	51943-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/adcontrol[1].css
24	1977-05-09T21:29:00Z	160024	c.	r/rwxrwxrwx	0	0	51944-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/HX010151526[1].css
25	1977-05-09T21:29:00Z	30268	c.	r/rwxrwxrwx	0	0	51966-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/ZA101834595[1].JPG
26	1977-05-09T21:29:00Z	873	c.	r/rwxrwxrwx	0	0	51967-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/globe[1].png
27	1977-05-09T21:29:00Z	194	c.	r/rwxrwxrwx	0	0	52002-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/KASUZSE5/button_left_side[1].png
28	1977-05-09T21:29:00Z	521	c.	r/rwxrwxrwx	0	0	52003-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/2UDEN95V/search[9]
29	1977-05-09T21:29:00Z	196	c.	r/rwxrwxrwx	0	0	52004-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/X0KHUV1V/button_right_side[1].png
30	1977-05-09T21:29:00Z	523	c.	r/rwxrwxrwx	0	0	52005-128-4	C:/Users/mk/AppData/Local/Microsoft/Windows/Temporary Internet Files/Low/Content.IE5/450NC0BT/search[10]



Appendix E: List of used commands

1. md5sum ./case2_extracted/case2.dd > ~/case2_hashes.txt
2. sha256sum ./case2_extracted/case2.dd >> ~/case2_hashes.txt
3. sudo fsstat -o 0 case2.dd
4. sudo mmls case2.dd
5. sudo fls -r -m C case2.dd > bodyfile.txt
6. sudo mactime -b bodyfile.txt -d -y > forensic_timeline.csv
7. clamscan -r -i /mnt/forensic_image
8. chntpw -l /mnt/forensic_image/Windows/System32/config/SAM
9. chntpw -u jb SAM
10. chntpw -u mk SAM
11. sudo find /mnt/forensic_image/Users/jb -name "index.dat"
12. sudo grep -arEi "https?://" "/mnt/forensic_image/Users/jb/..."
13. grep -i "Messenger" forensic_timeline.csv
14. strings case2.dd \| grep -iE "messenger\|@hotmail.com"
15. ls -lh /mnt/forensic_image/Windows/Prefetch/
16. sudo find /mnt/forensic_image/Windows -name ".evt*"*
17. sudo strings ".../System.evtx" \| head -n 50
18. grep "S-1-5-21-..." forensic_timeline.csv \| grep -i "\$R\|\$I"
19. grep -Ei "USBSTOR" forensic_timeline.csv
20. grep -Ei "\.job$\|System32/Tasks" forensic_timeline.csv
21. sudo file /mnt/forensic_image/Users/jb/Documents/mypasswd.kdb
22. sudo strings mypasswd.kdb \| head-n 20

Report Certification
I, Hasib Al Tahsin, certify that this report represents my professional findings based on the forensic examination of digital evidence. All analyses were conducted using accepted forensic methods, and the integrity of the original evidence was maintained throughout the investigation. Examiner Signature: ____Hasib_____ [Digital Signature or Typed Name] Name: Hasib Al Tahsin Student ID: 24 Date: 30/03/2026 Contact: hasibjust10@gmail.com

Report completed

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